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(54) **NEEDLE ACTUATOR ASSEMBLY FOR
DRUG DELIVERY SYSTEM**

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(51) **Int. Cl.**

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A61M 5/142 (2006.01)

A61M 5/158 (2006.01)

A61M 5/20 (2006.01)

A61M 5/34 (2006.01)

(52) **U.S. Cl.**

CPC *A61M 5/32* (2013.01); *A61M 5/14248*

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2005/14252 (2013.01); *A61M 2005/1585*

(2013.01); *A61M 2005/206* (2013.01); *A61M*

5/348 (2013.01)

(71) Applicant: **Becton, Dickinson and Company,**
Franklin Lakes, NJ (US)

(72) Inventors: **Michael Vincent Quinn**, East Hanover,
NJ (US); **Mohammed Yassir Mahdi**,
Stamford, CT (US); **Oliver Charles**
Gazeley, Leamington Spa (GB); **James**
Alexander Senior, Warwick (GB);
Malcolm Stanley Boyd, Wellesbourne
(GB)

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(57)

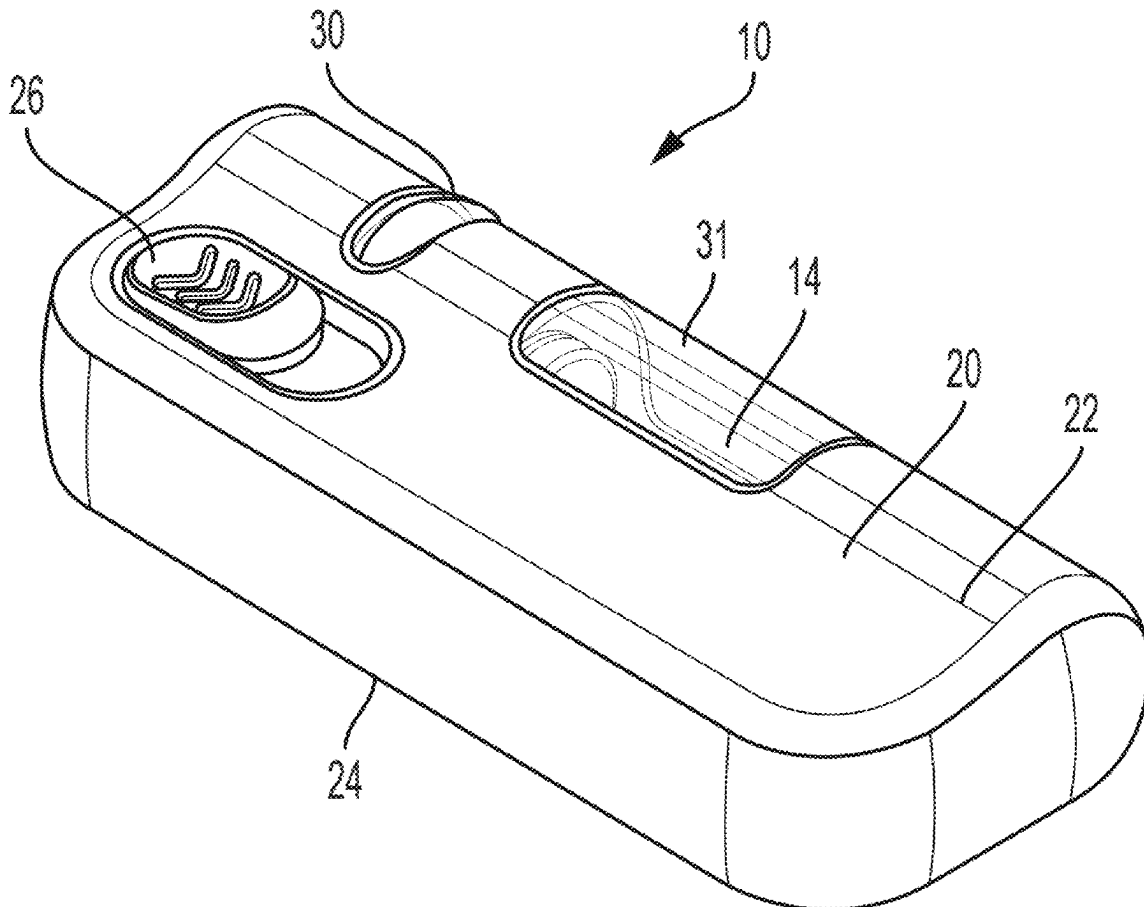
ABSTRACT

Related U.S. Application Data

(63) Continuation of application No. 18/134,841, filed on
Apr. 14, 2023, now Pat. No. 12,329,950, which is a
continuation of application No. 16/805,996, filed on
Mar. 2, 2020, now Pat. No. 11,654,248, which is a
continuation of application No. 15/616,212, filed on
Jun. 7, 2017, now Pat. No. 10,603,445.

(60) Provisional application No. 62/347,921, filed on Jun.
9, 2016.

A needle actuator assembly for a drug delivery system includes a needle actuator body having a guide surface, a needle shuttle having a cam surface, with the needle shuttle moveable along a vertical axis between a first position and a second position. The needle shuttle is configured to move between the first and second position through engagement between the guide surface of the needle actuator body and the cam surface of the needle shuttle. The assembly further including a needle received by the needle shuttle.



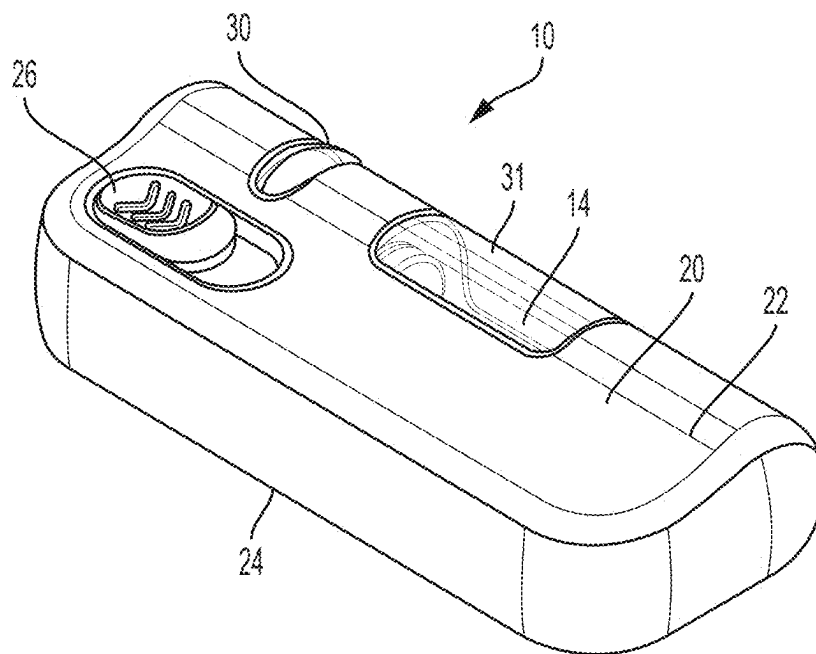


FIG. 1

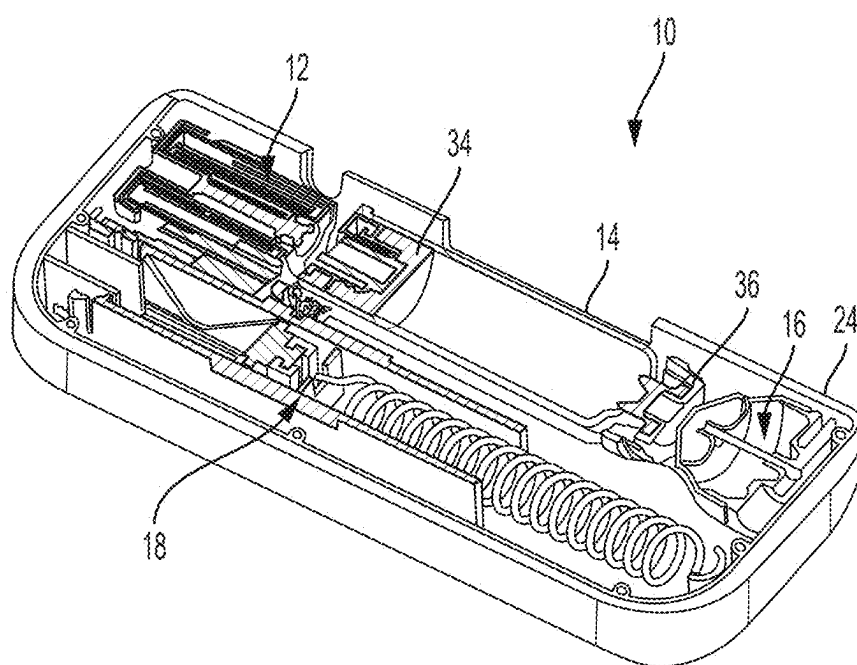


FIG. 2

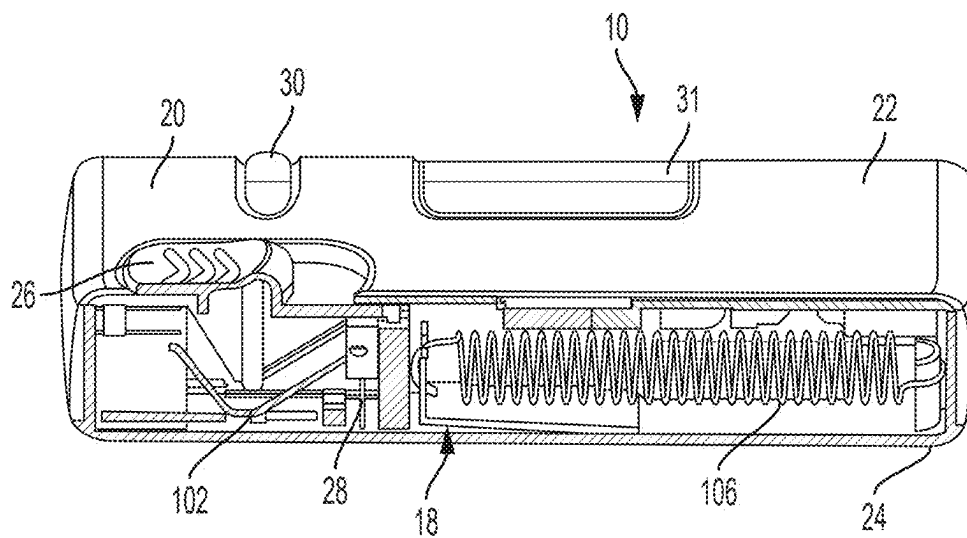


FIG. 3

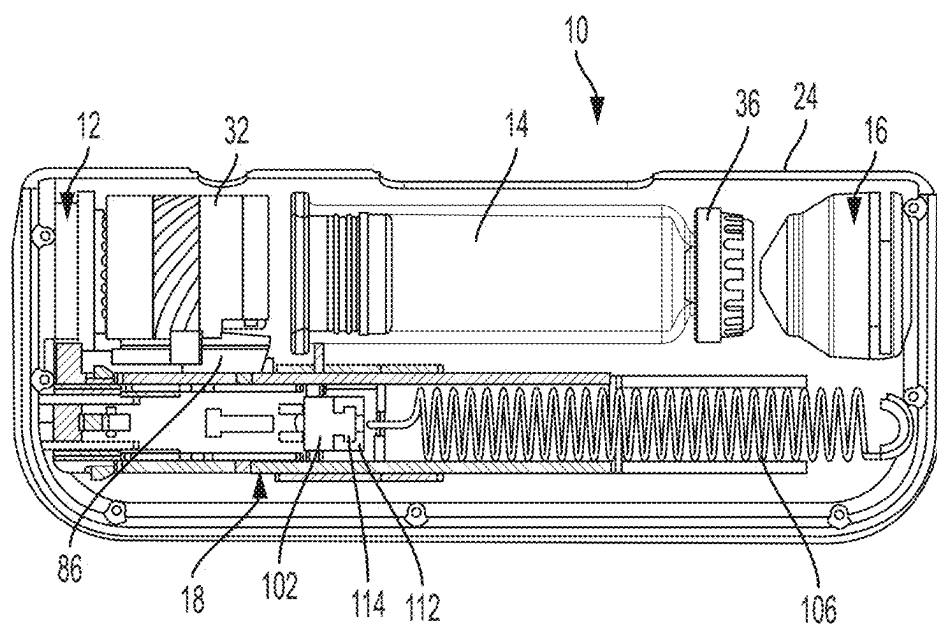


FIG. 4

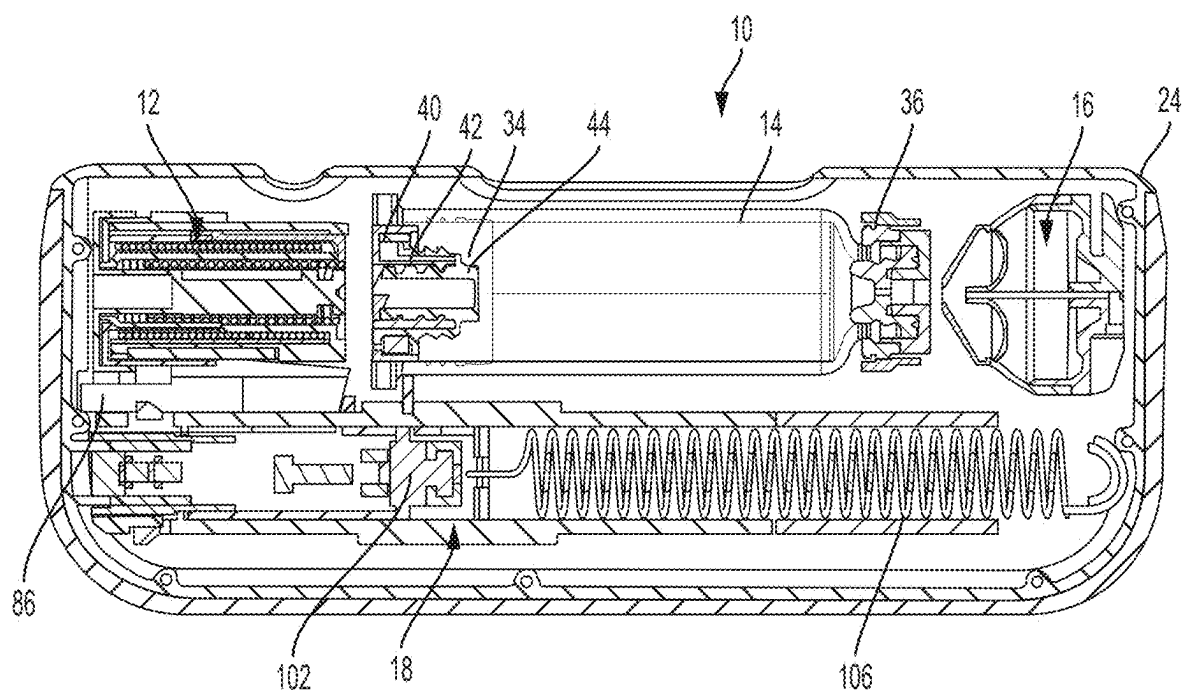


FIG. 5

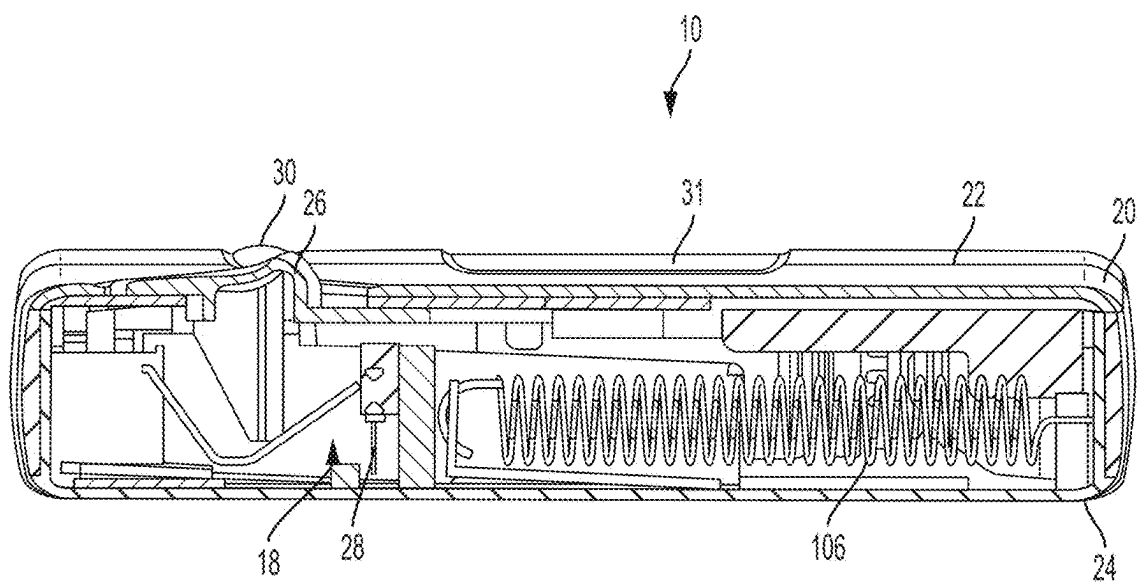


FIG. 6

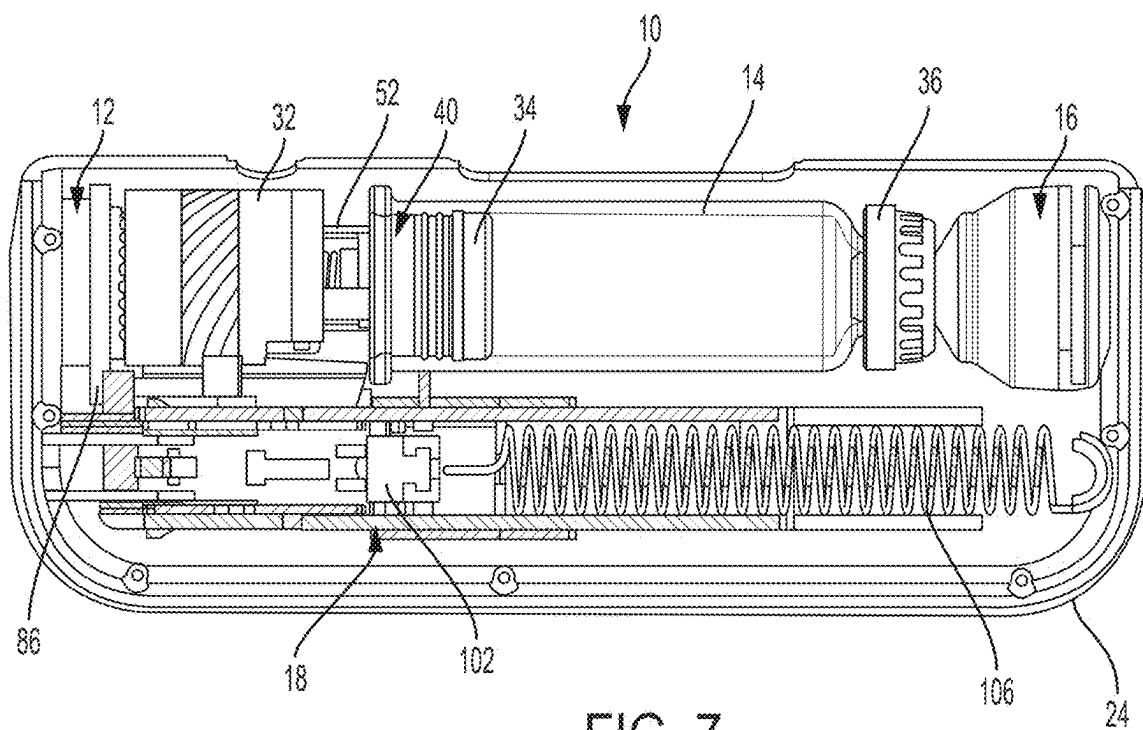


FIG. 7

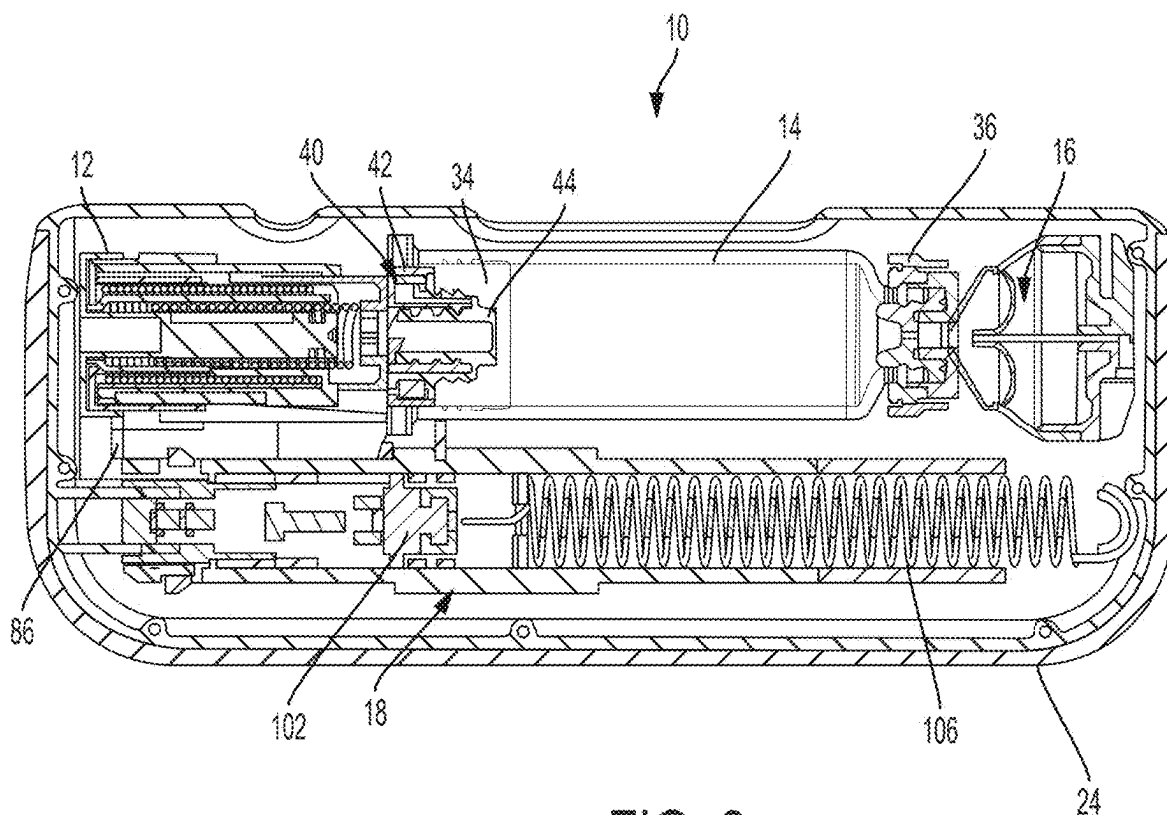


FIG. 8

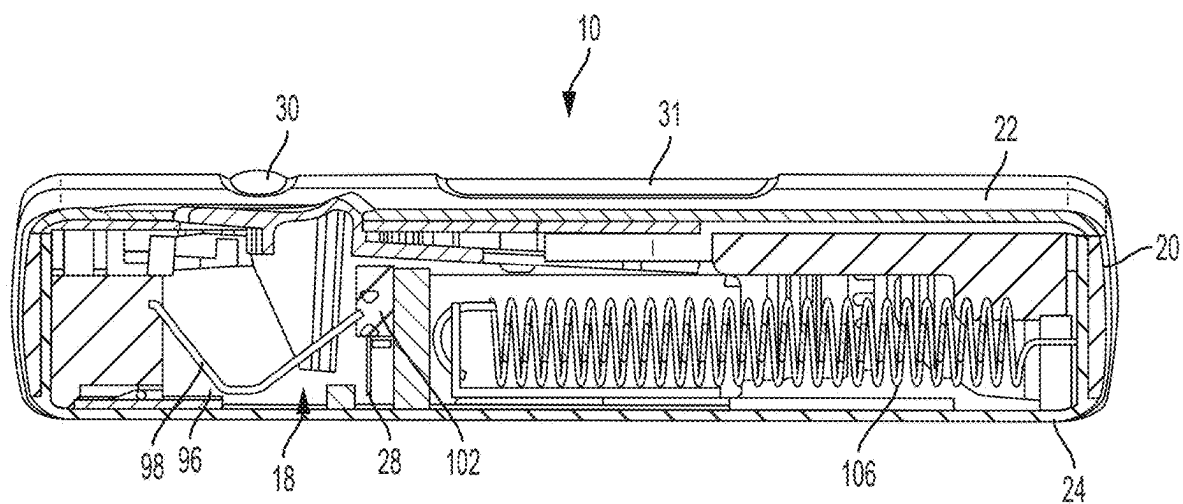


FIG. 9

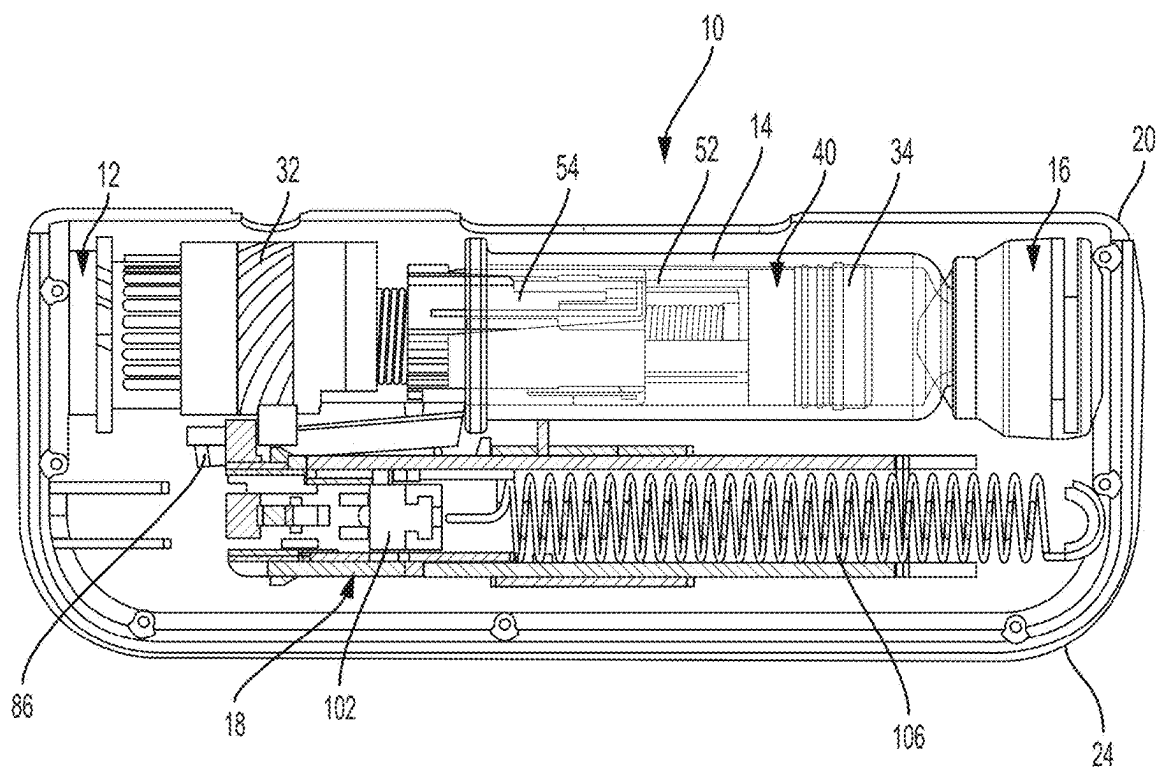


FIG. 10

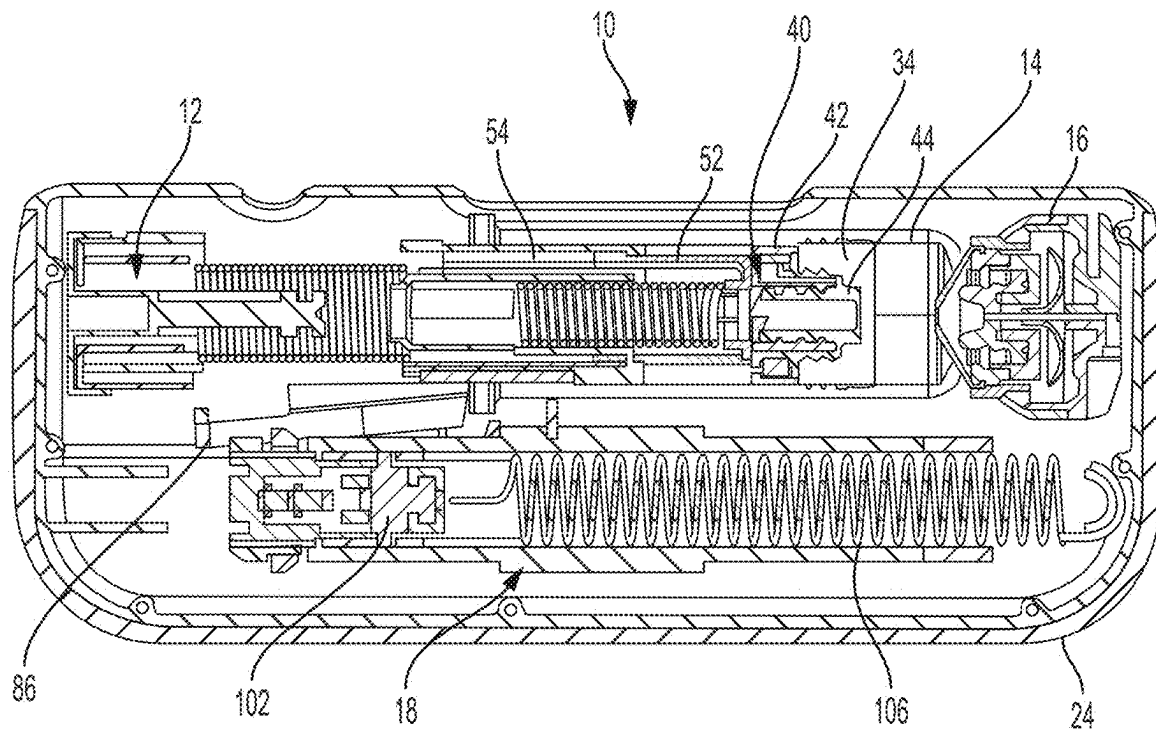


FIG. 11

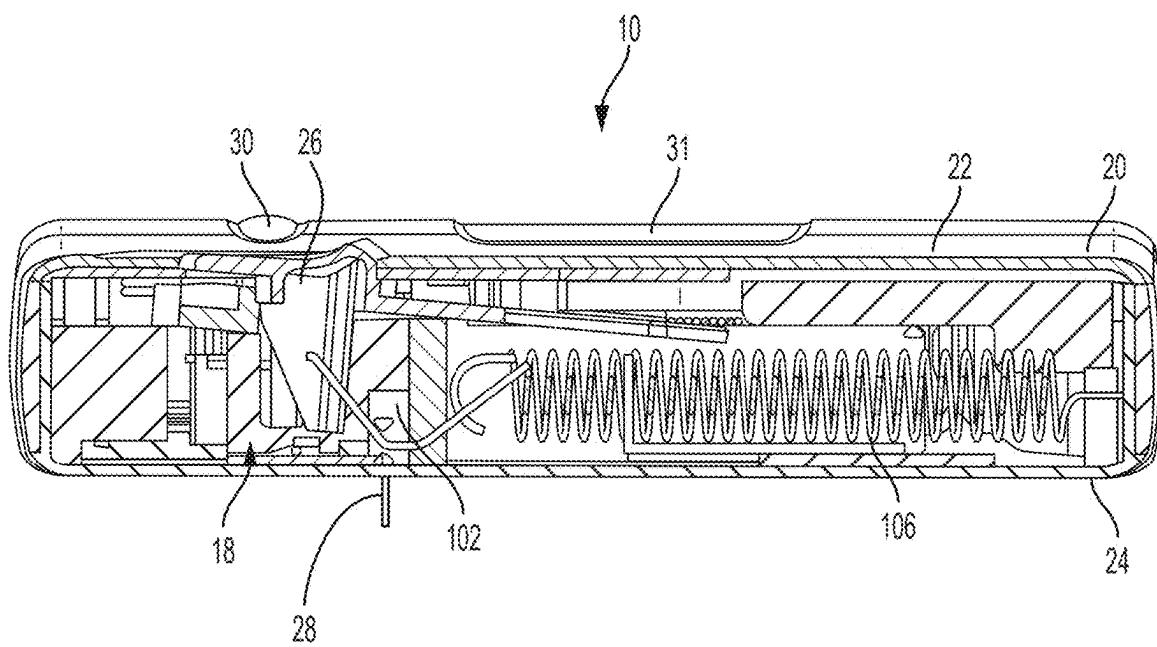


FIG. 12

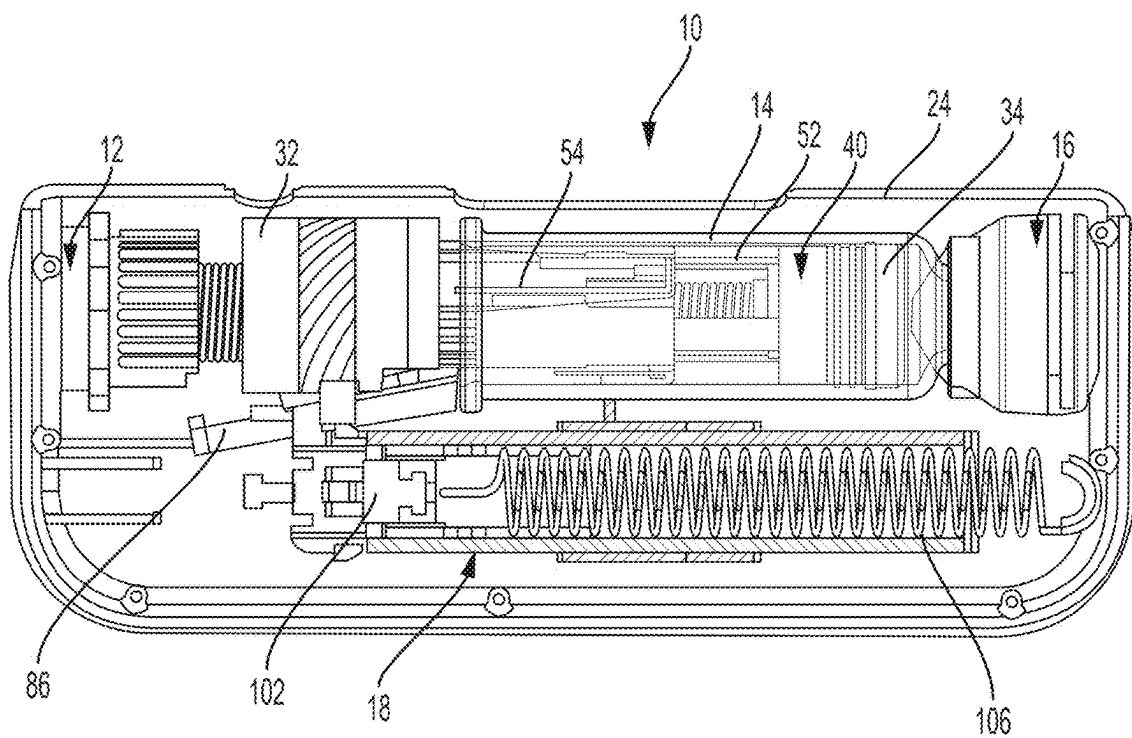


FIG. 13

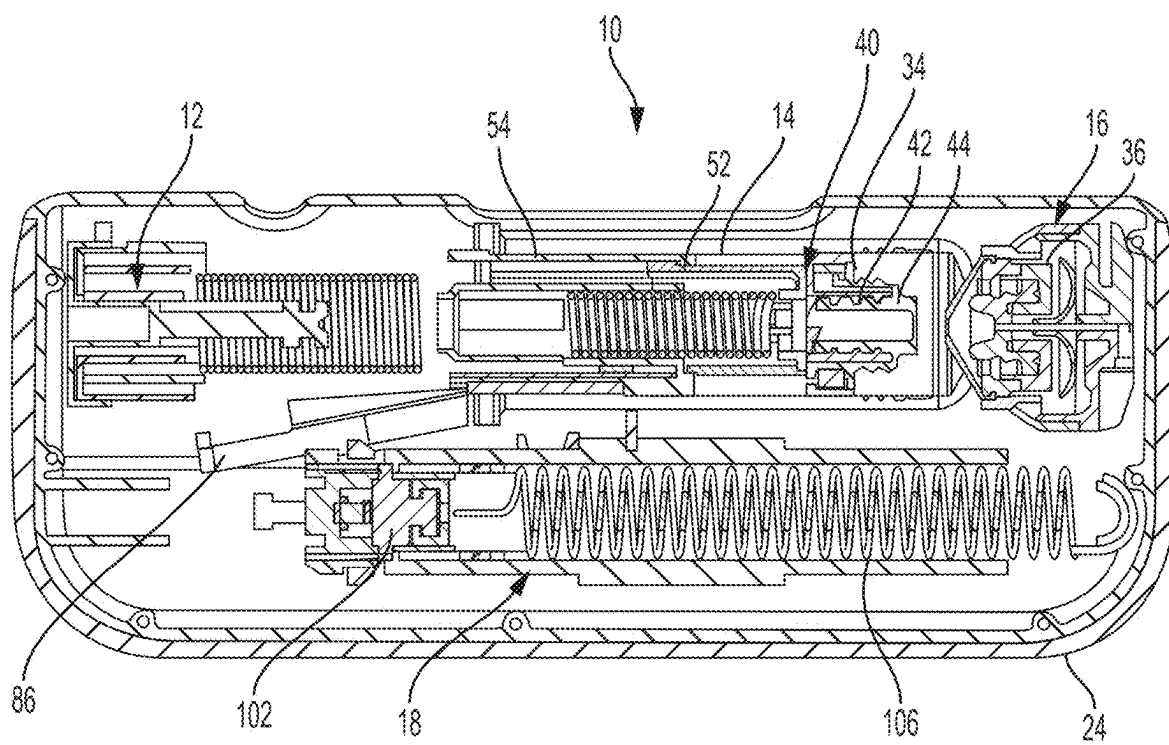


FIG. 14

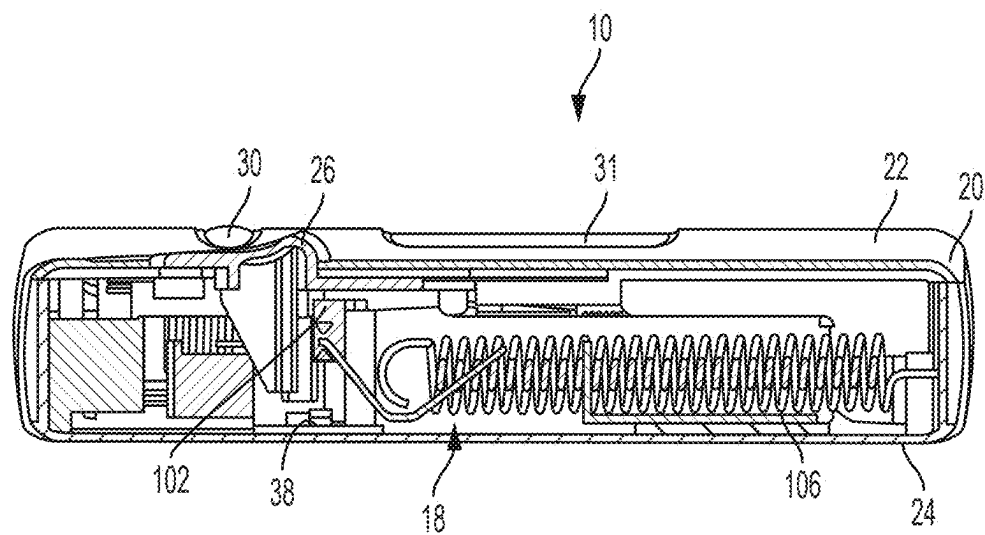


FIG. 15

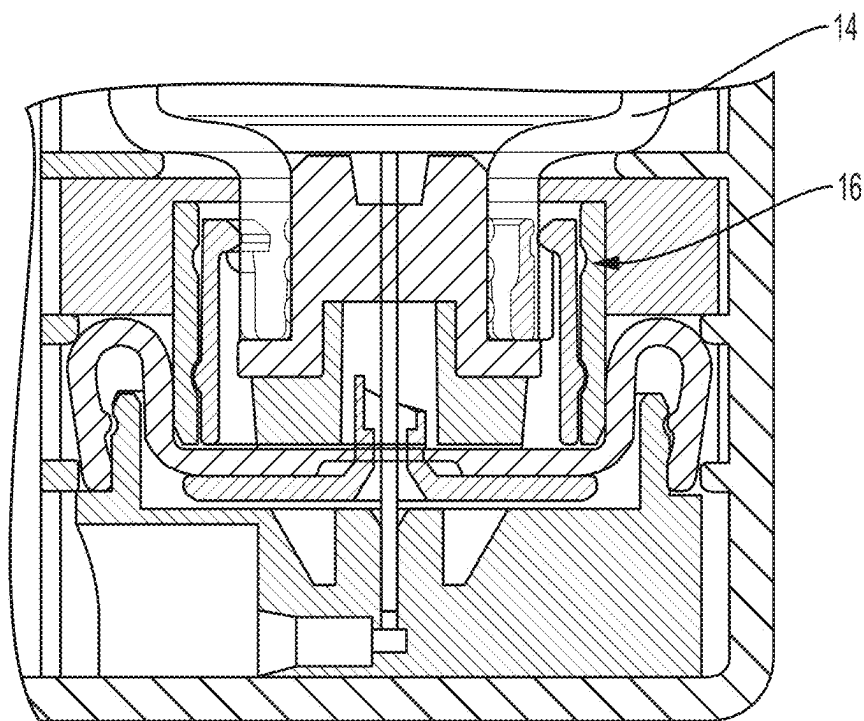
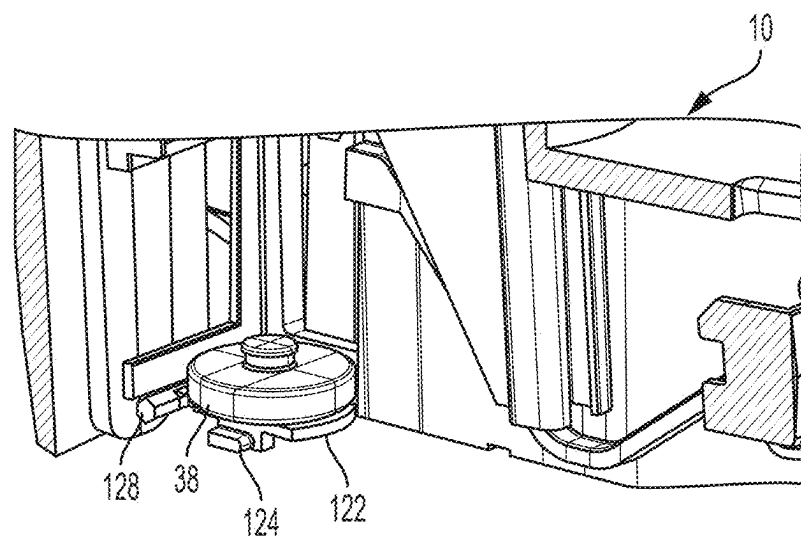
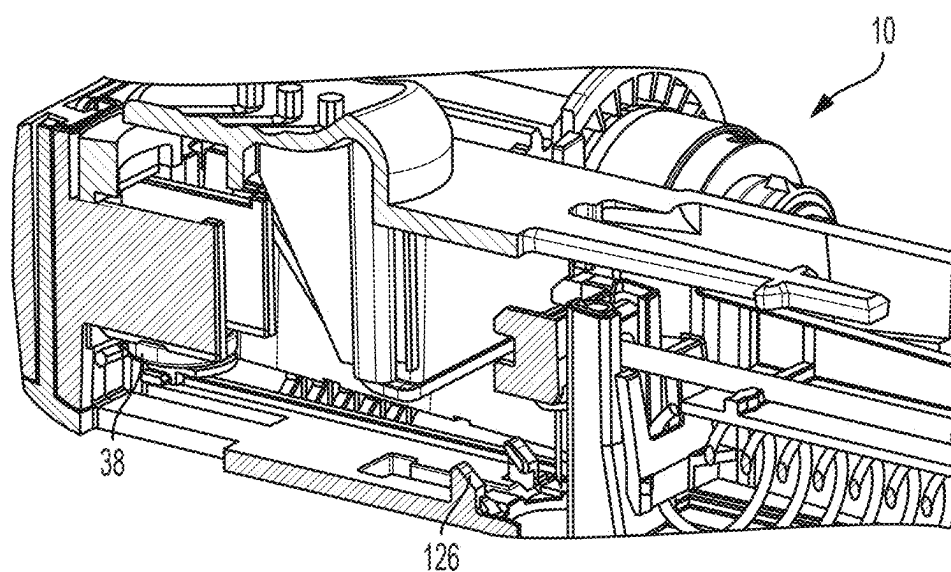
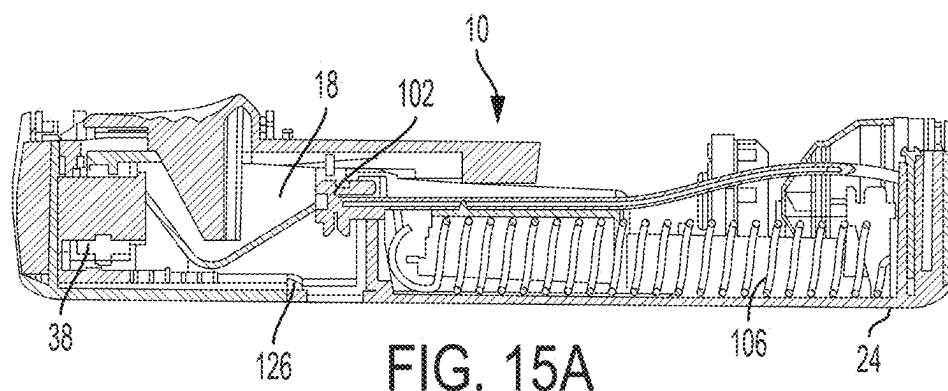


FIG. 16



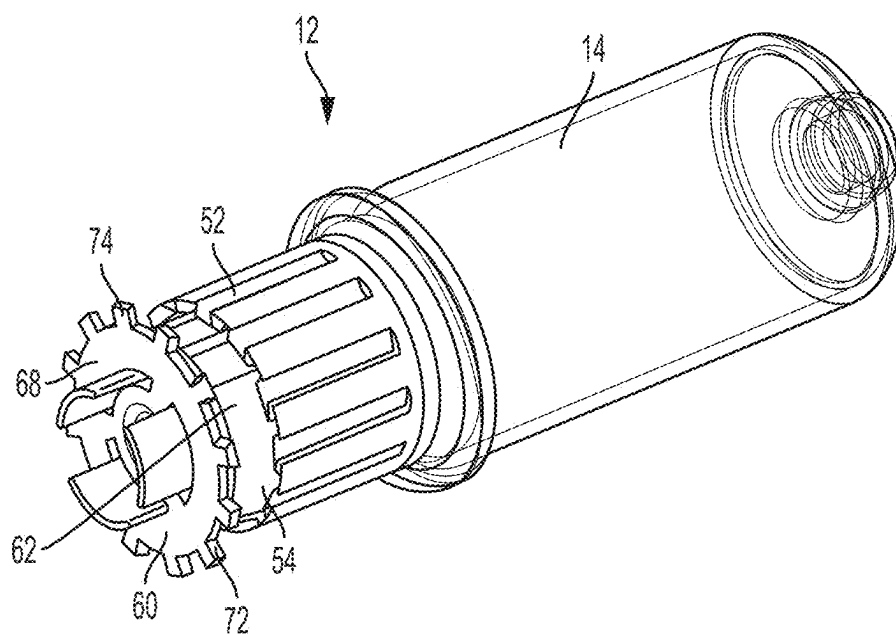


FIG. 17

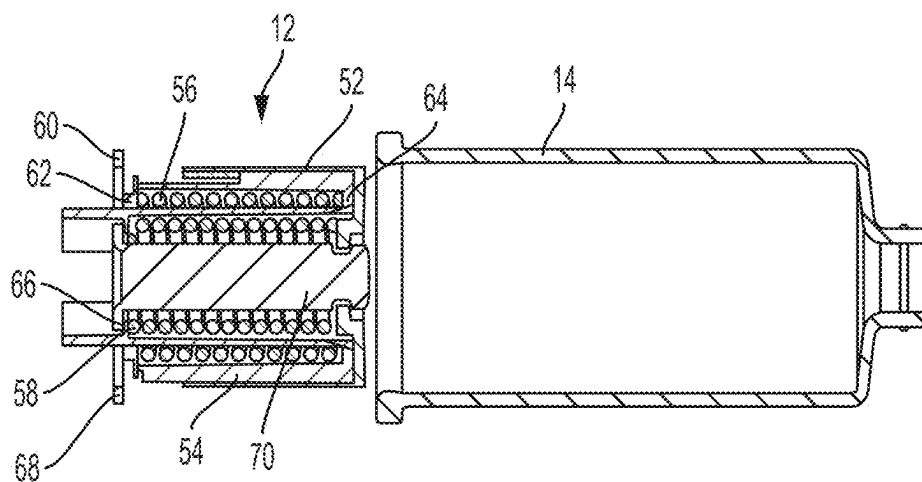


FIG. 18

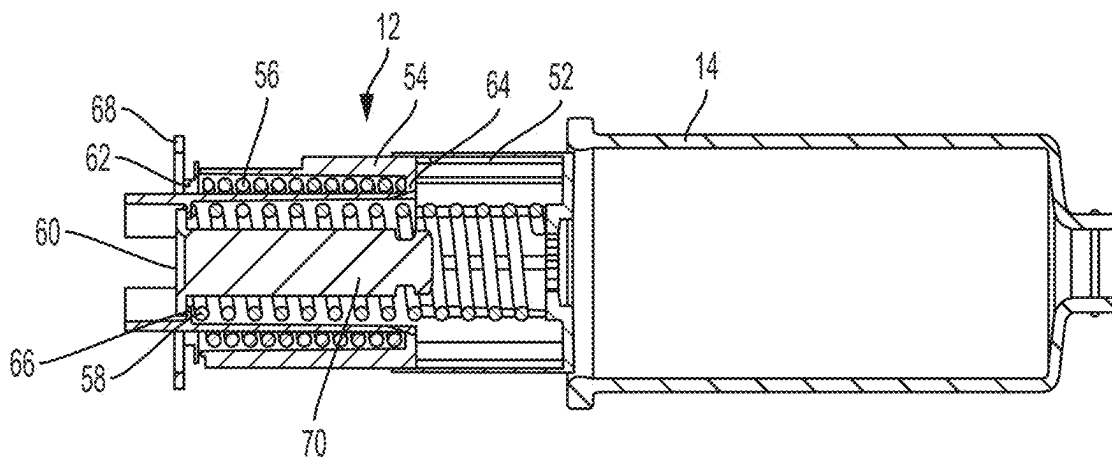


FIG. 19

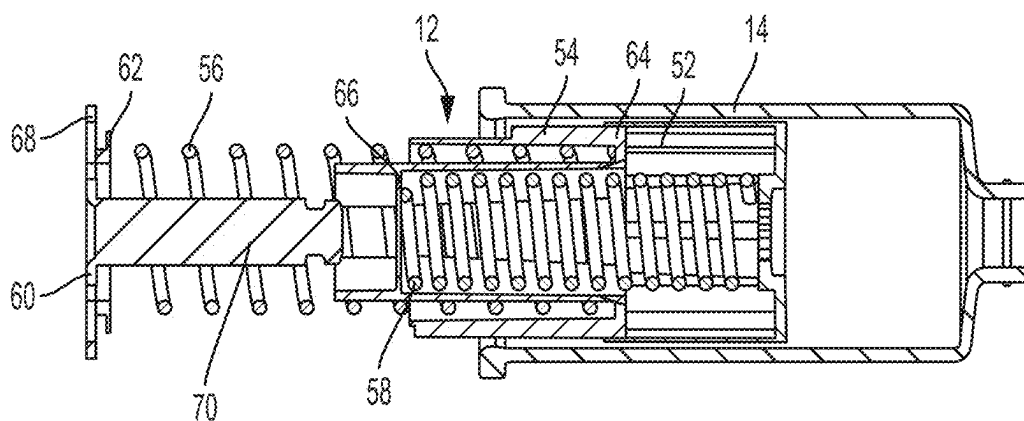


FIG. 20

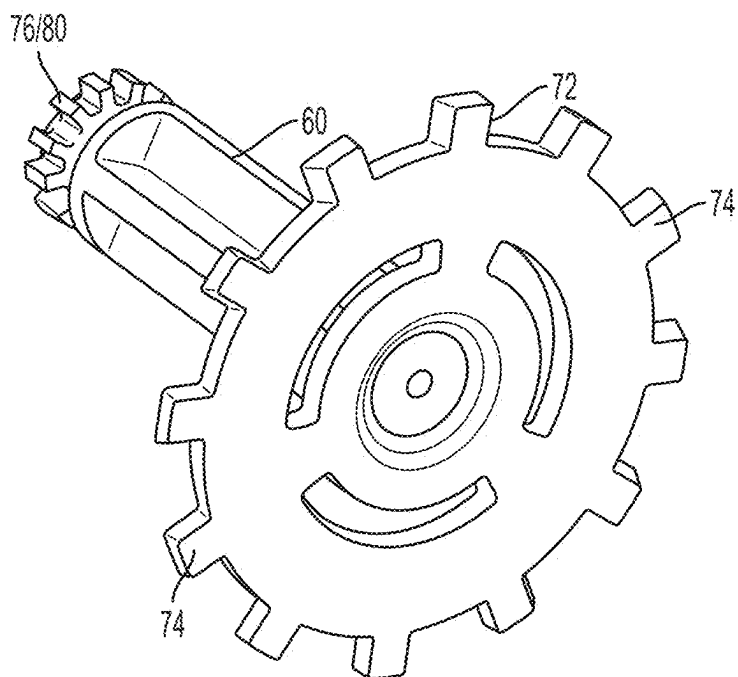


FIG. 21

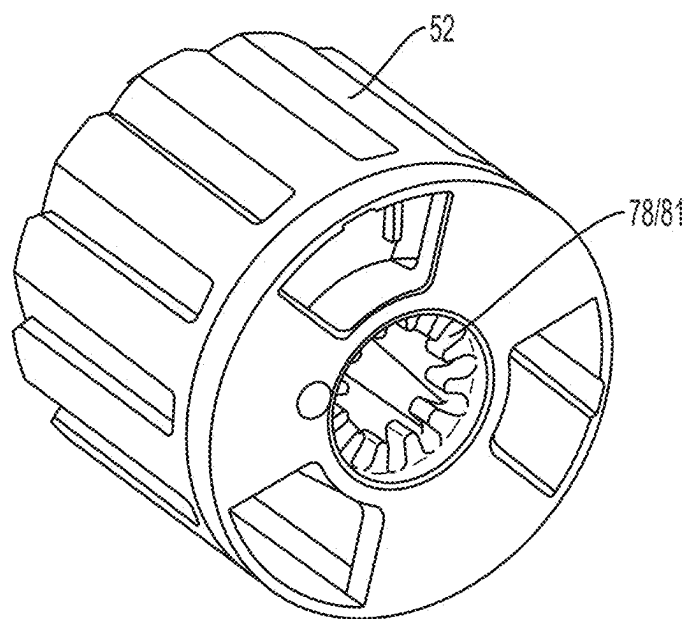


FIG. 22

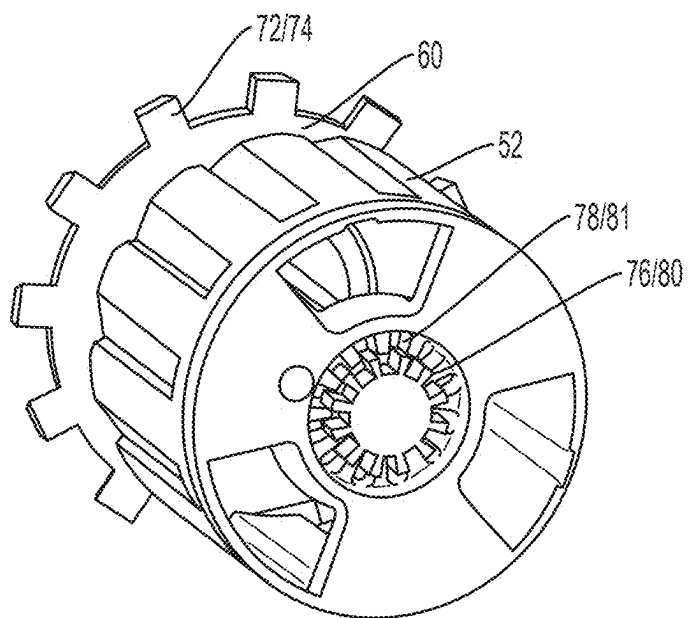


FIG. 23

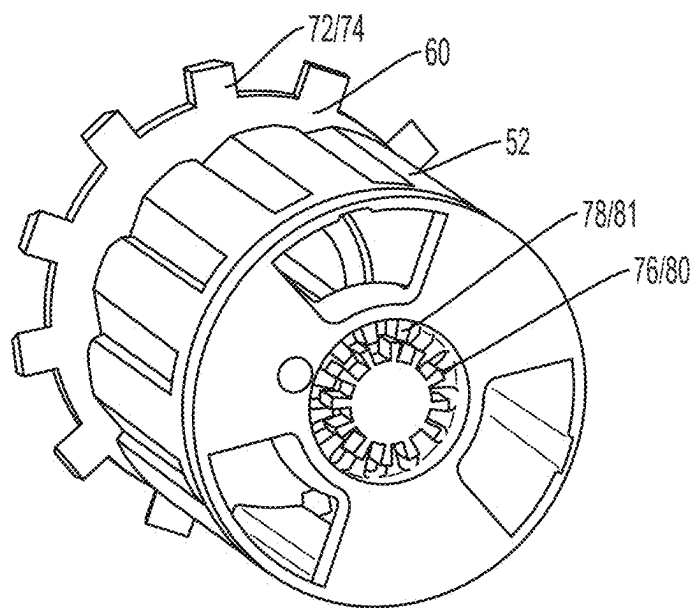


FIG. 24

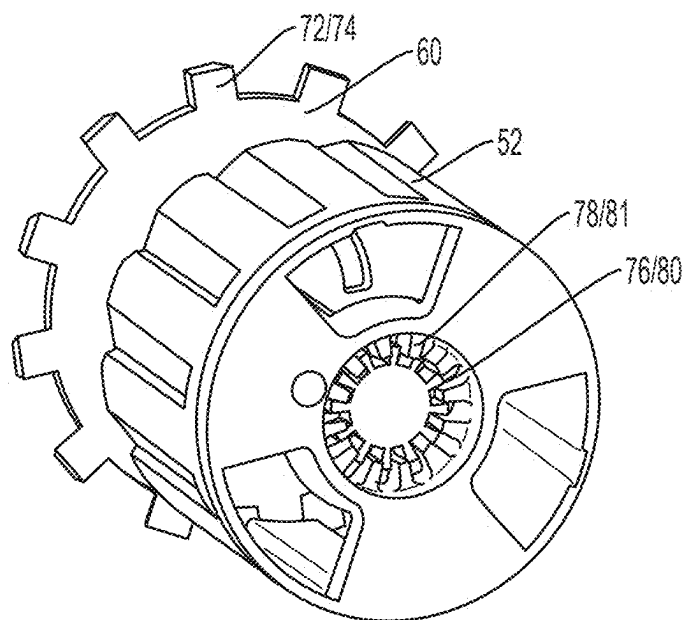


FIG. 25

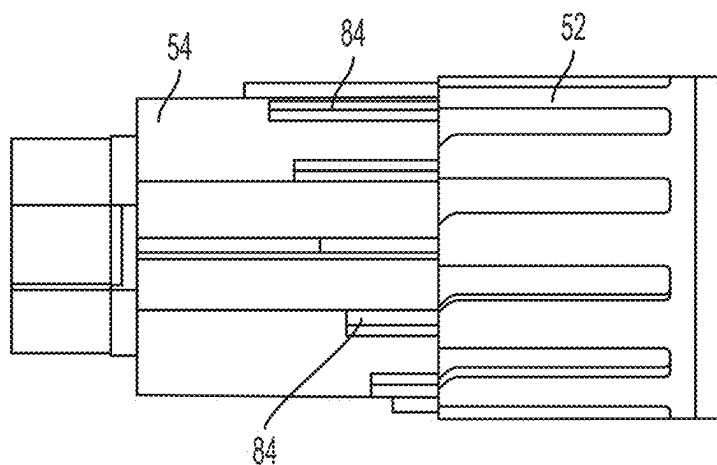


FIG. 26

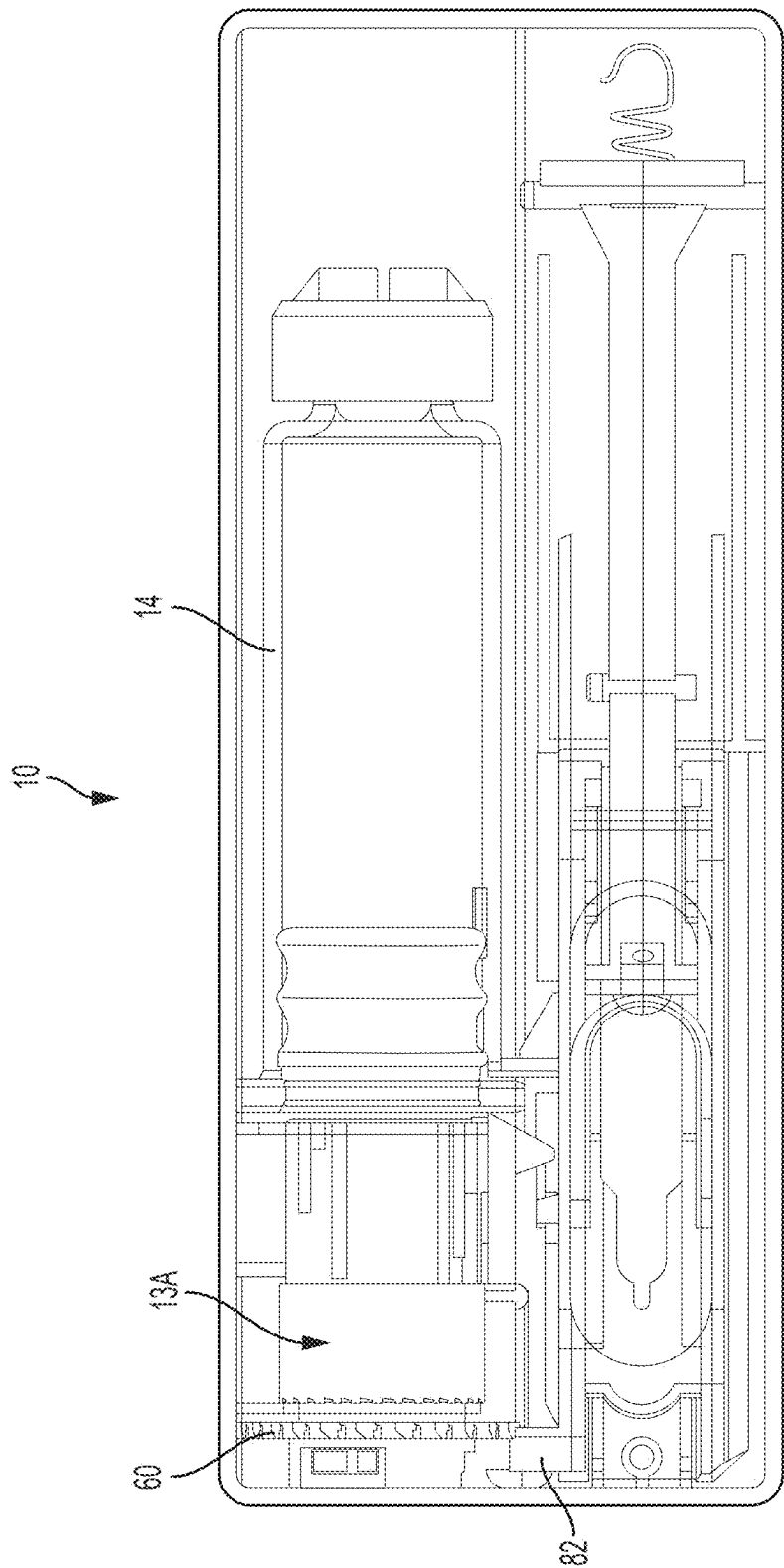


FIG. 27

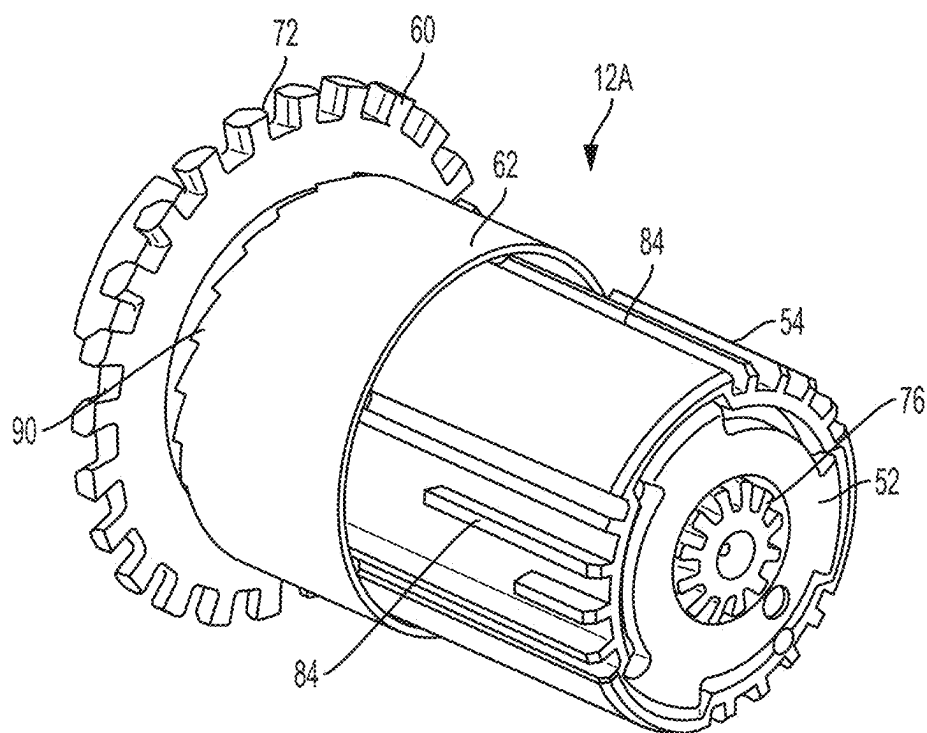


FIG. 28

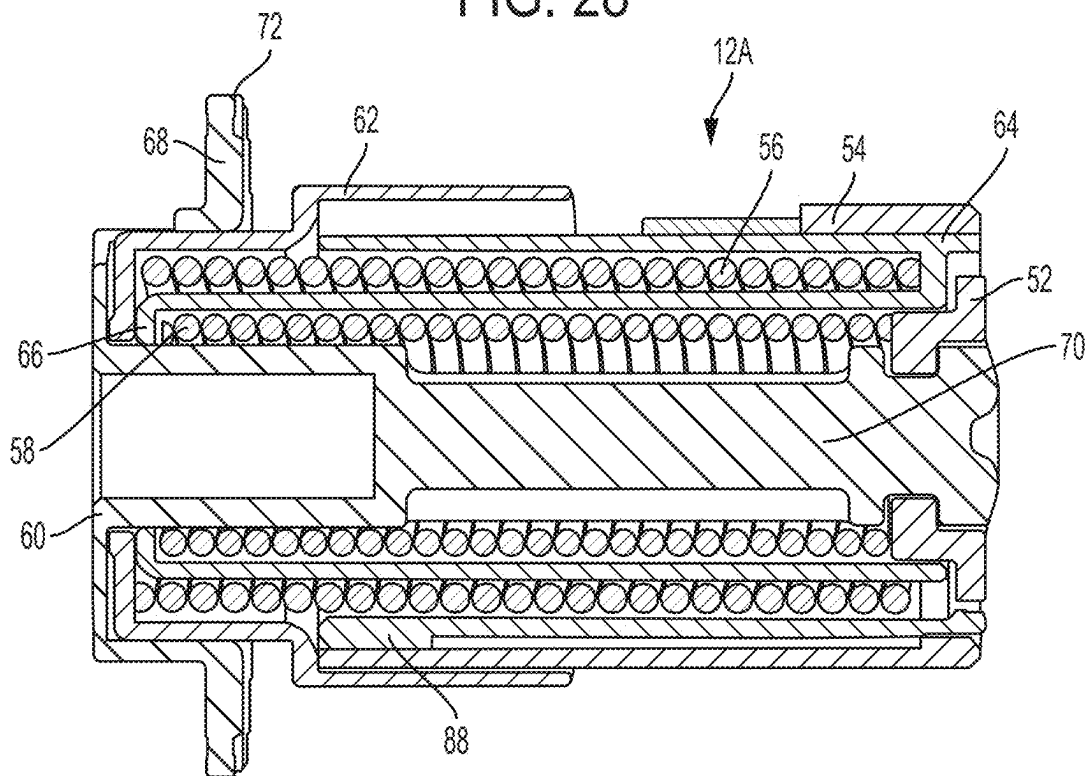


FIG. 29

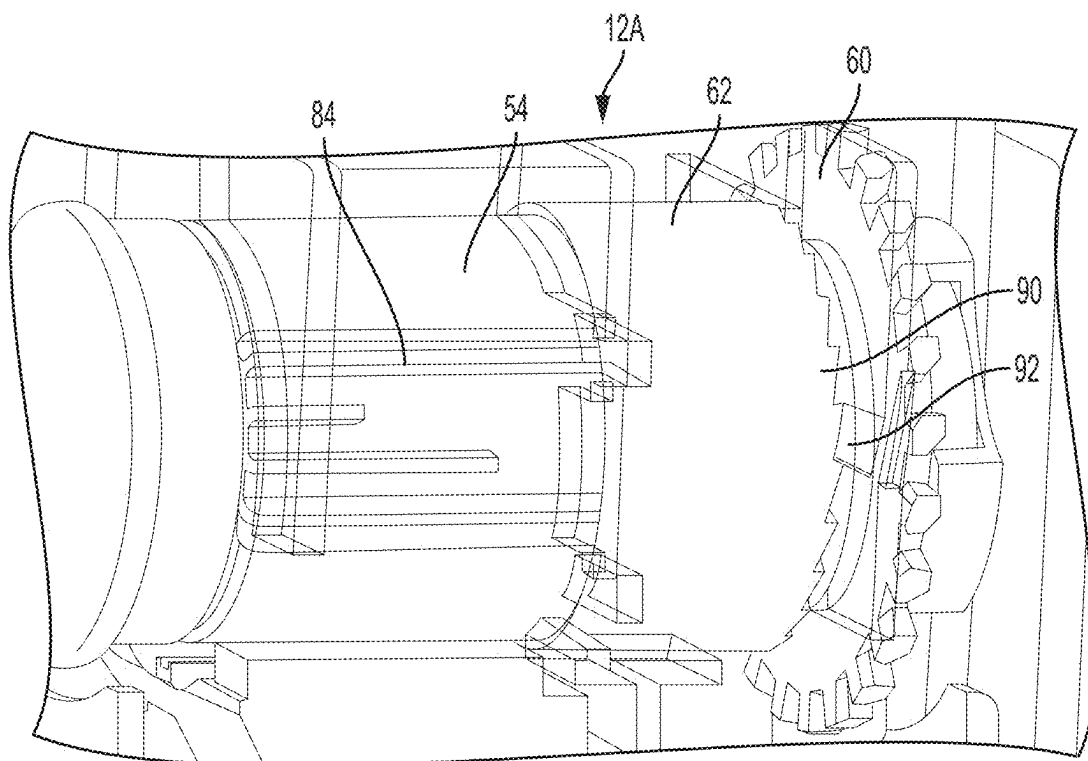


FIG. 30

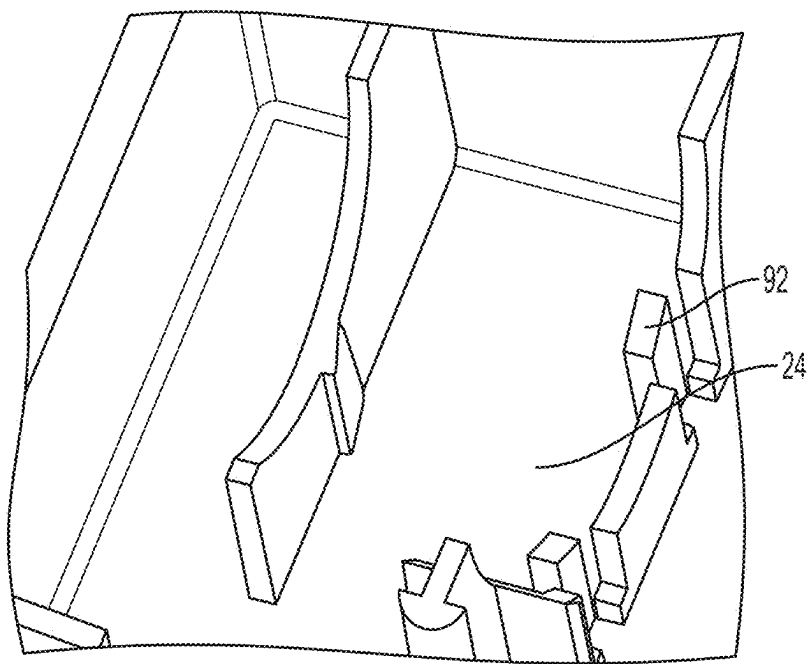


FIG. 31

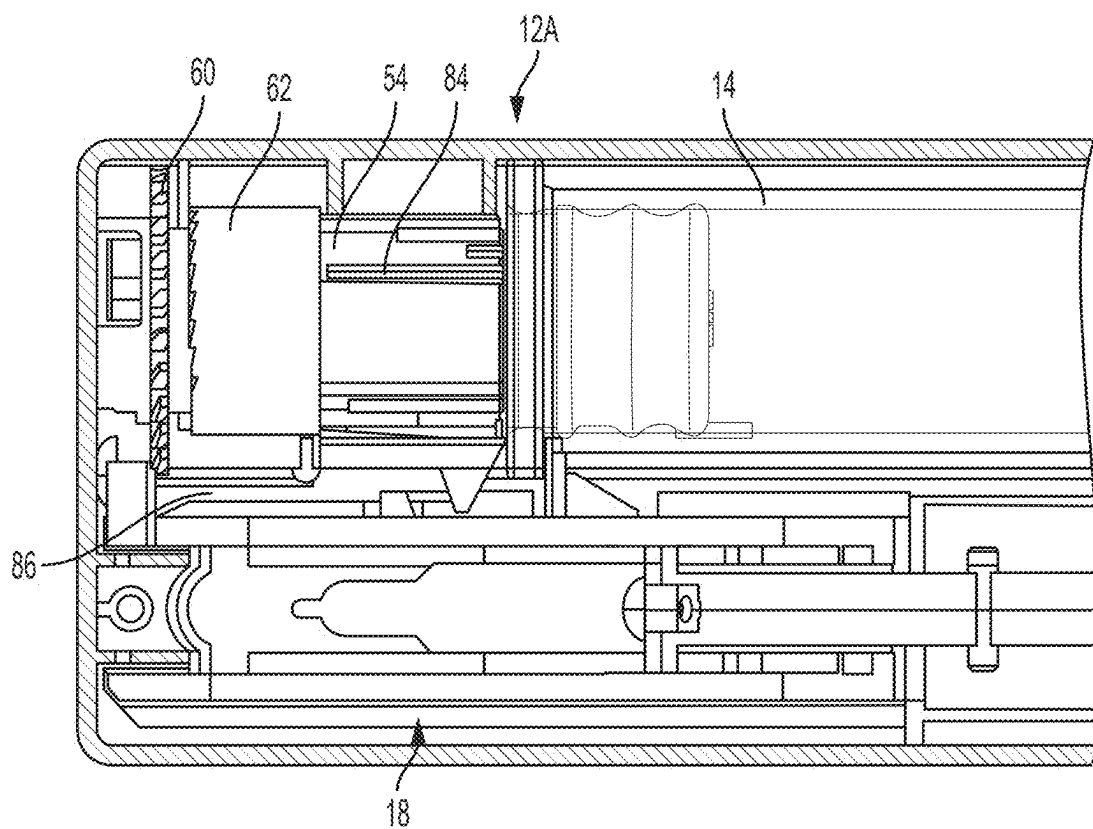


FIG. 32

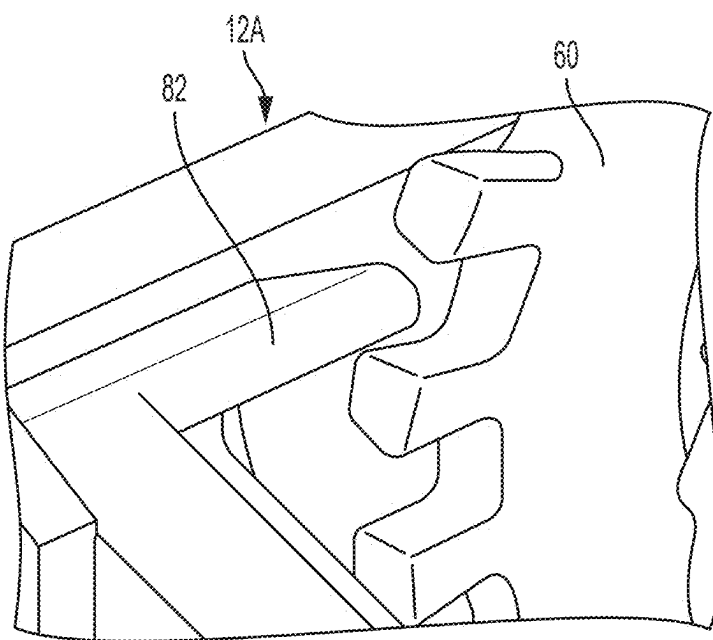


FIG. 33

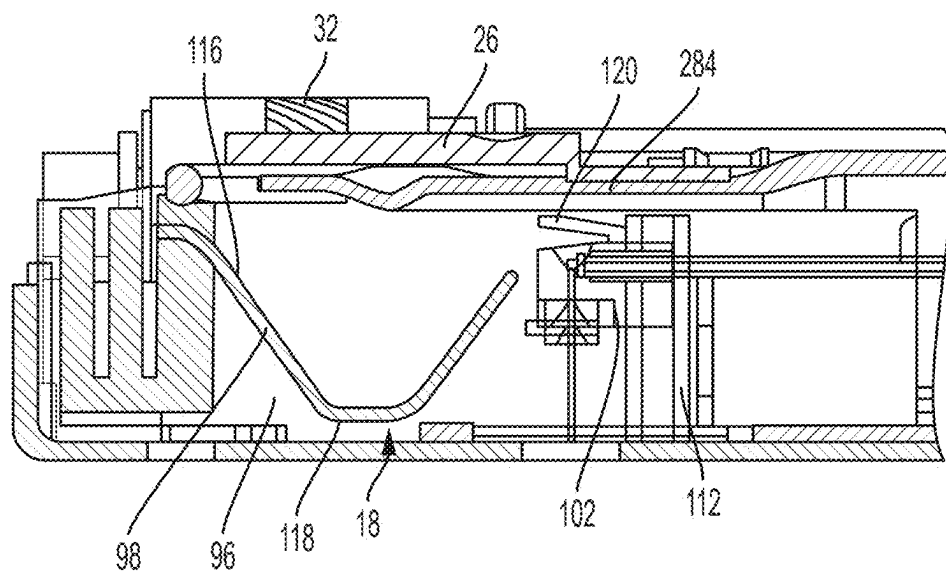


FIG. 34

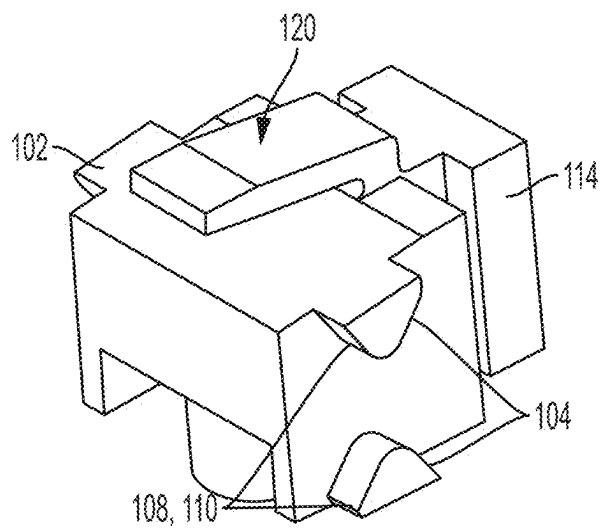


FIG. 35

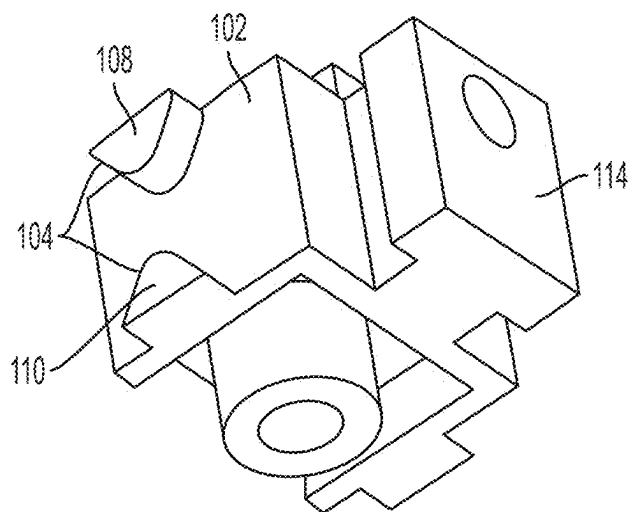


FIG. 36

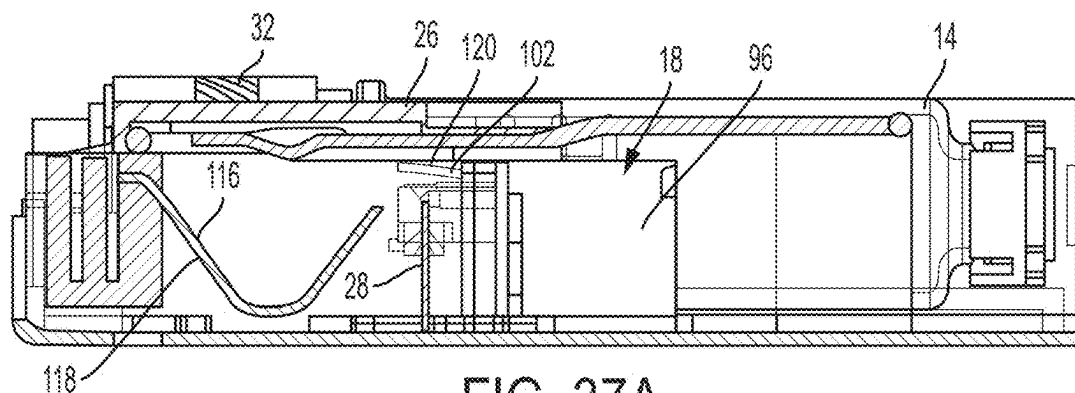


FIG. 37A

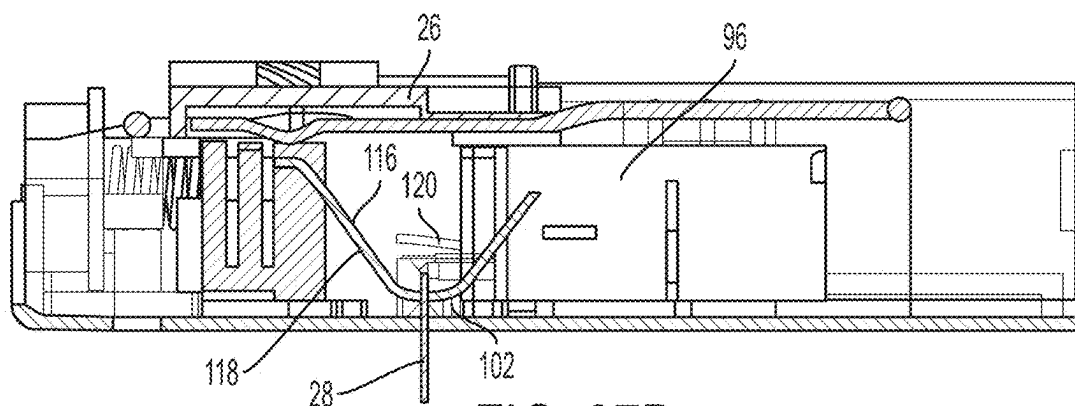


FIG. 37B

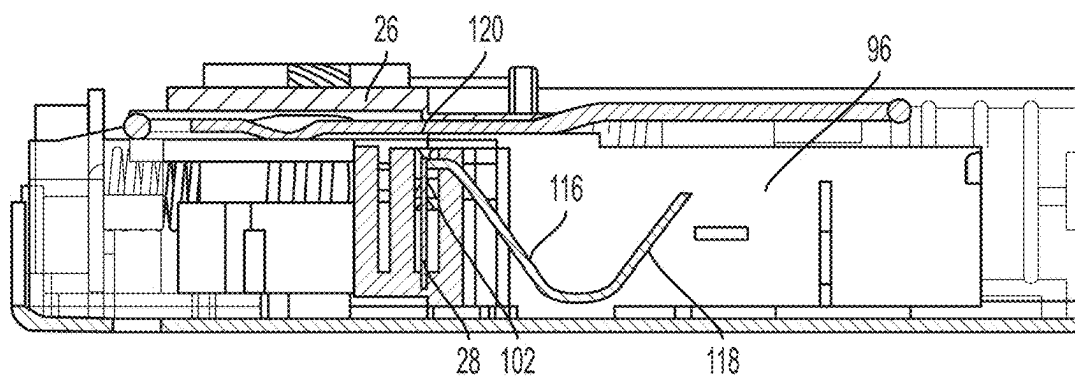


FIG. 37C

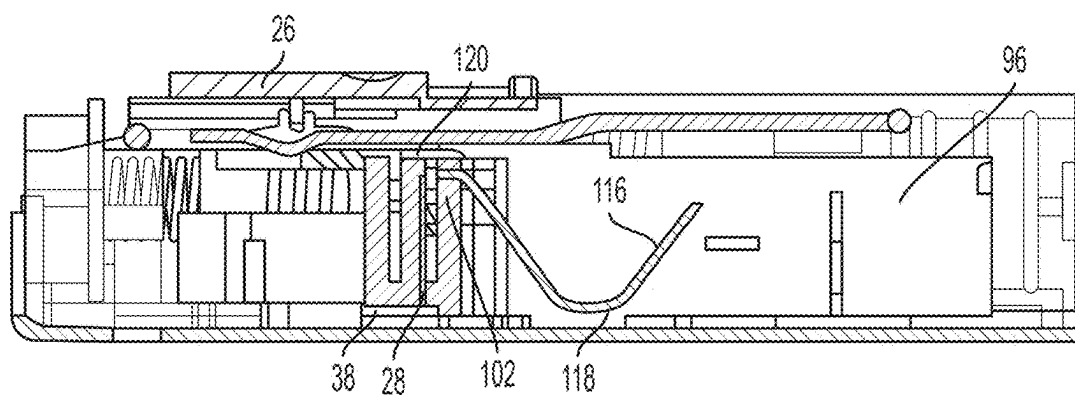


FIG. 37D

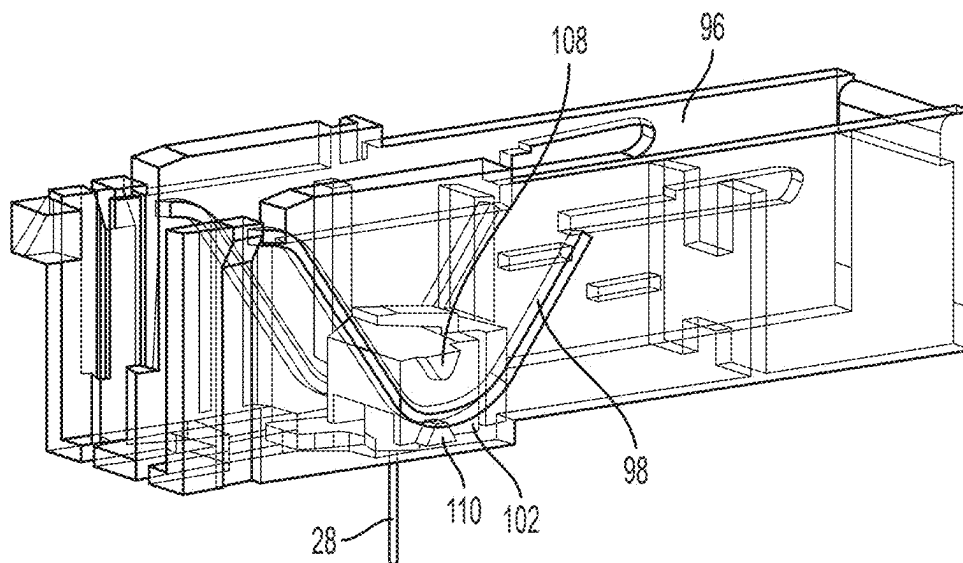


FIG. 38A

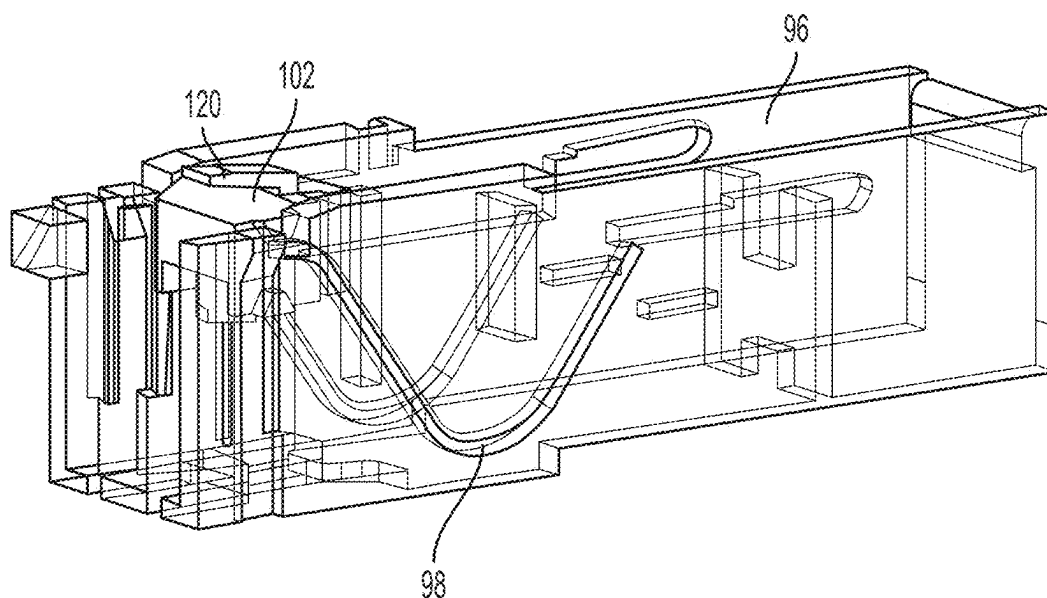


FIG. 38B

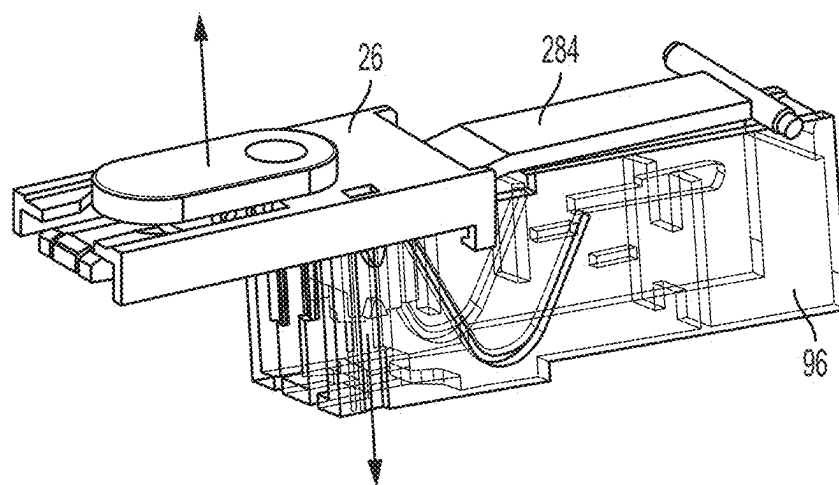


FIG. 39

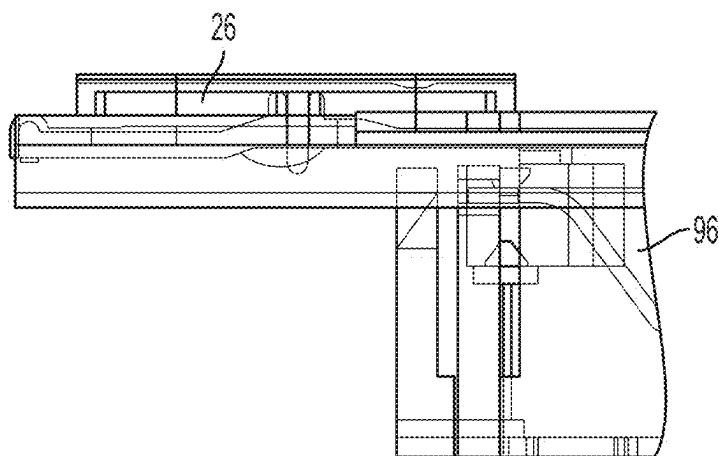


FIG. 40A

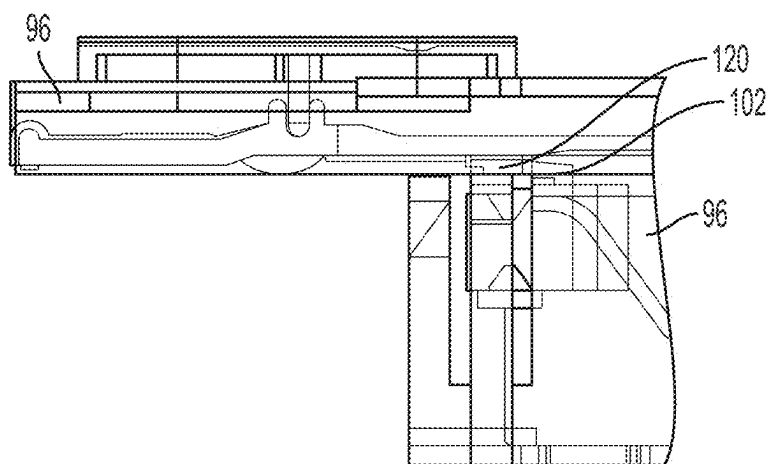


FIG. 40B

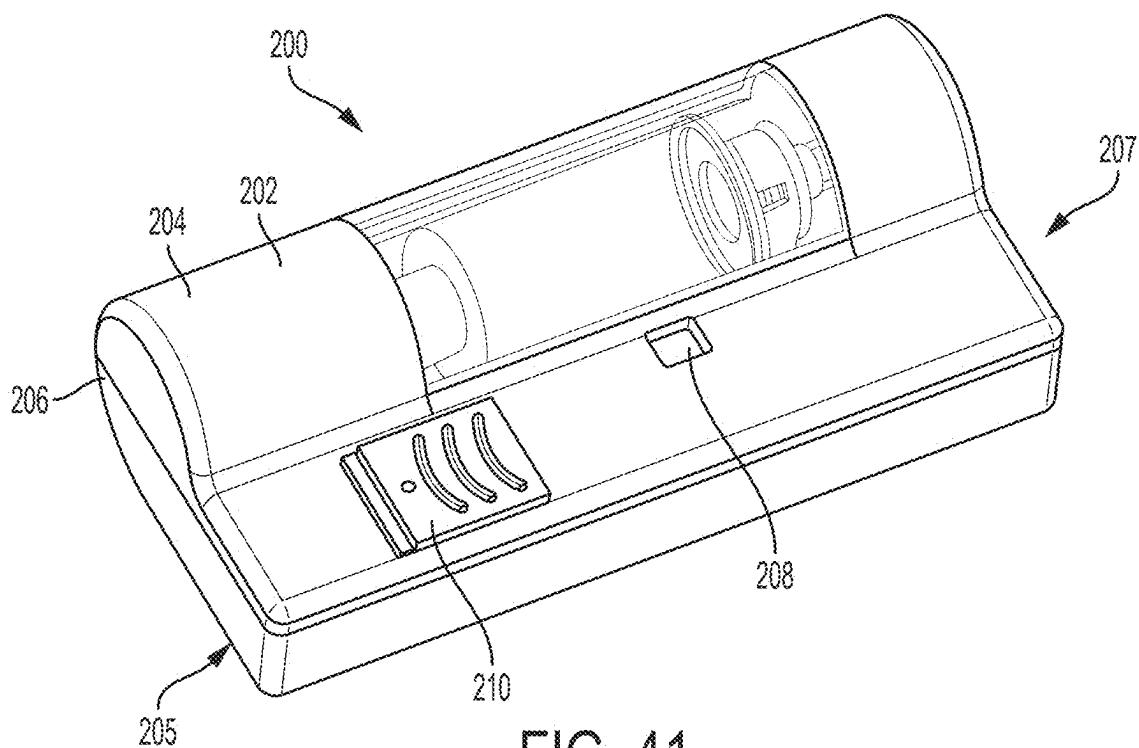


FIG. 41

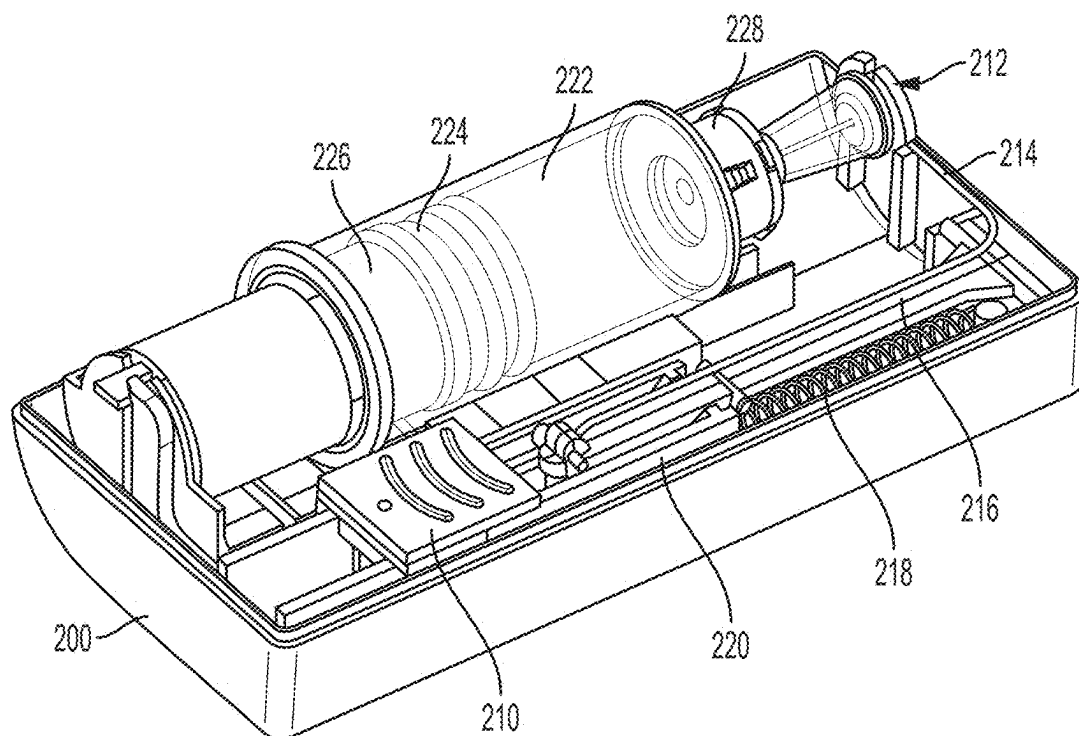


FIG. 42

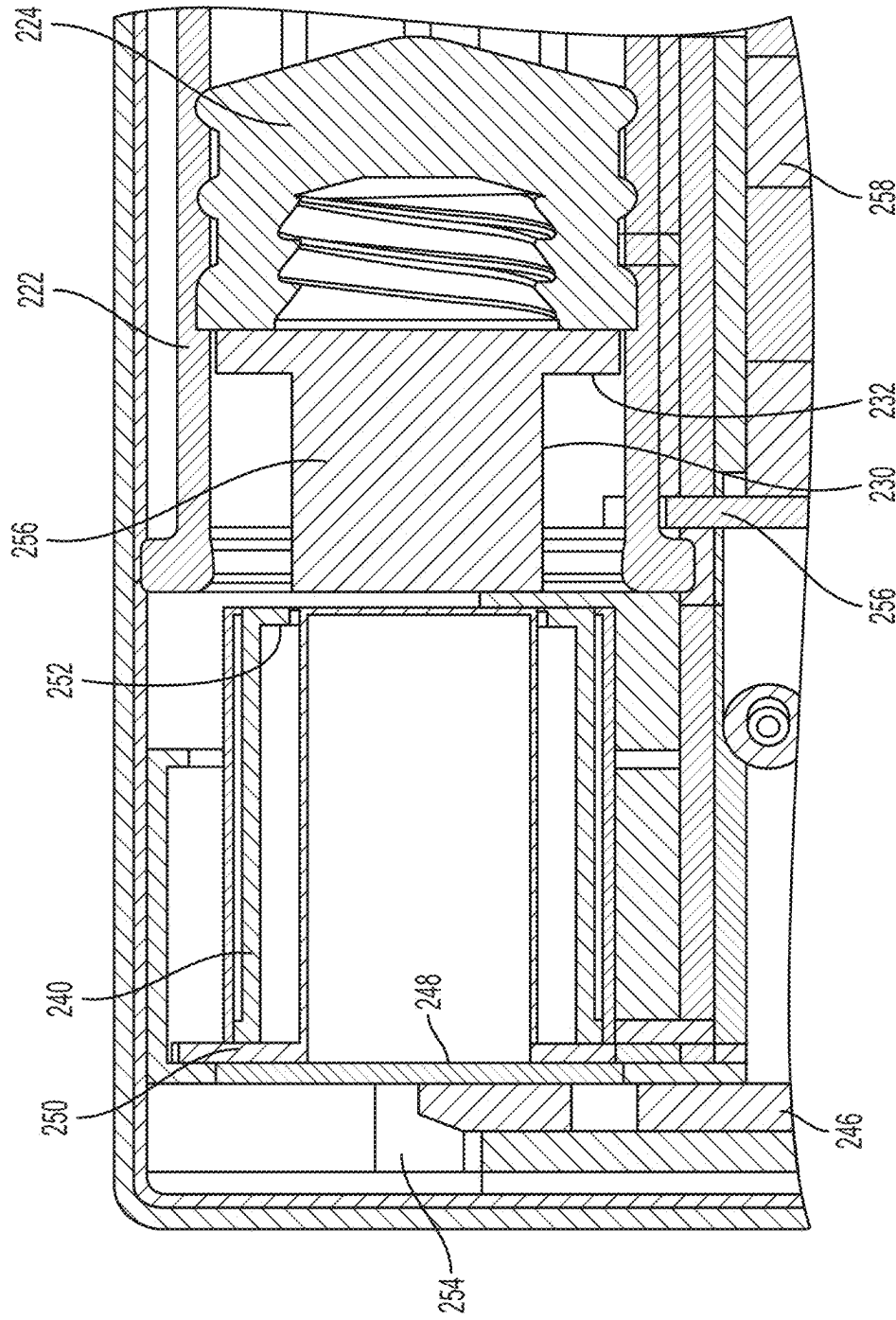


FIG. 43

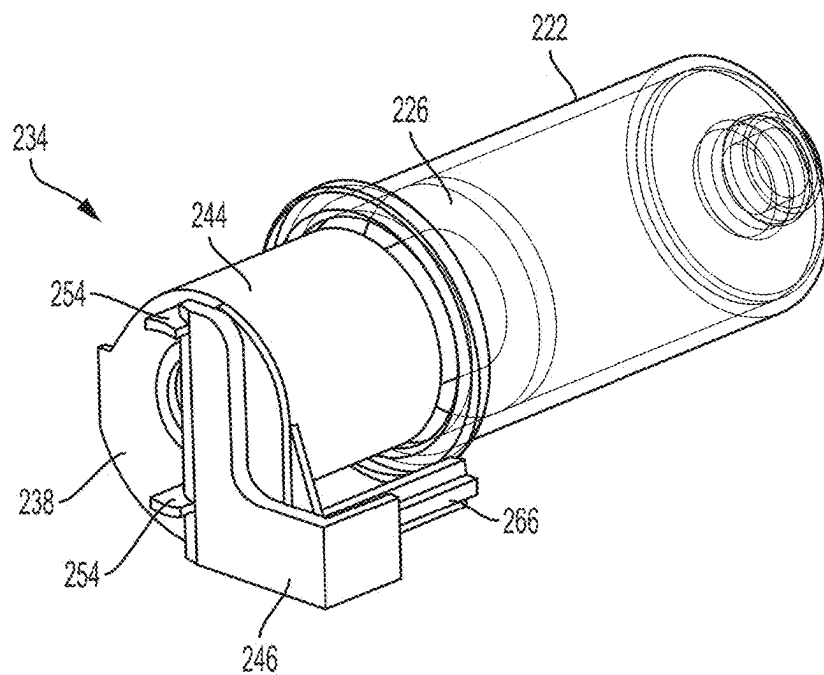


FIG. 44

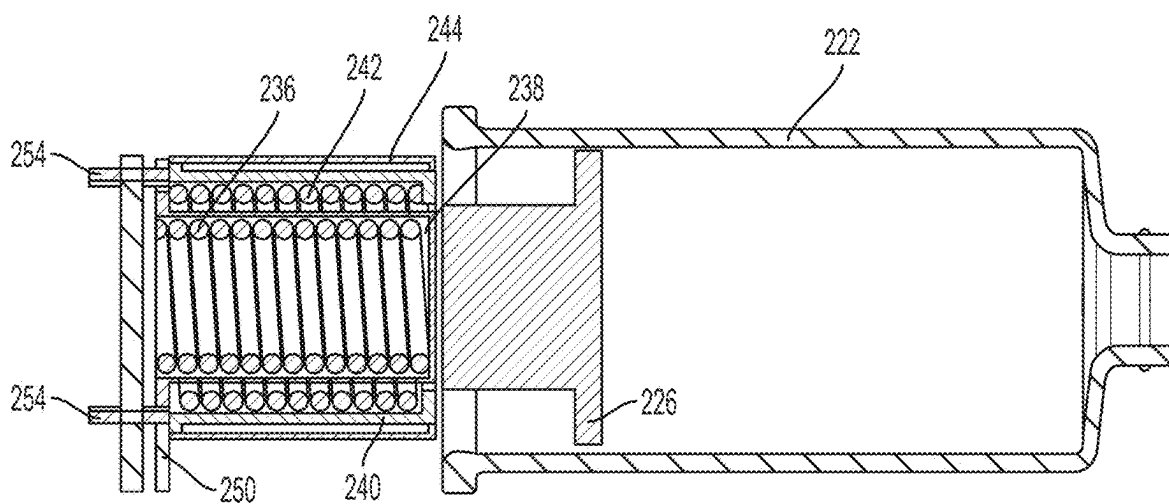


FIG. 45

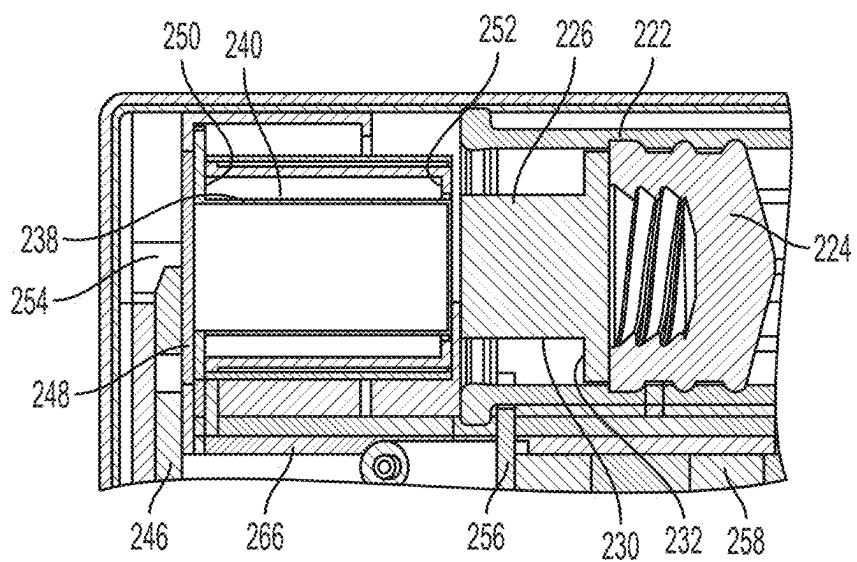


FIG. 46

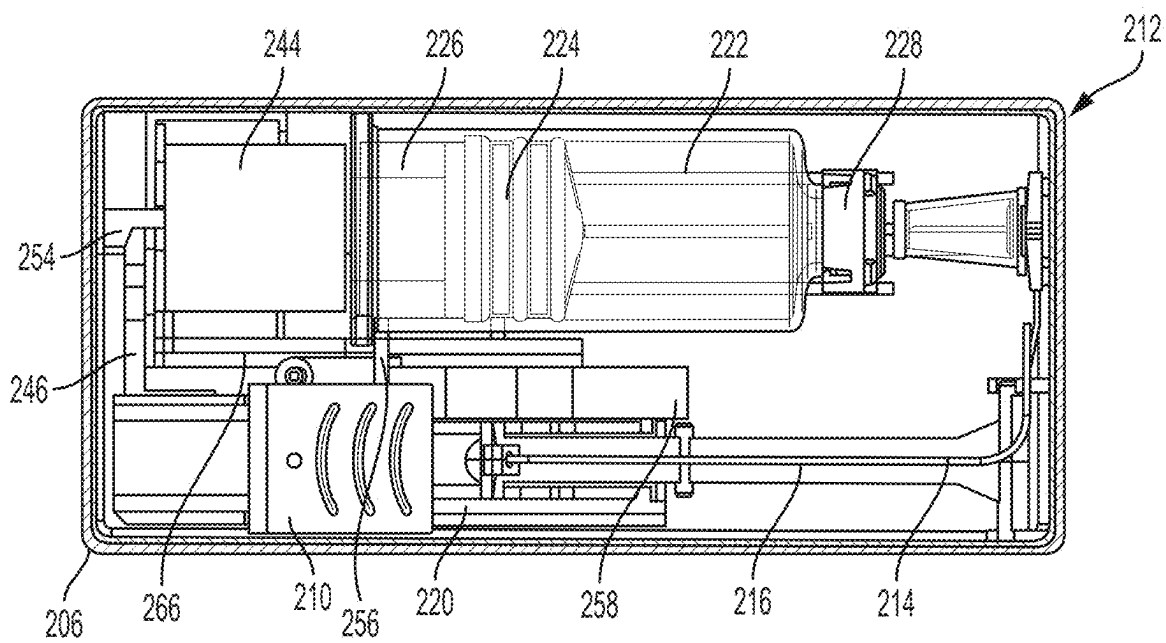


FIG. 47

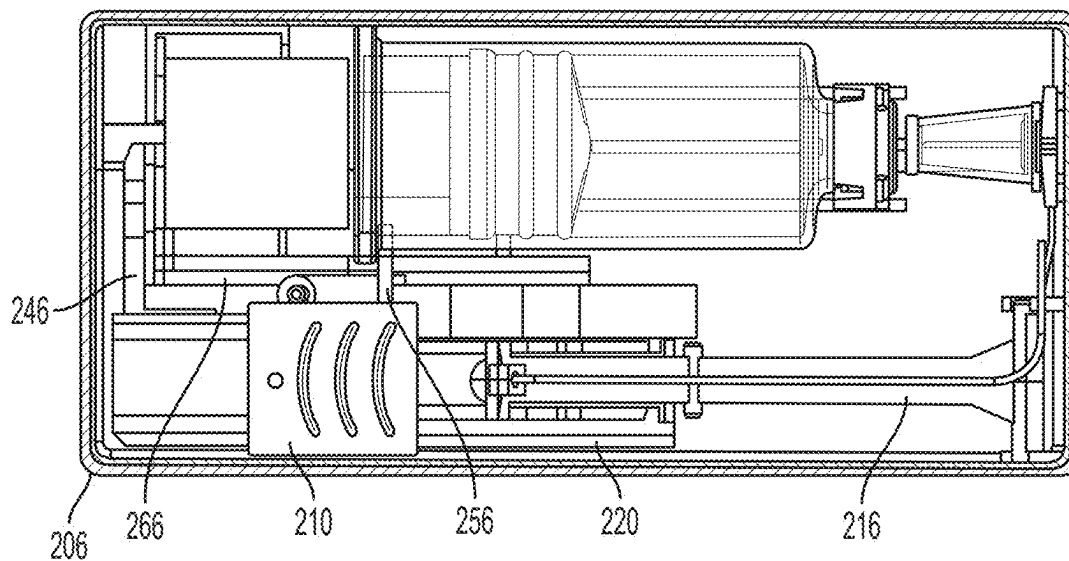


FIG. 48

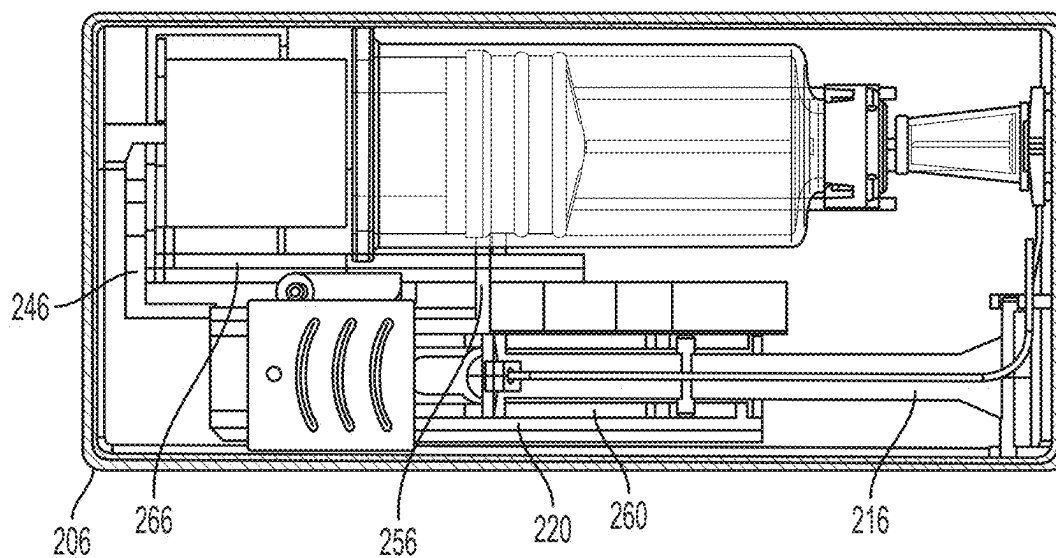


FIG. 49

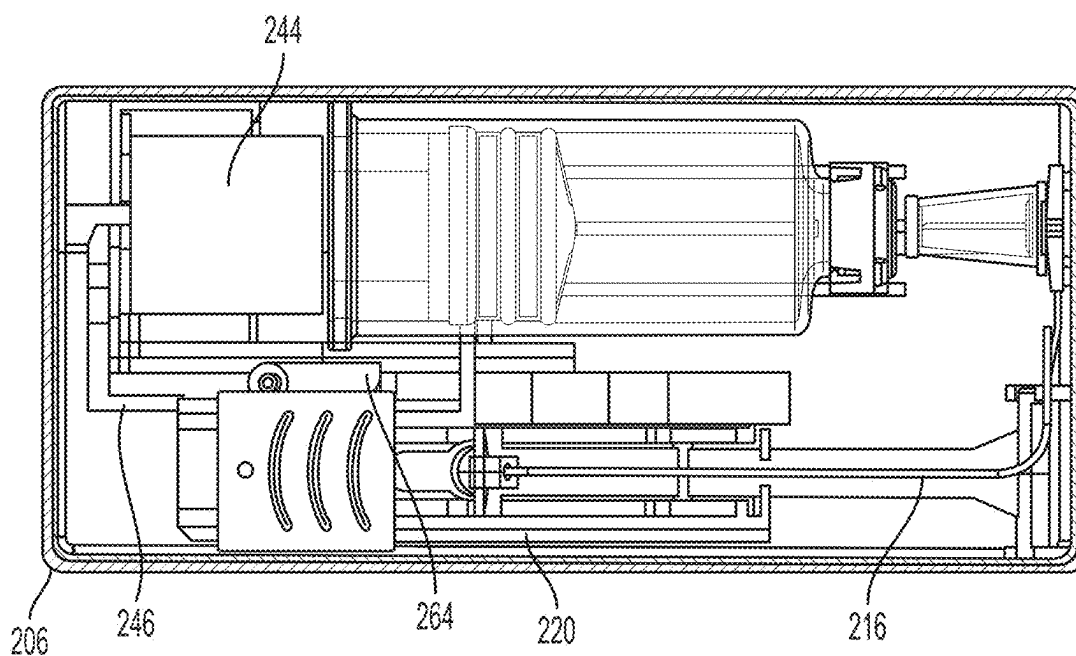


FIG. 50

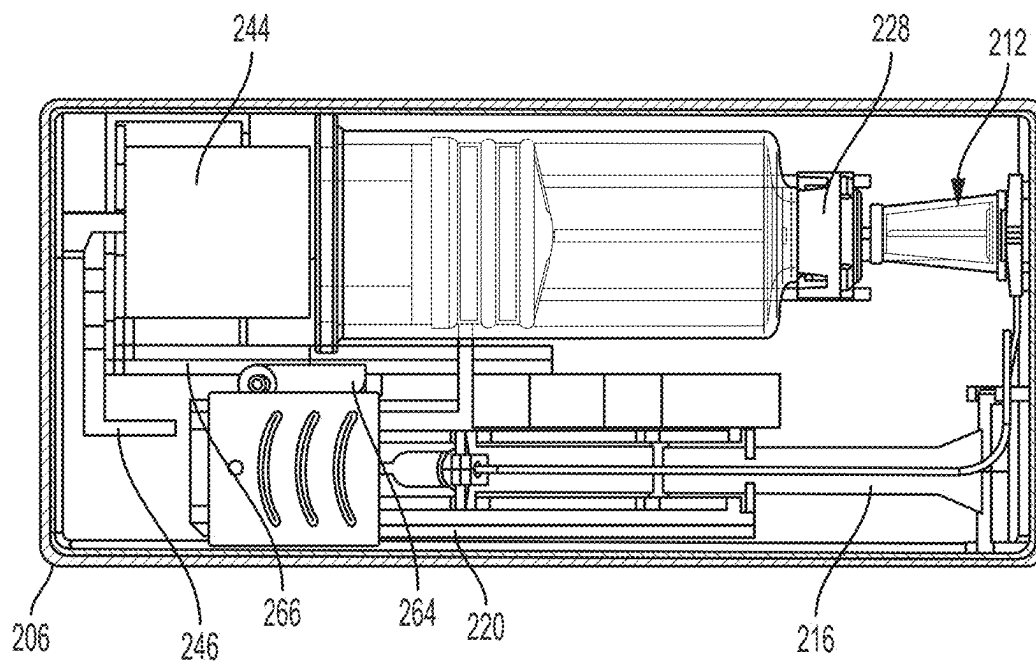


FIG. 51

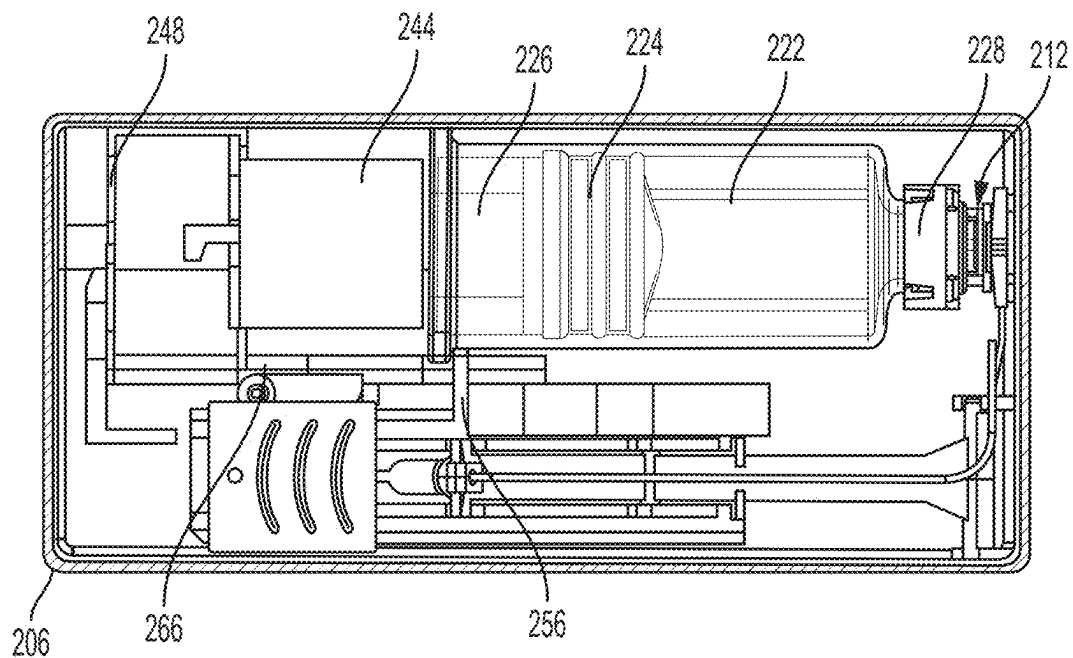


FIG. 52

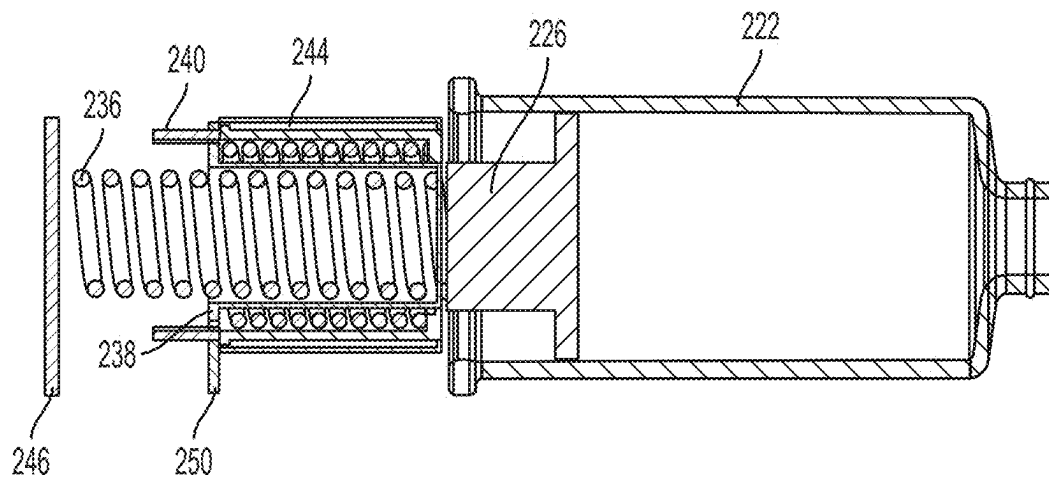


FIG. 53

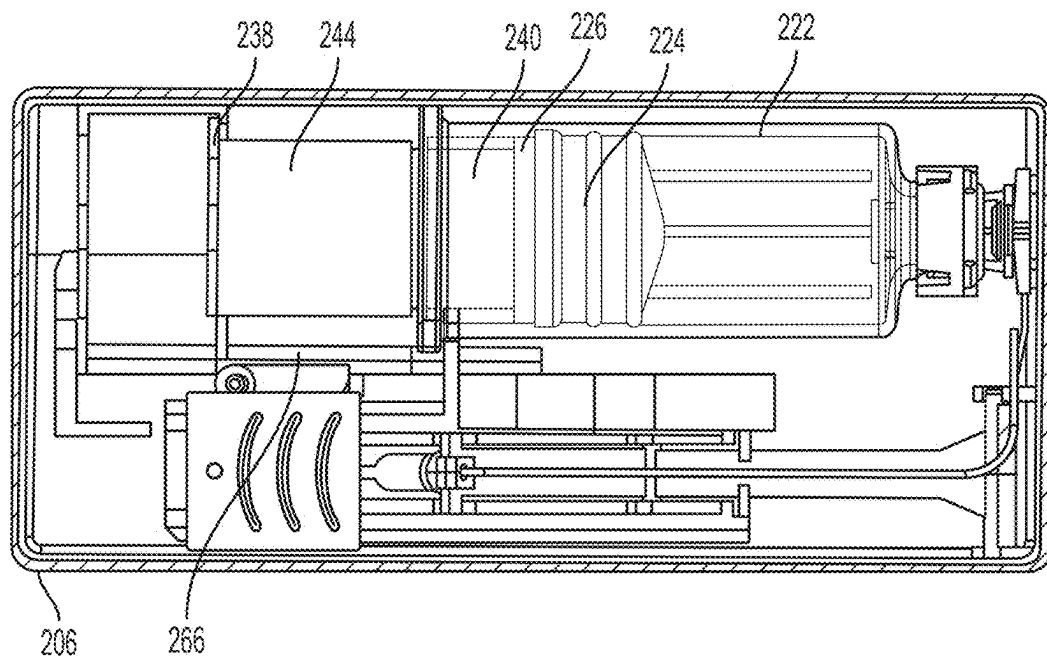


FIG. 54

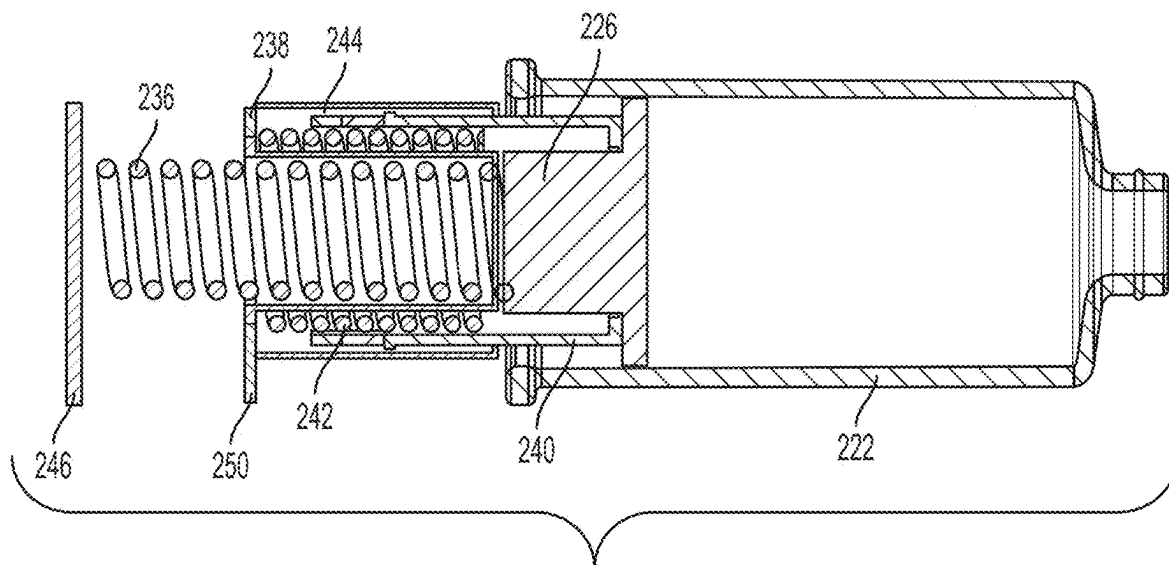


FIG. 55

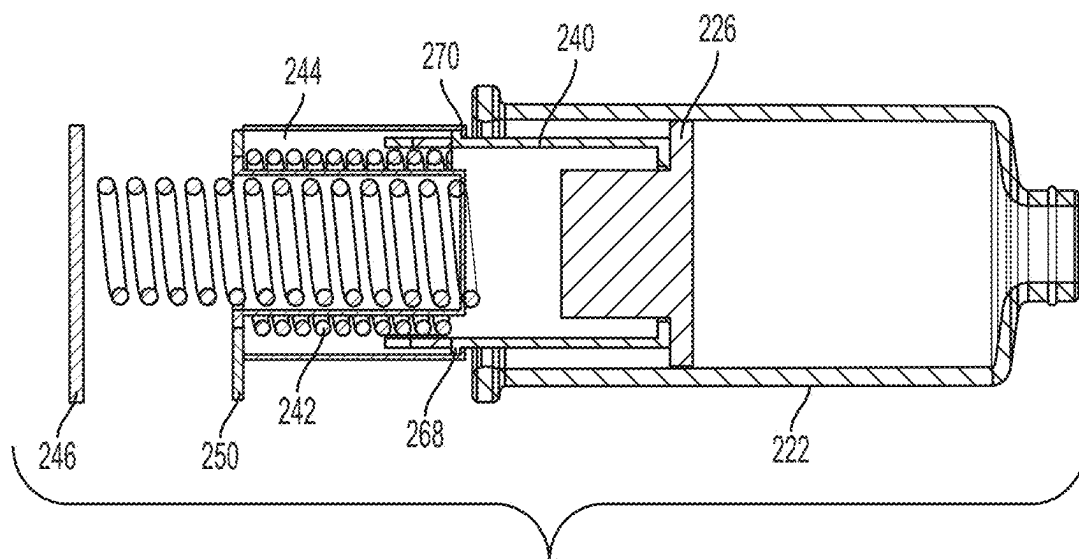


FIG. 56

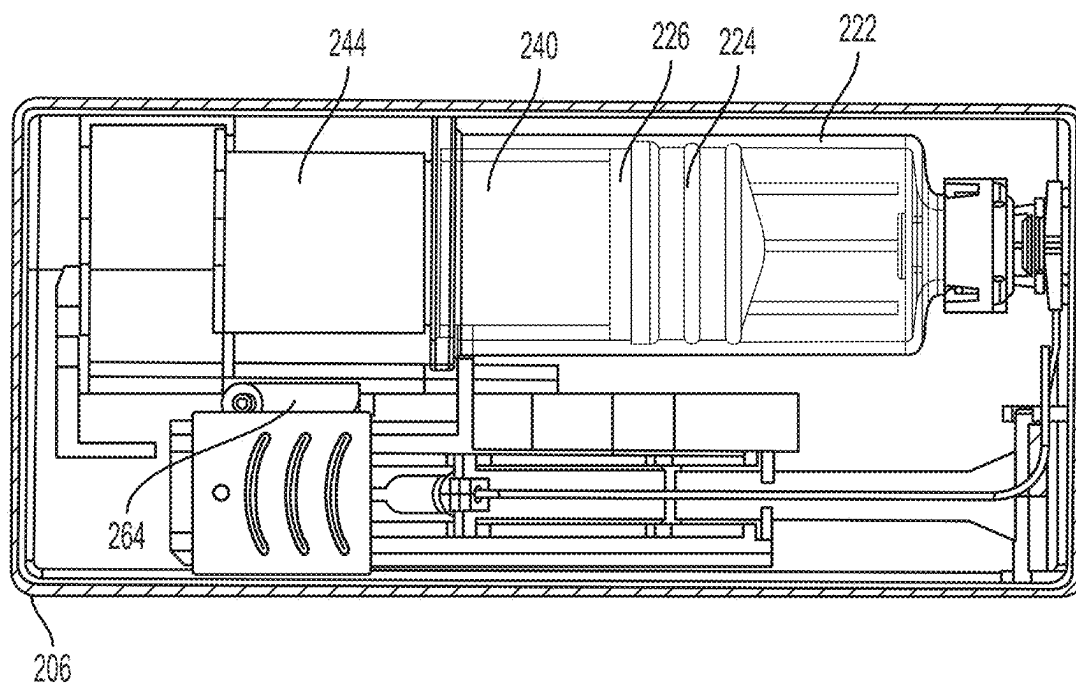


FIG. 57

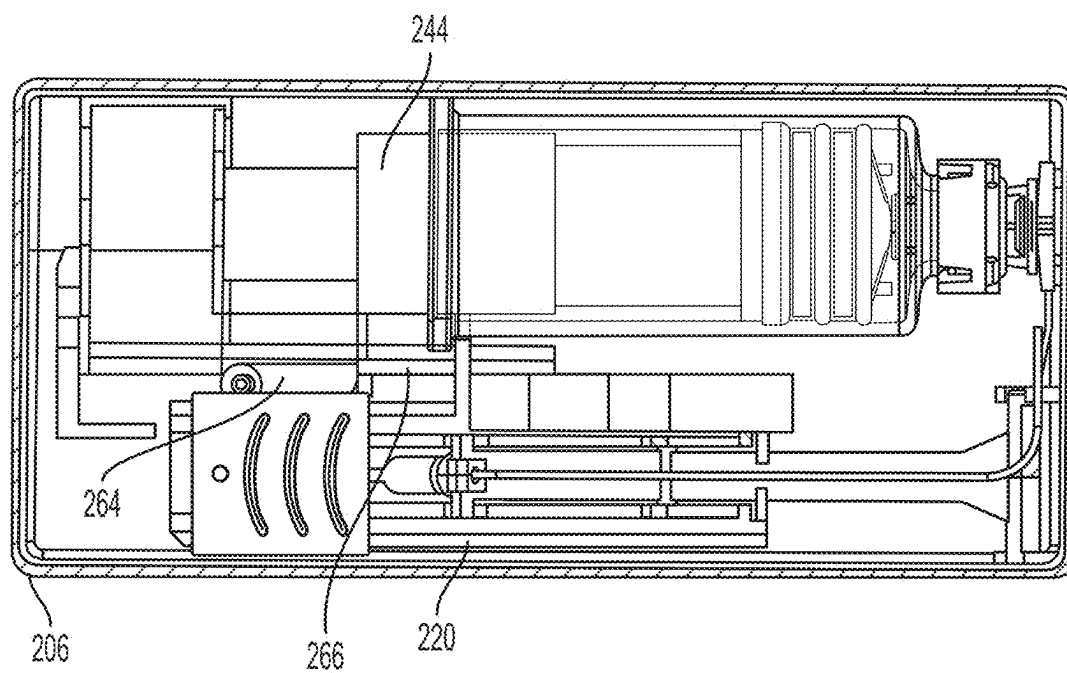


FIG. 58

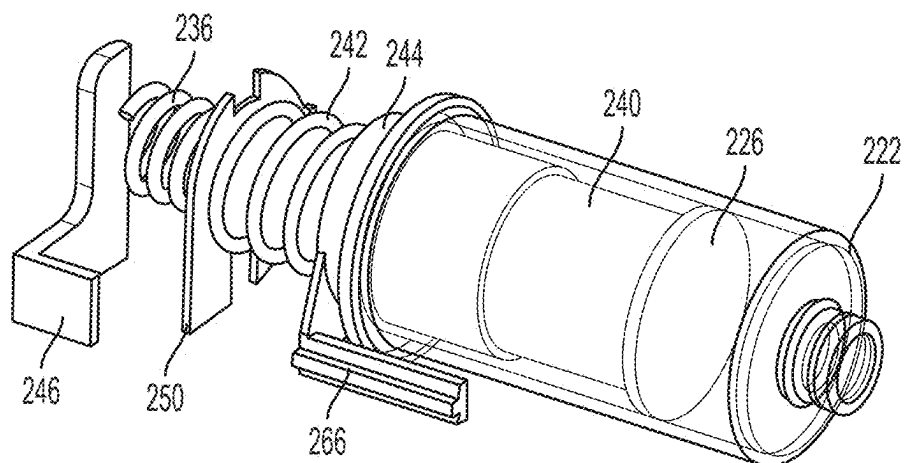


FIG. 59

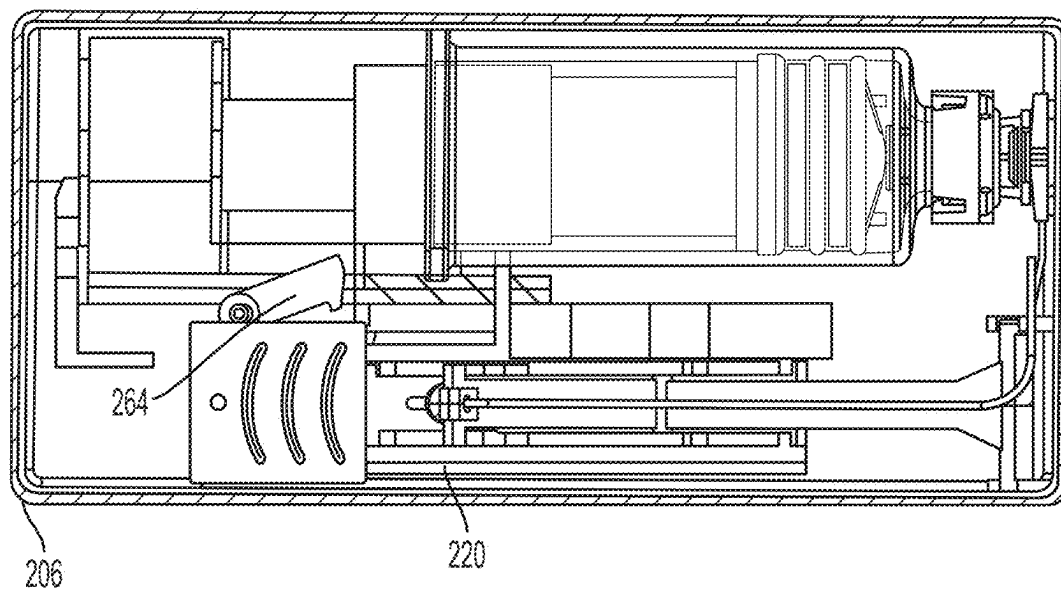


FIG. 60

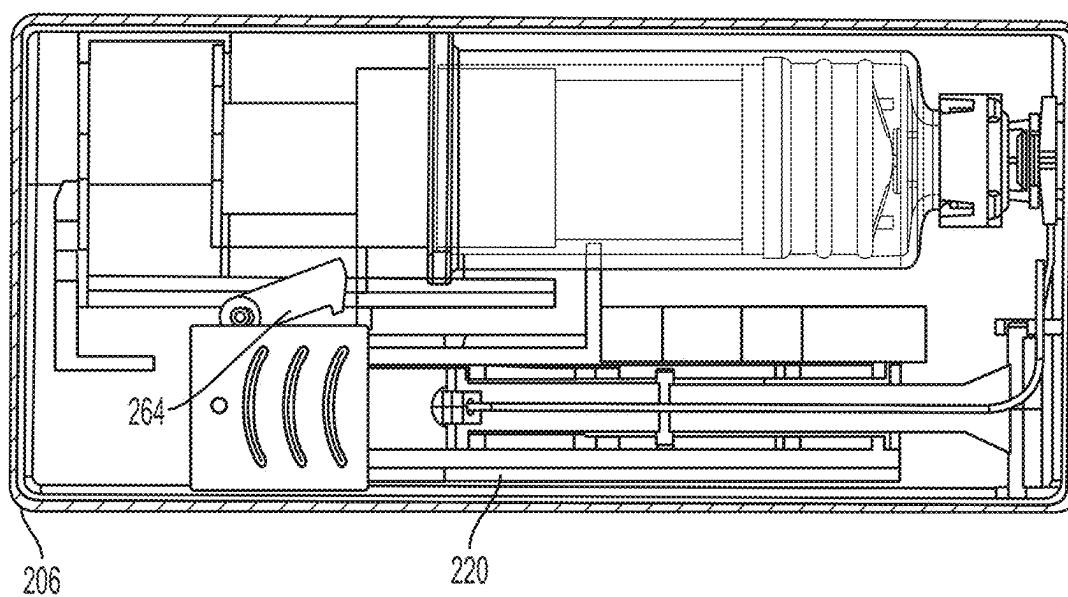


FIG. 61

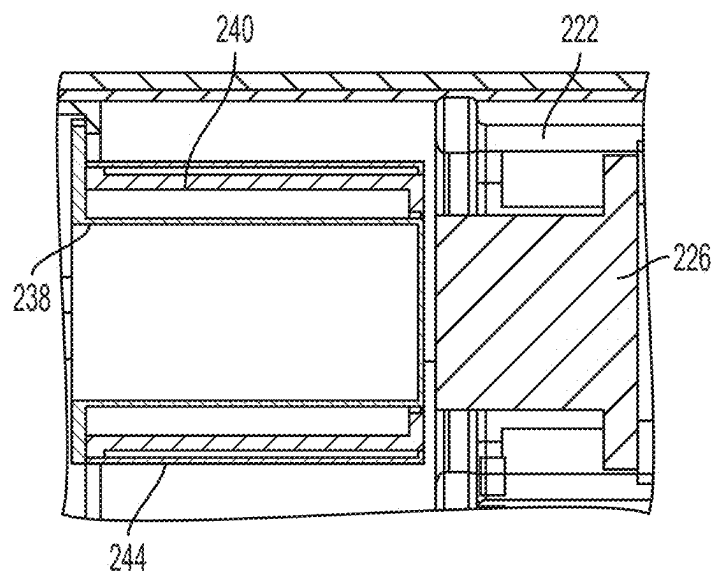


FIG. 62

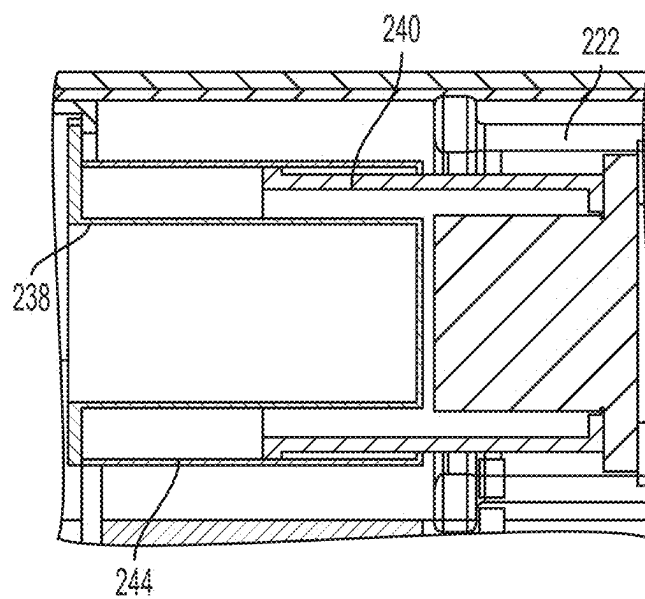


FIG. 63

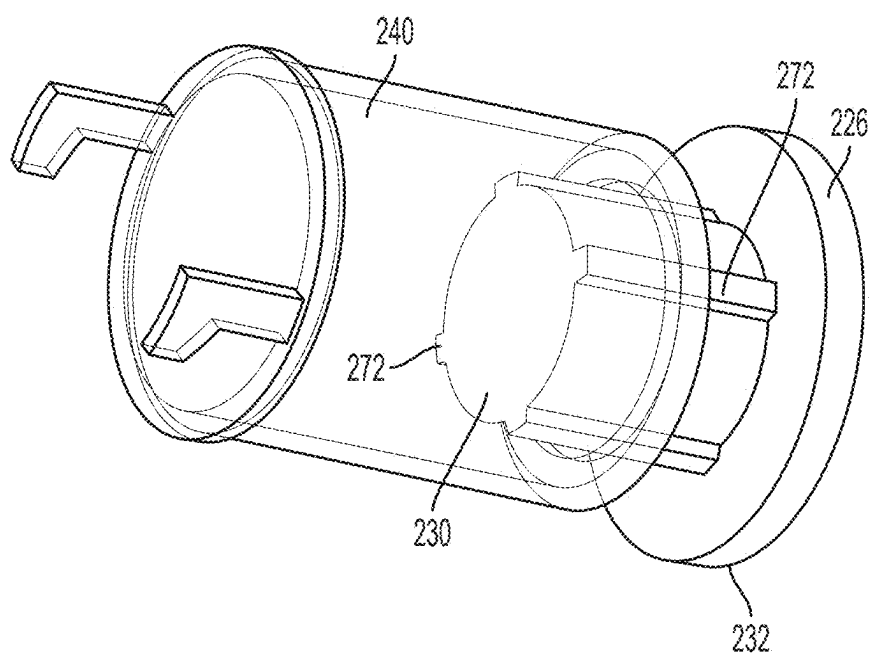


FIG. 64

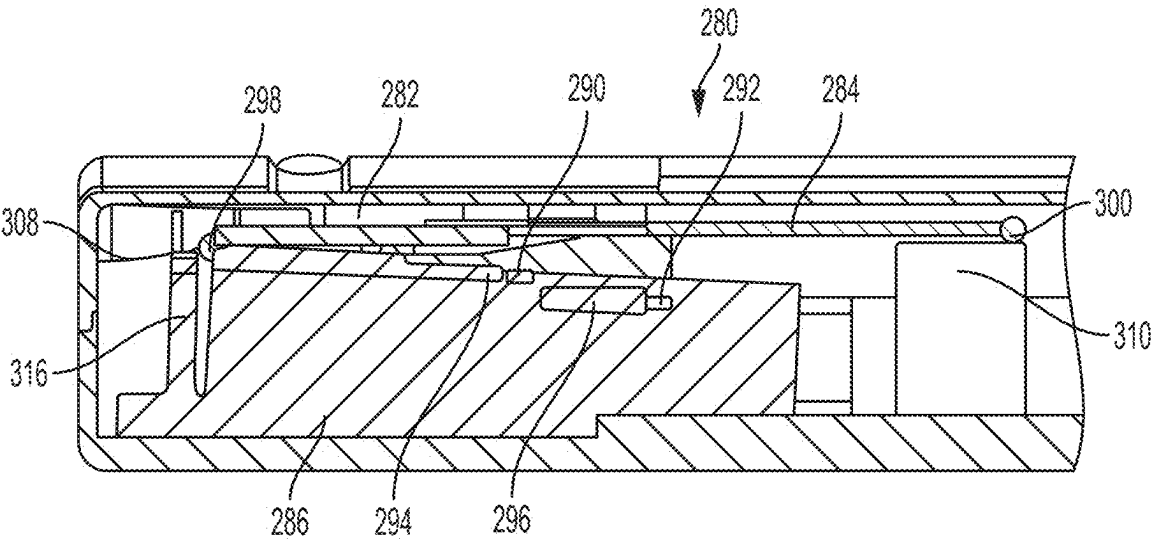


FIG. 65A

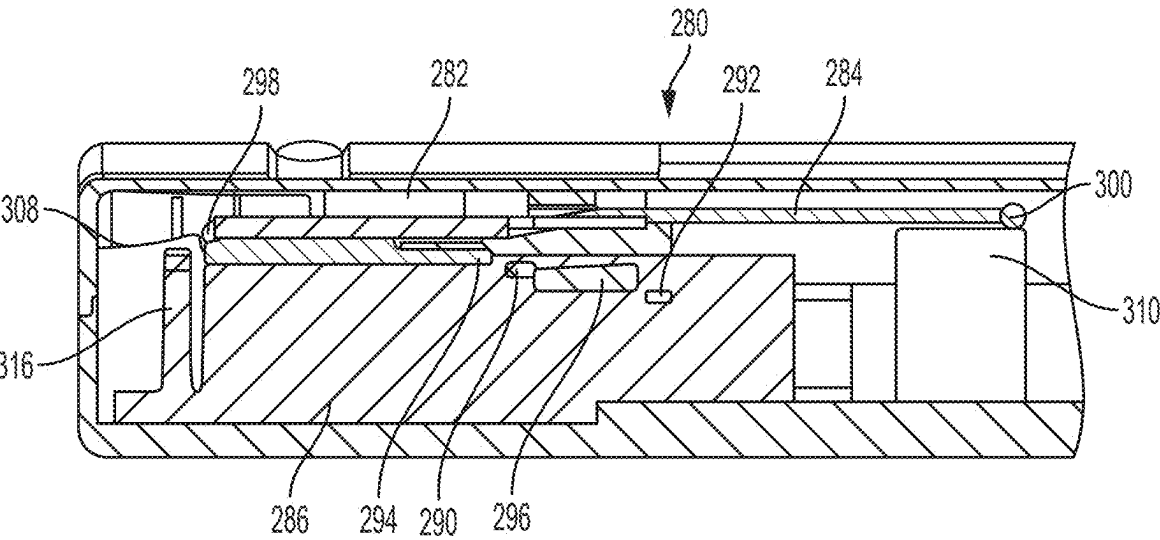


FIG. 65B

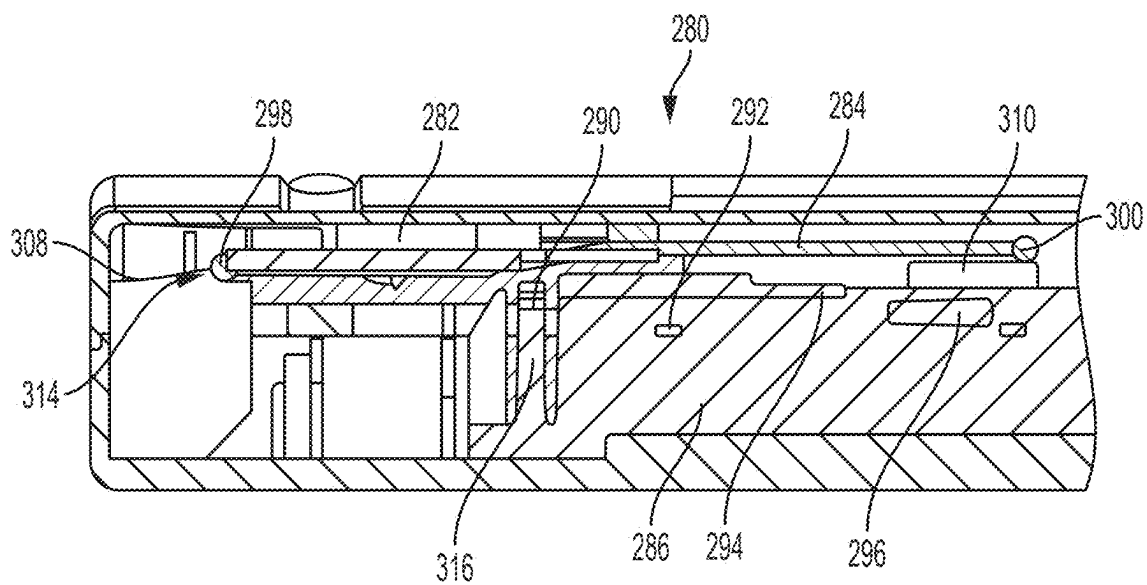


FIG. 65C

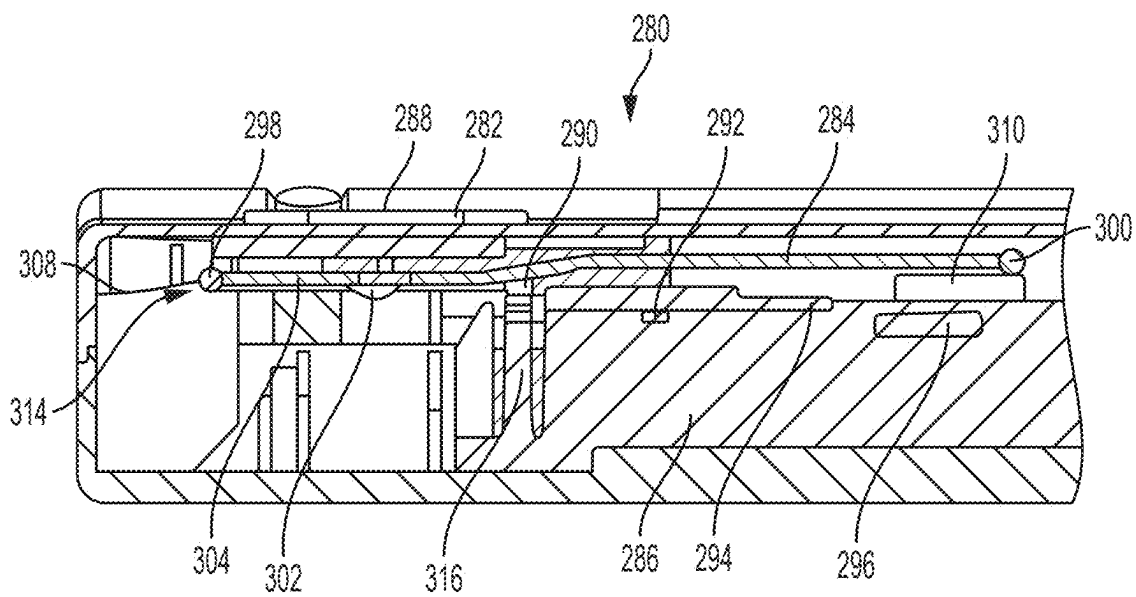


FIG. 65D

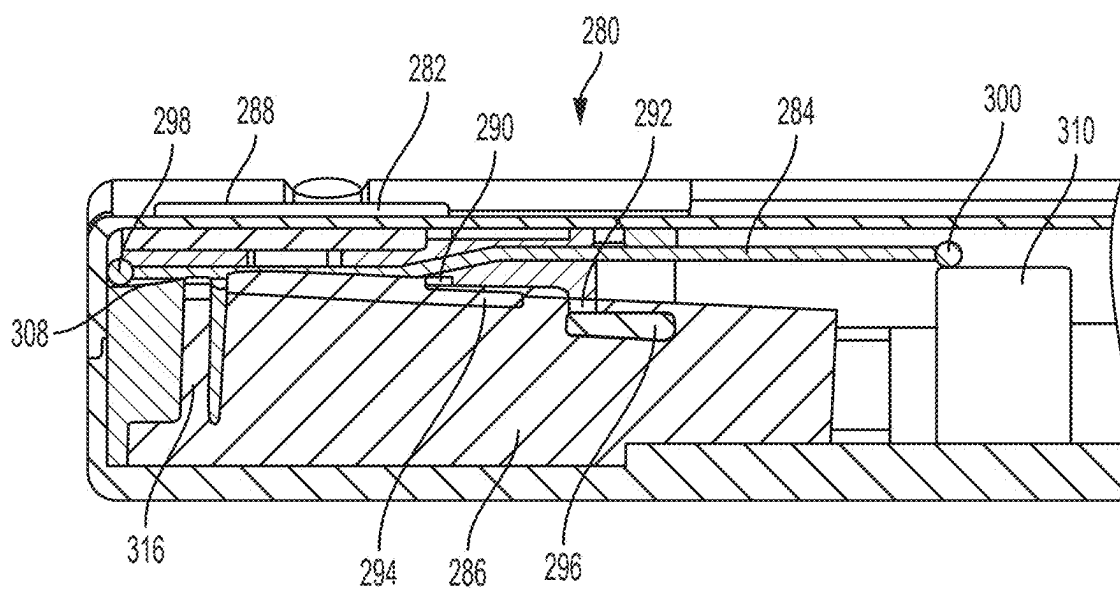


FIG. 65E

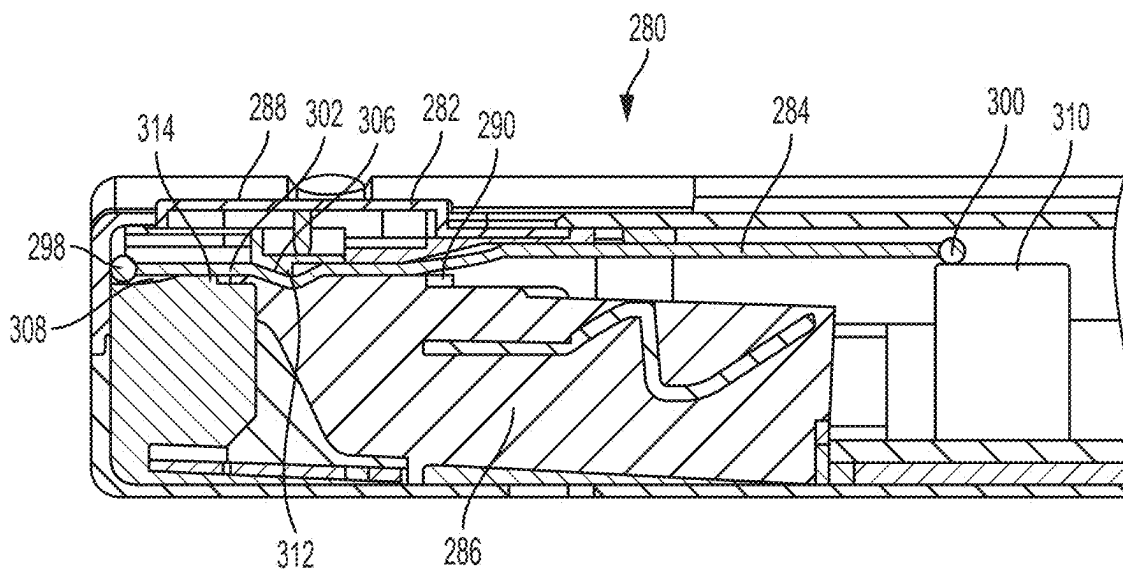


FIG. 65F

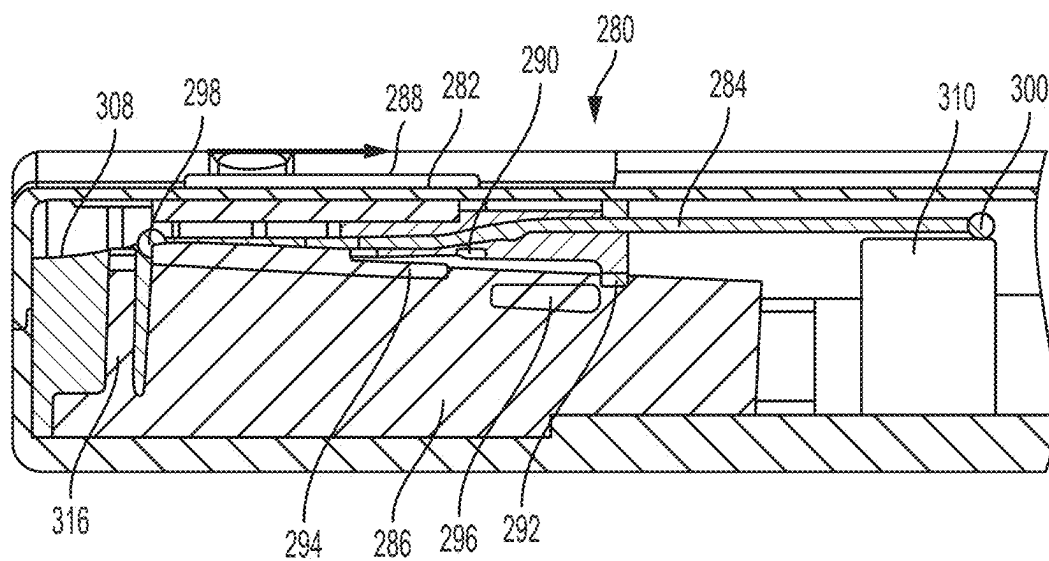


FIG. 65G

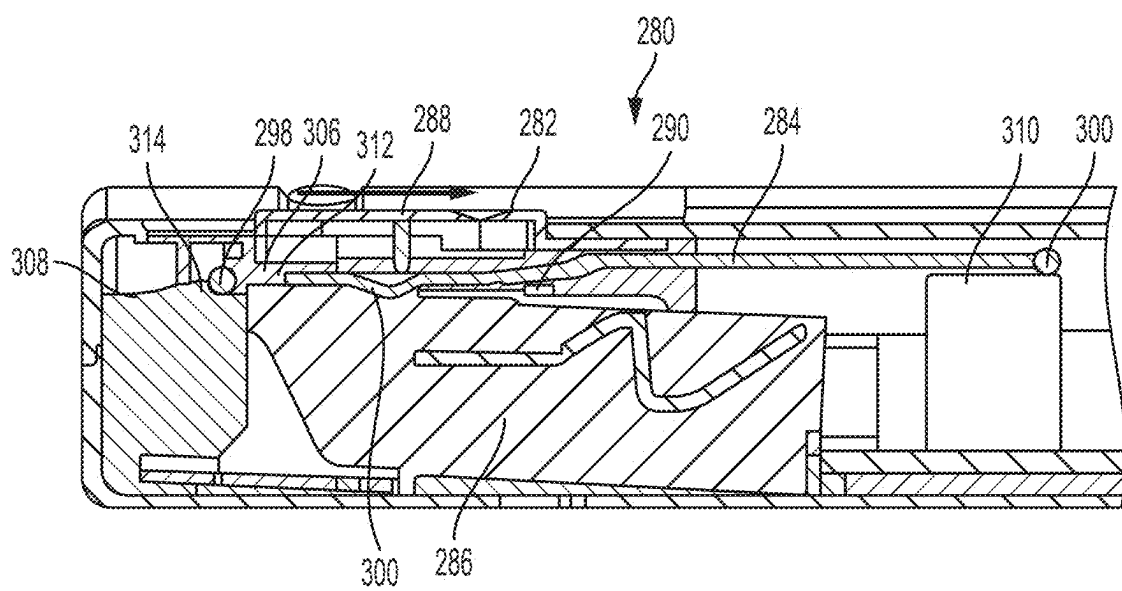


FIG. 65H

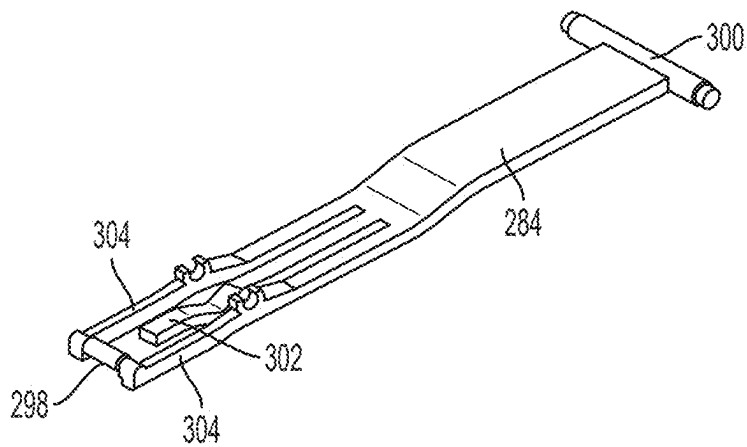


FIG. 66

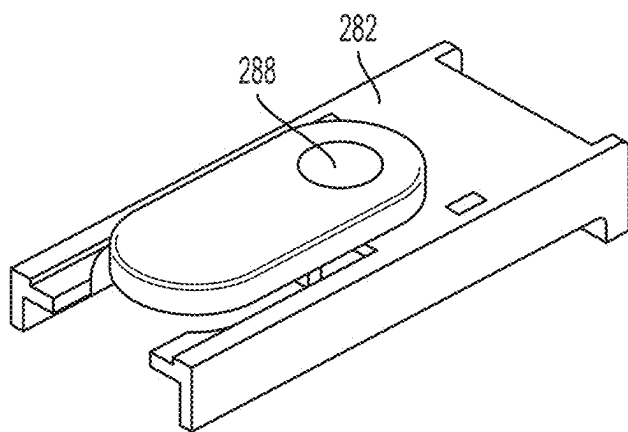


FIG. 67

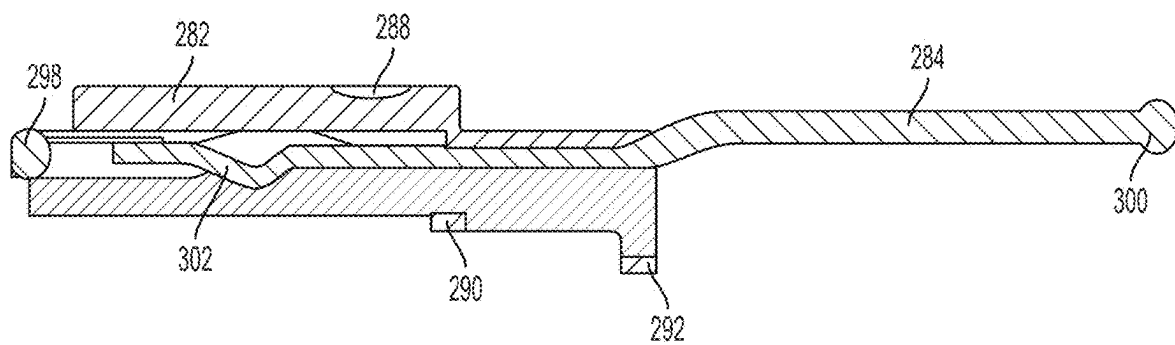


FIG. 68

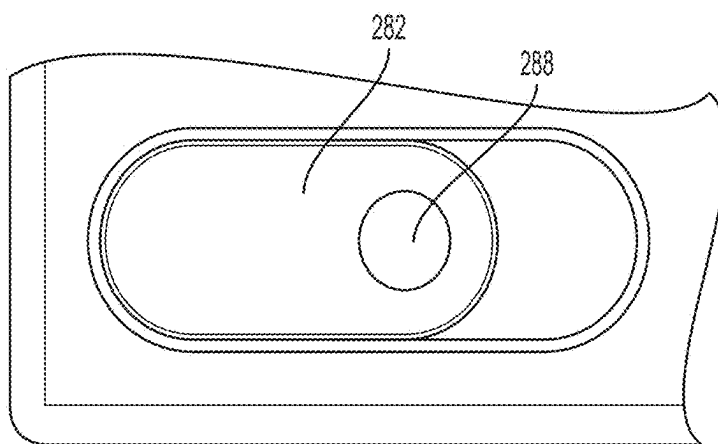


FIG. 69

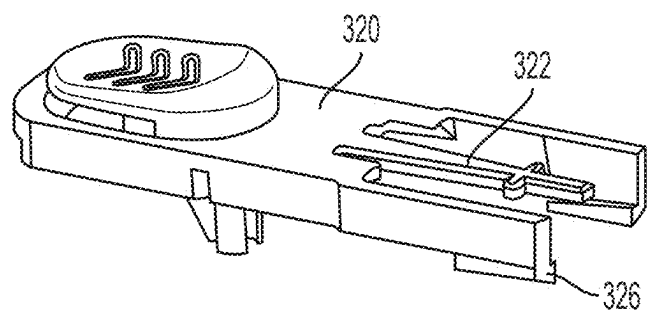


FIG. 68A

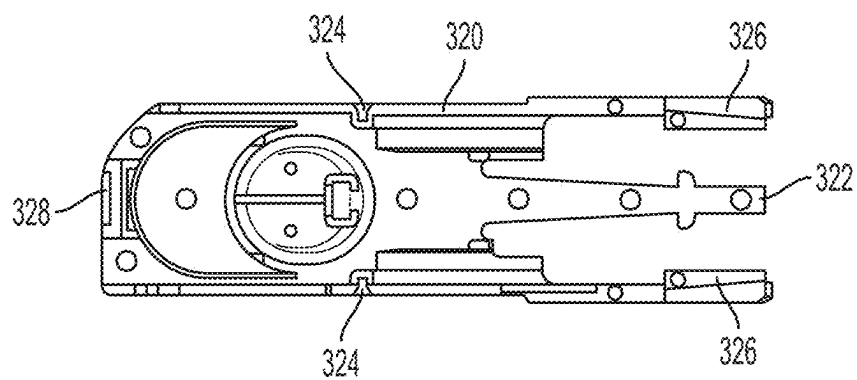


FIG. 68B

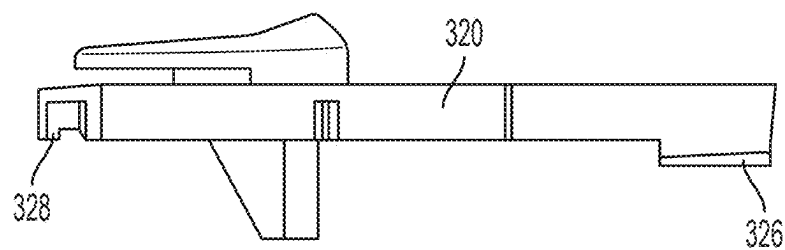


FIG. 68C

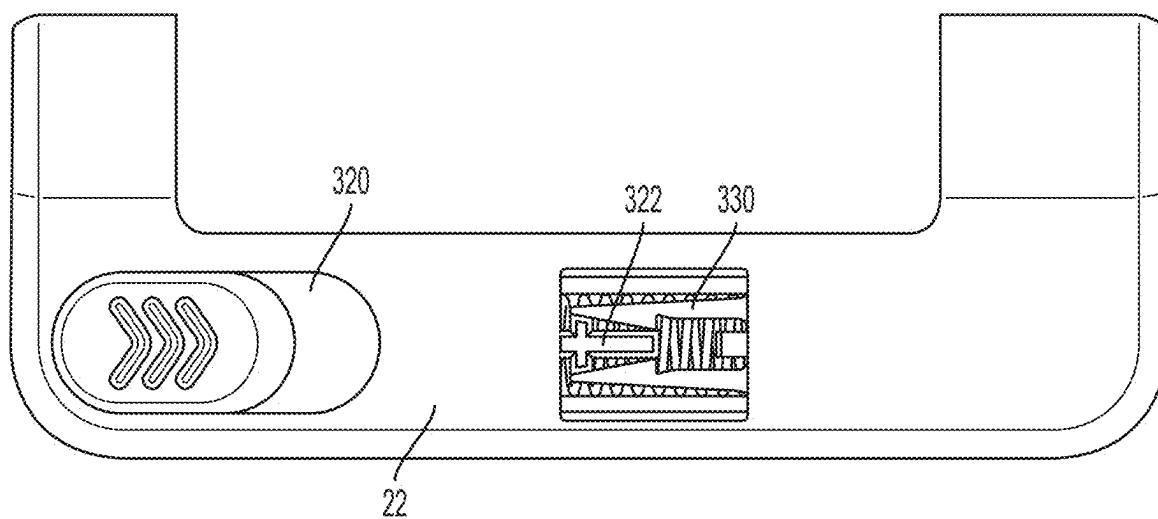


FIG. 68D

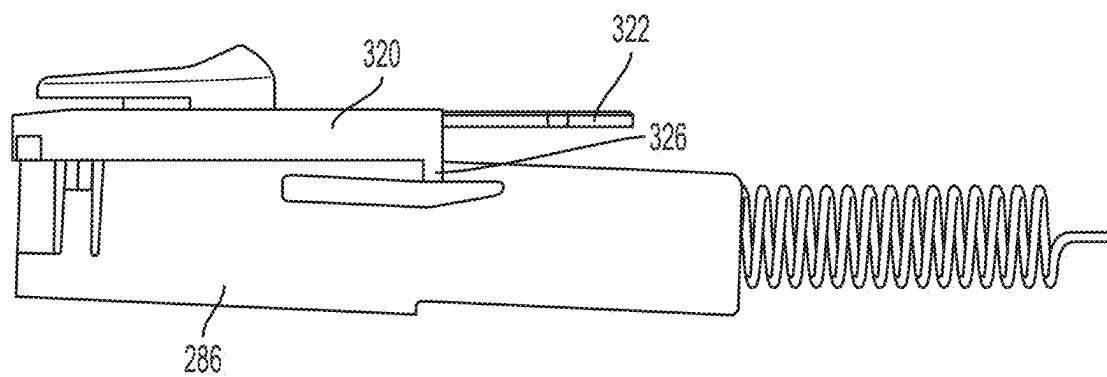


FIG. 68E

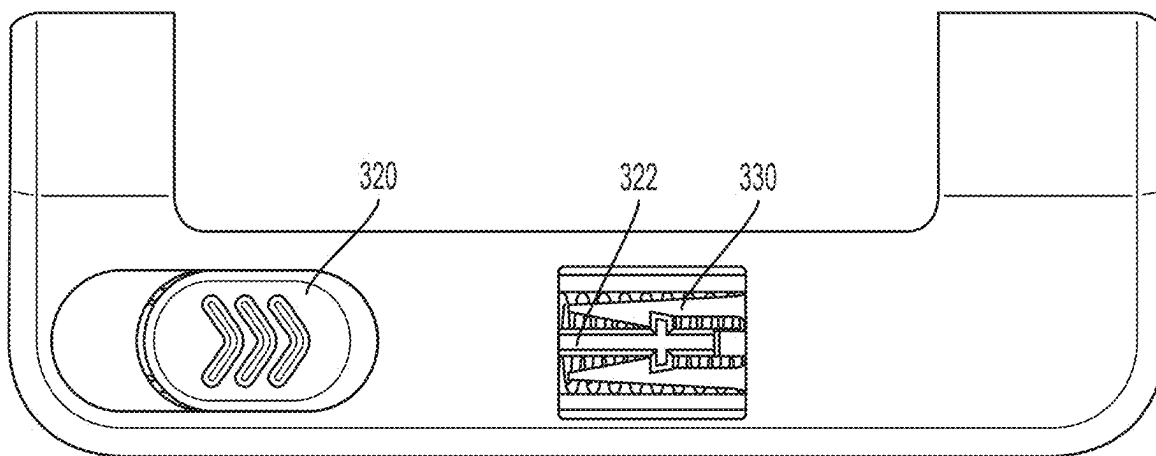


FIG. 68F

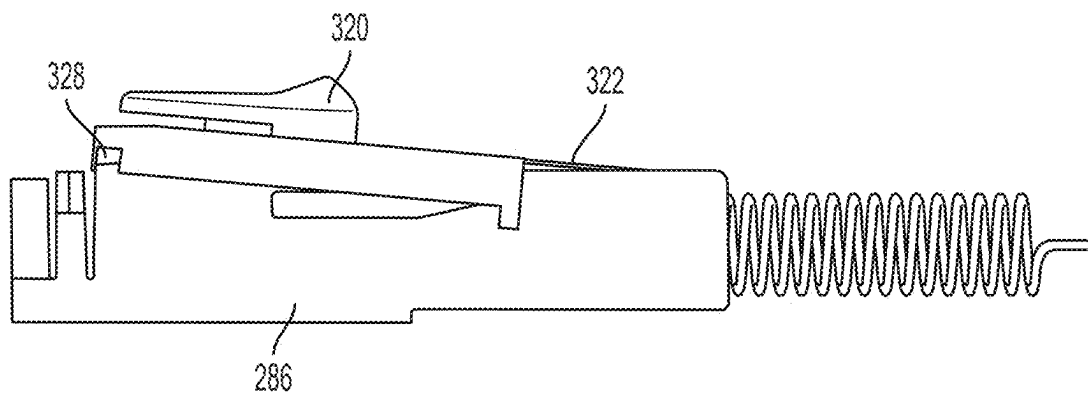


FIG. 68G

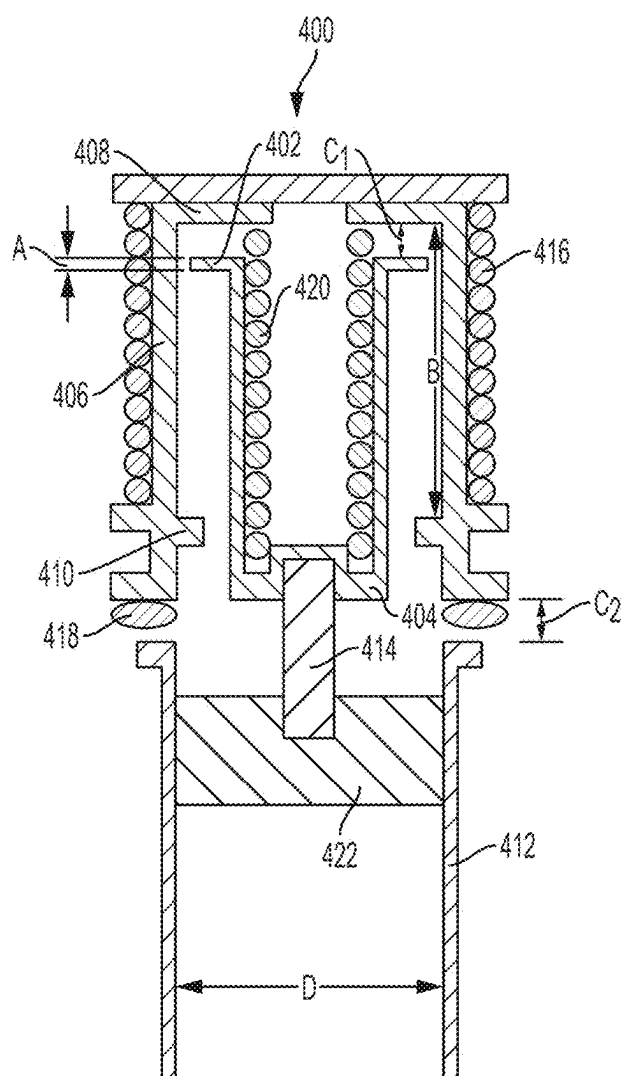


FIG. 70A

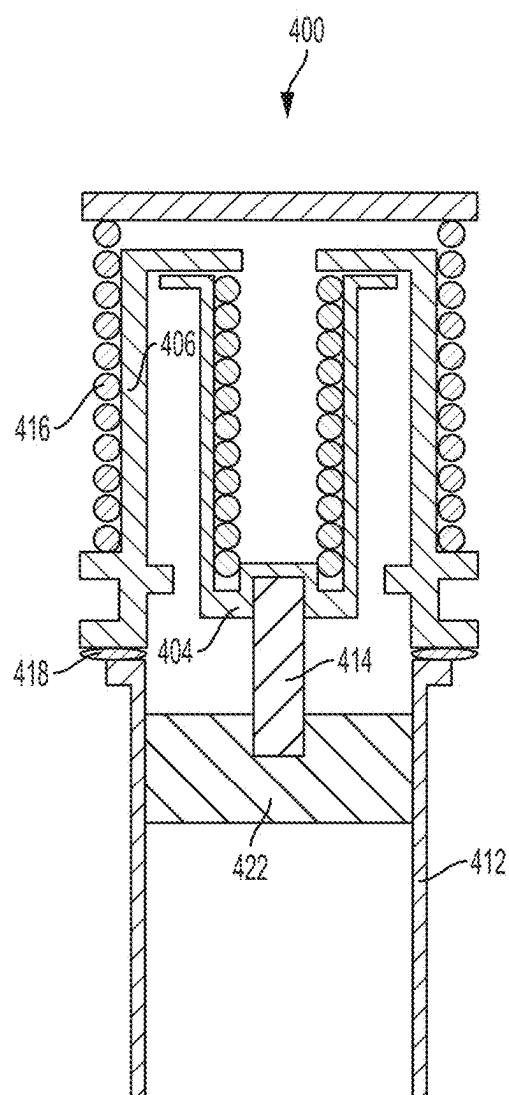


FIG. 70B

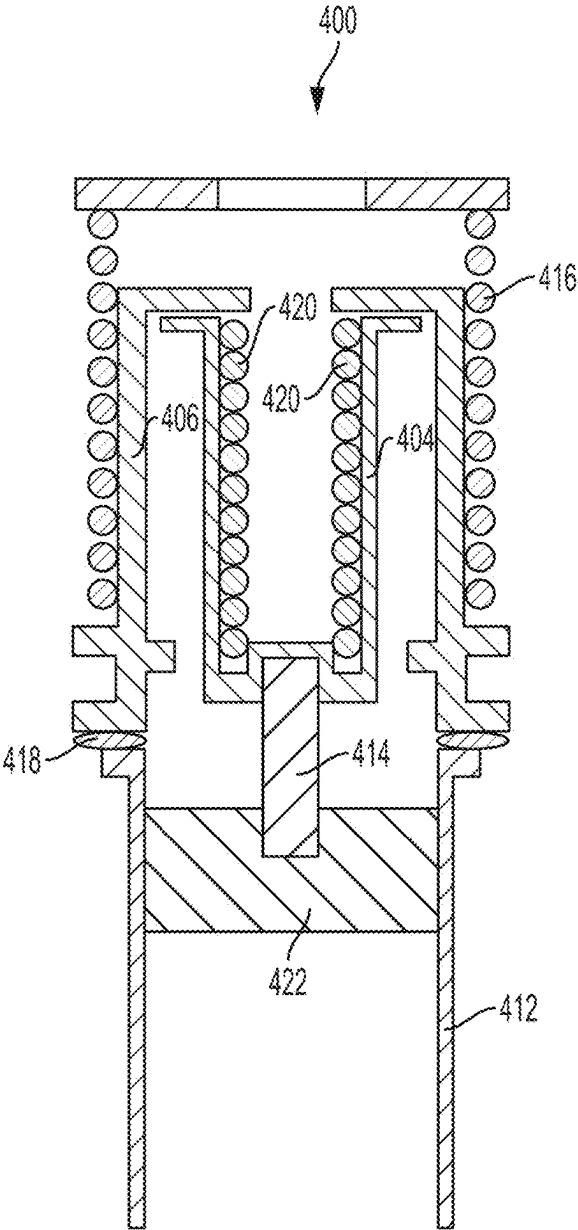


FIG. 70C

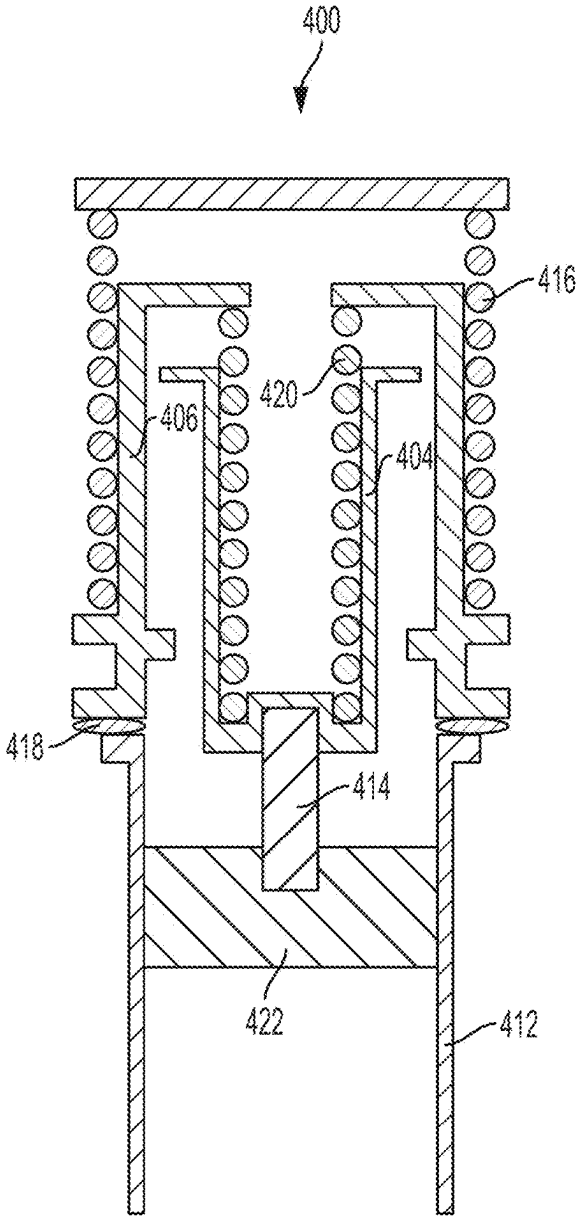


FIG. 70D

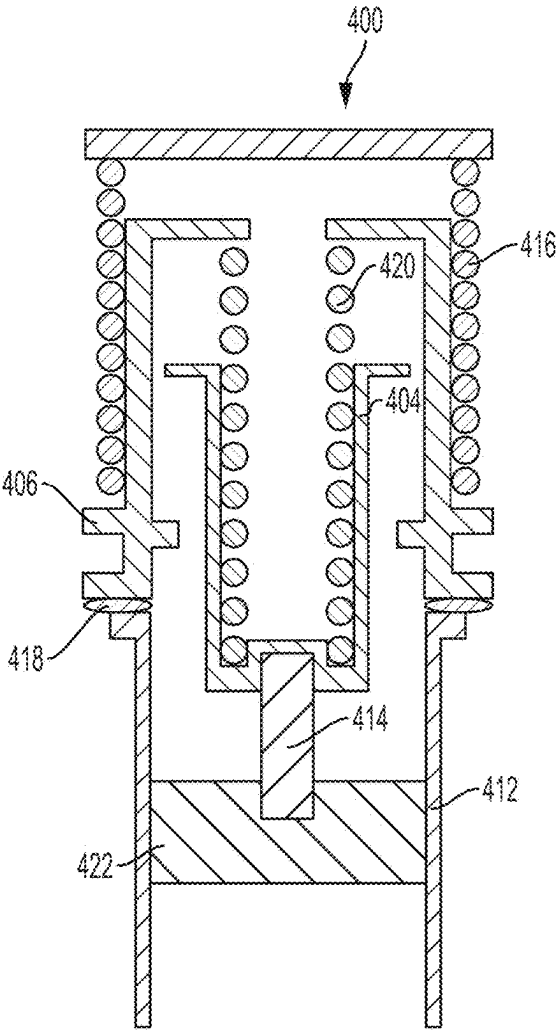


FIG. 70E

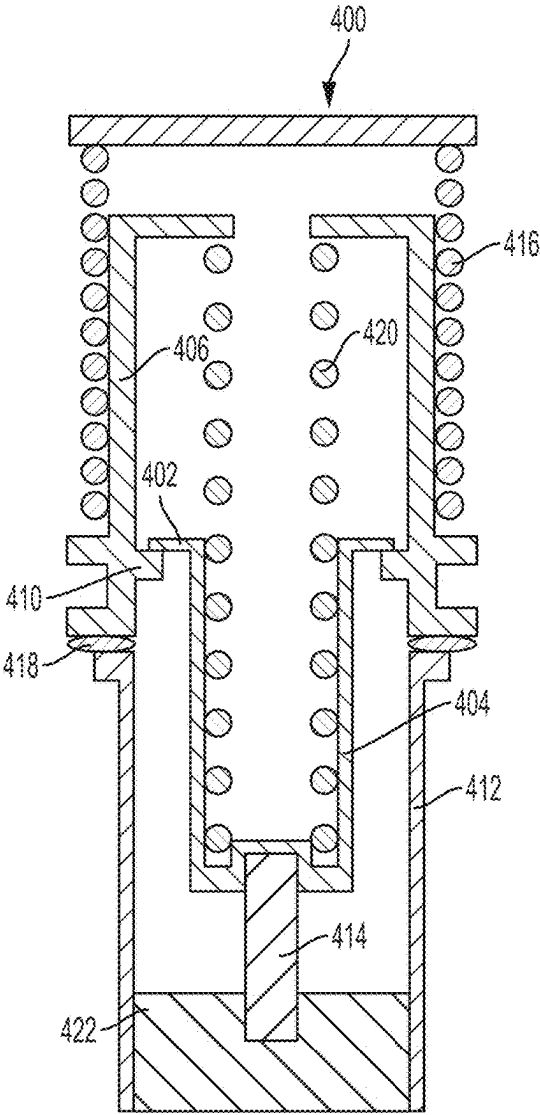


FIG. 70F

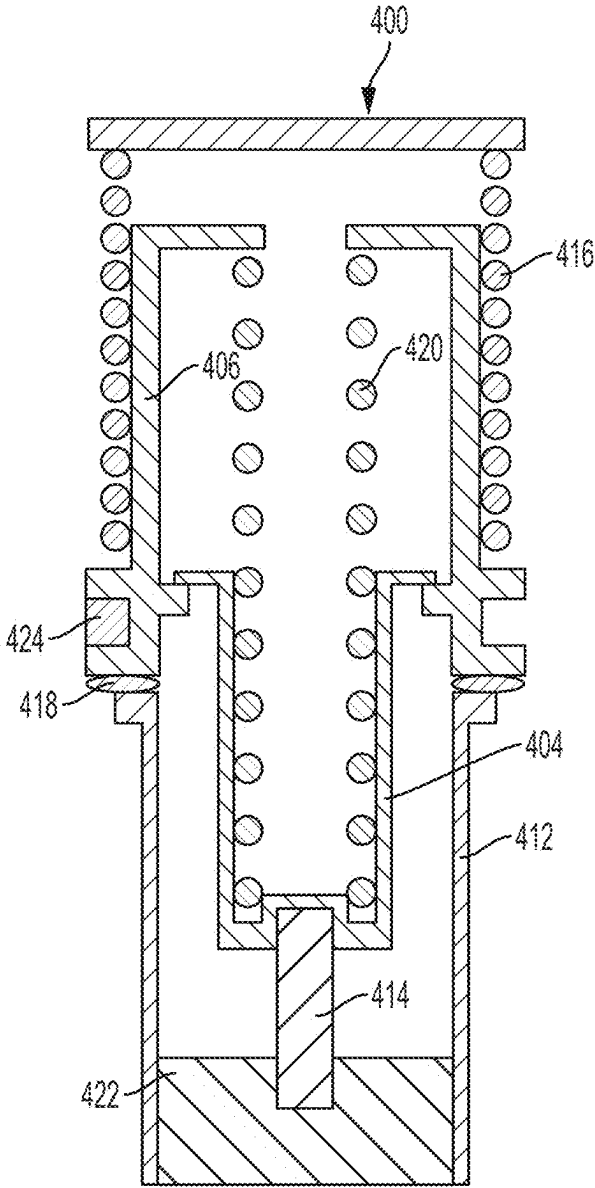


FIG. 70G

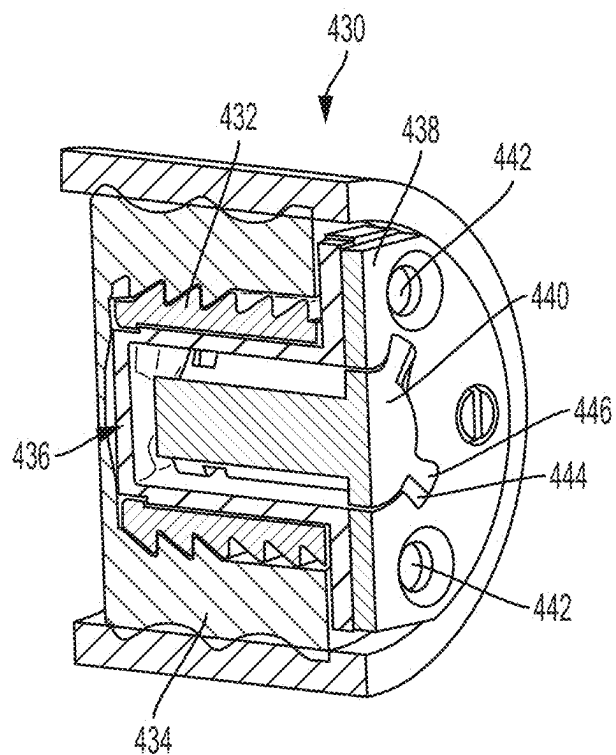


FIG. 71

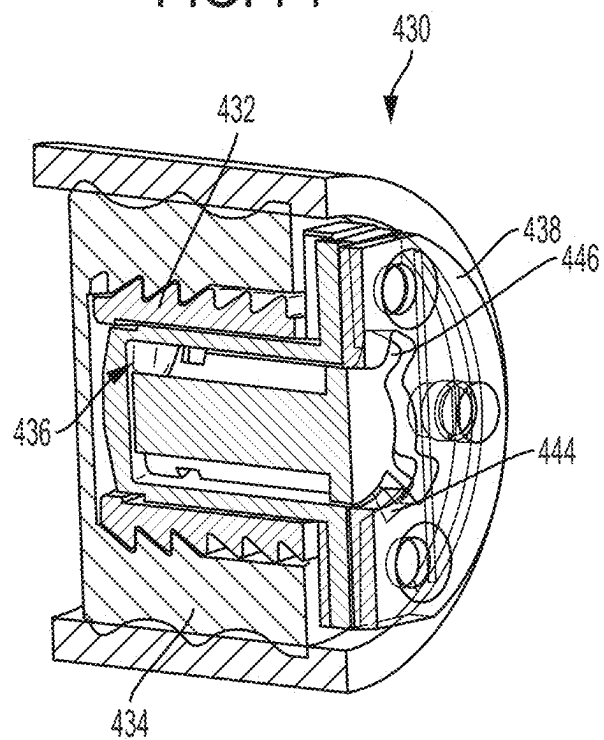


FIG. 72

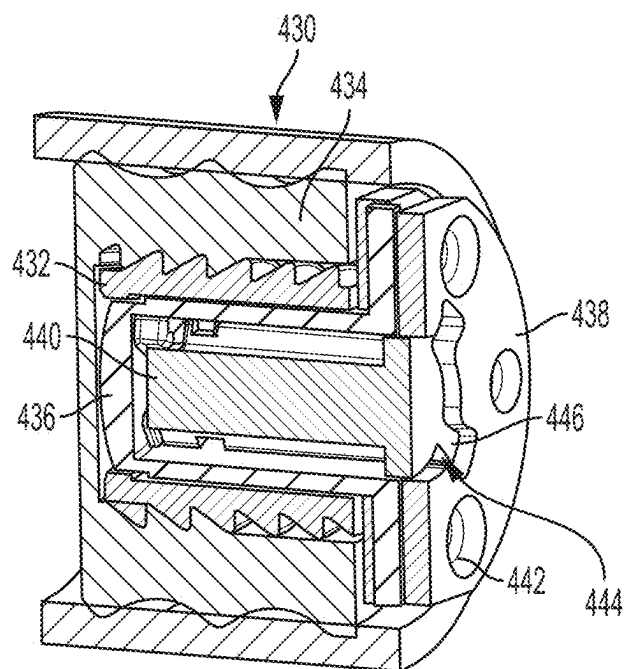


FIG. 73

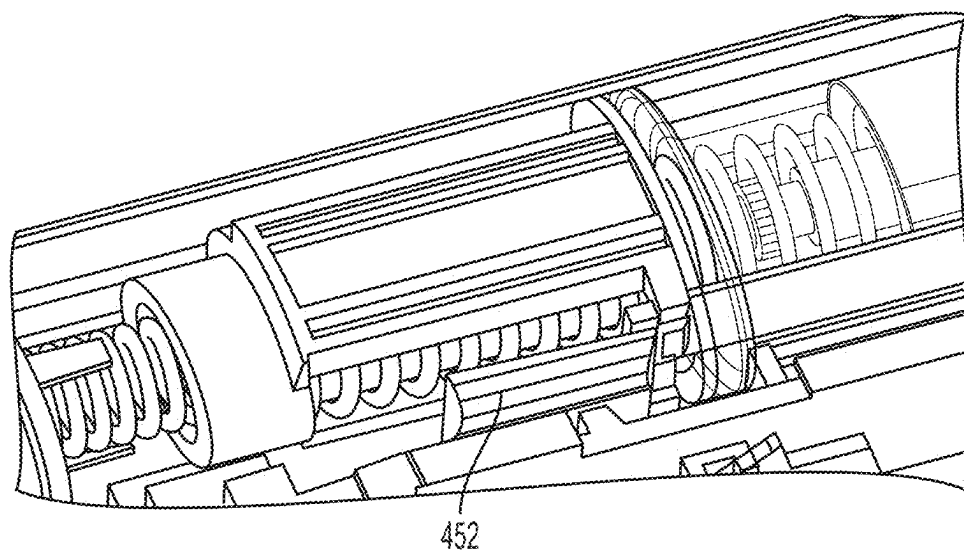


FIG. 74

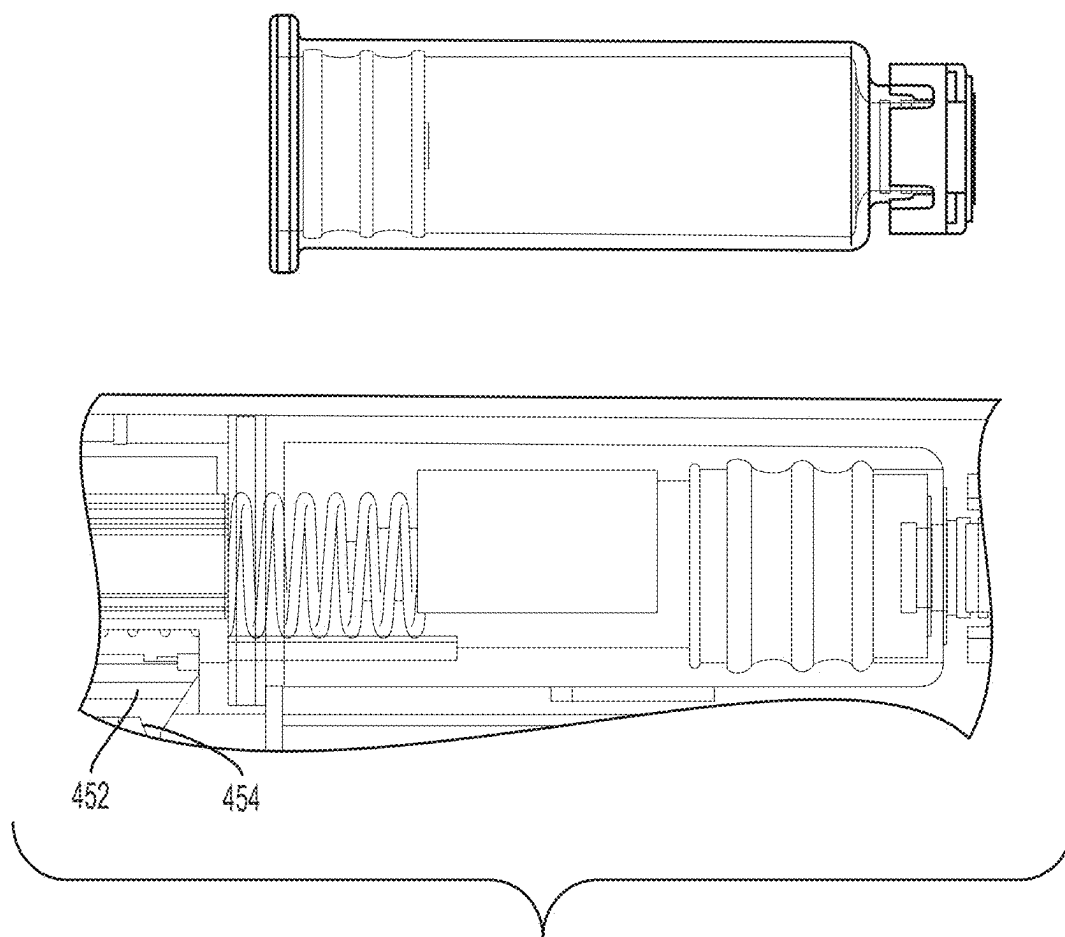


FIG. 75

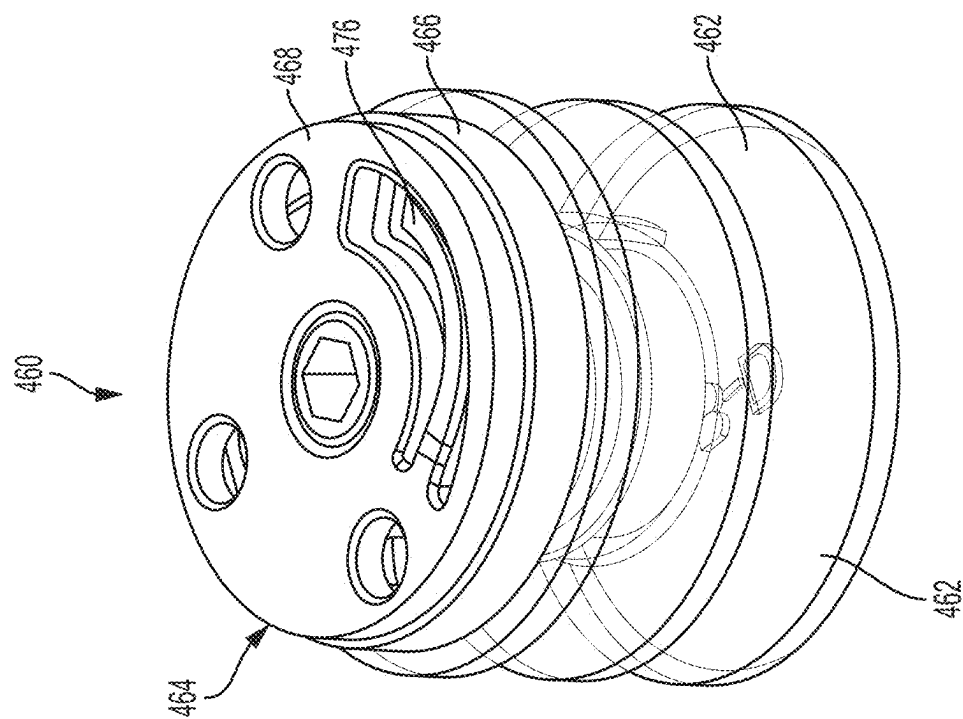


FIG. 77

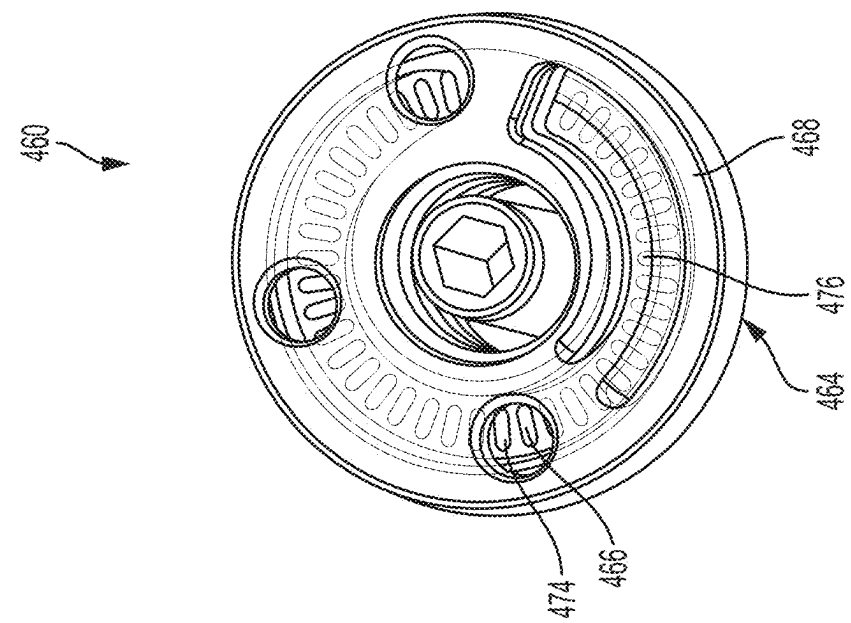


FIG. 76

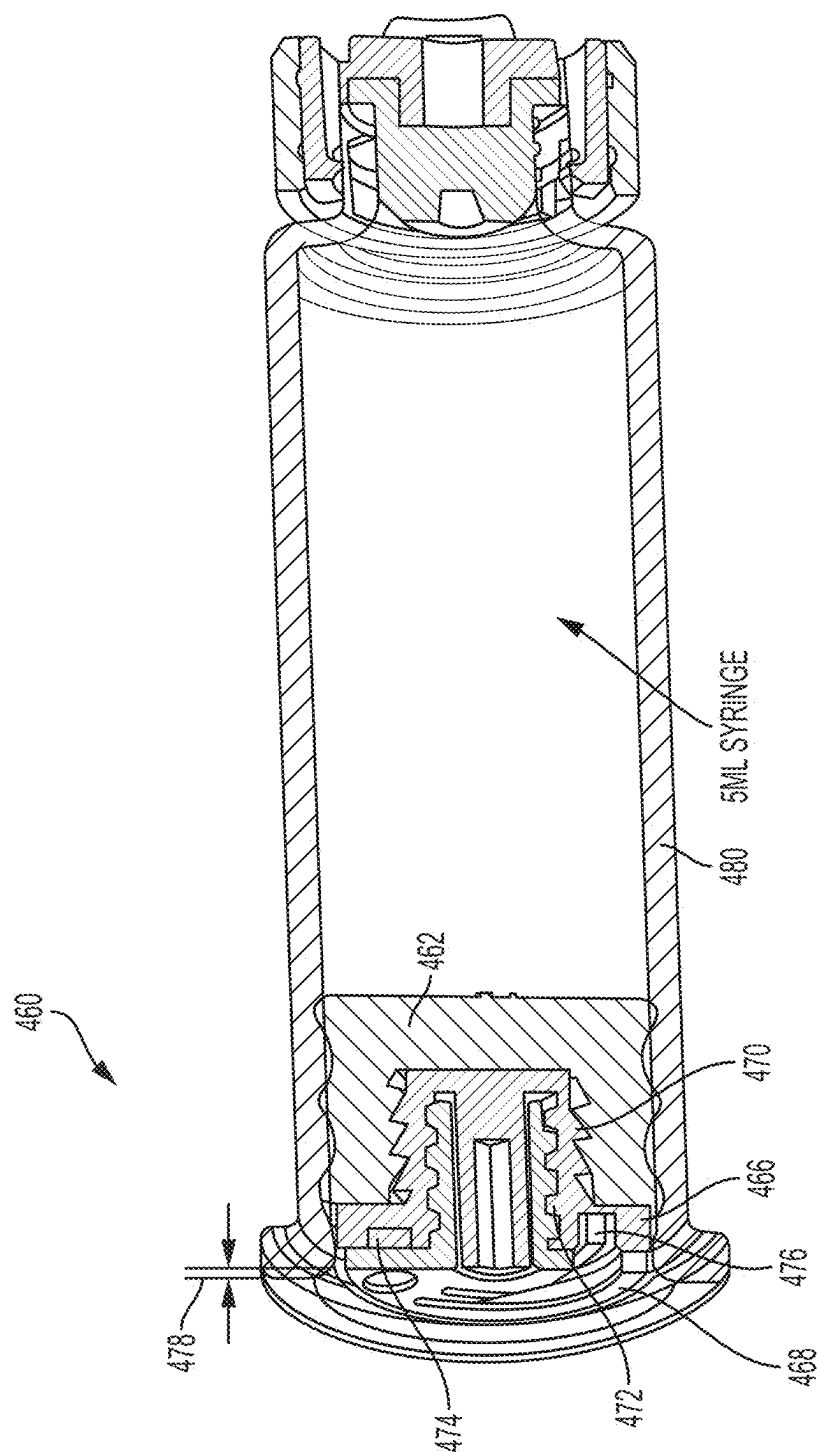


FIG. 78

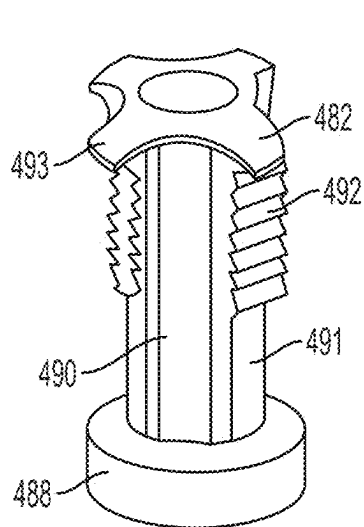


FIG. 79

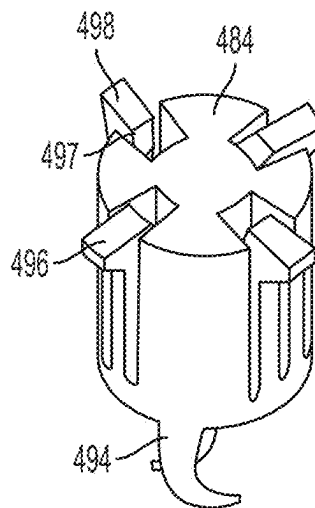


FIG. 80

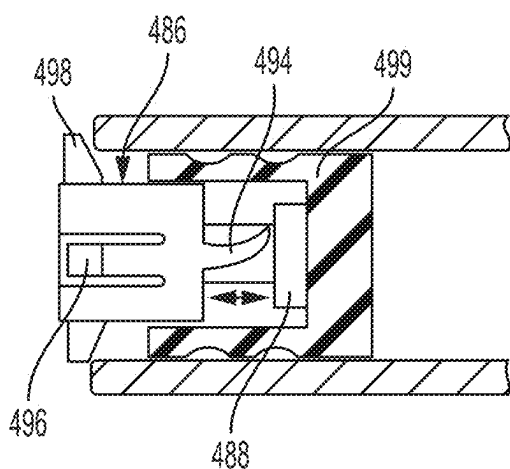


FIG. 81A

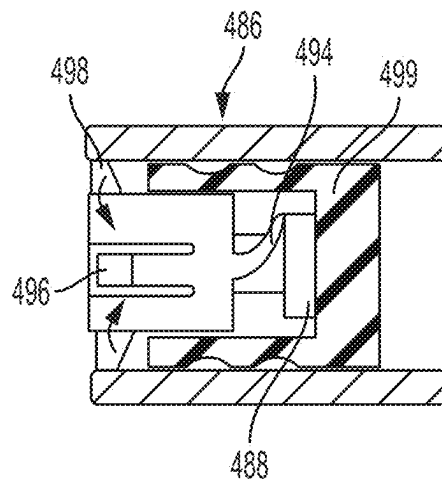


FIG. 81B

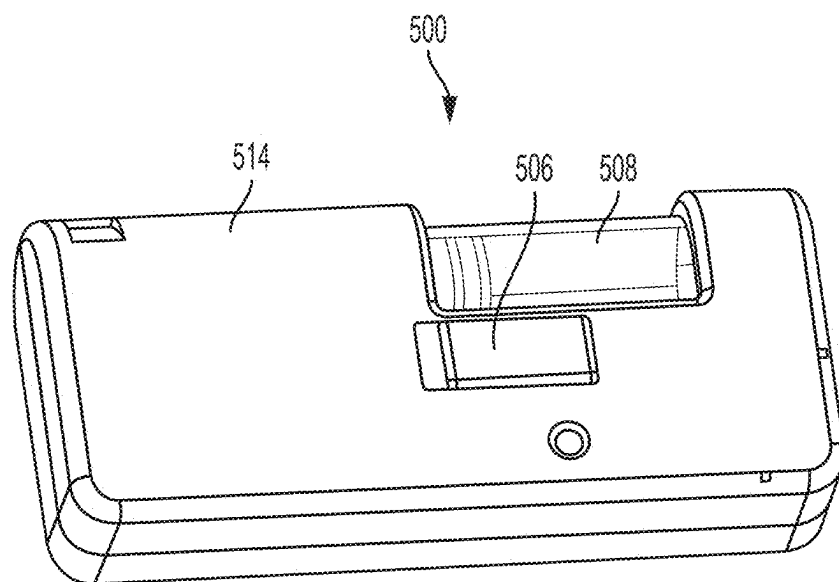


FIG. 82

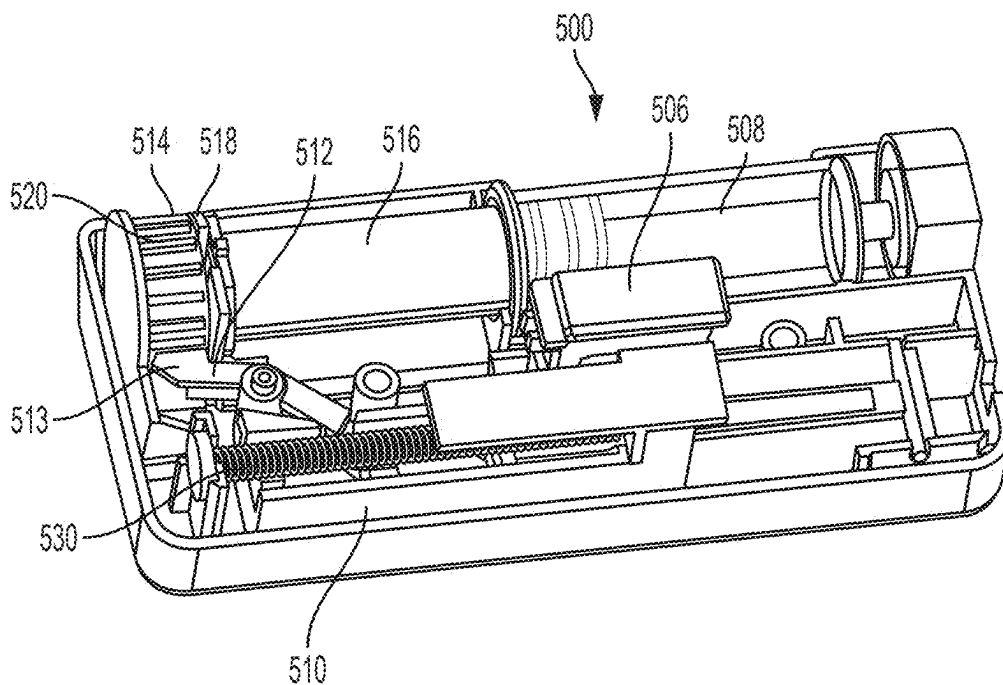


FIG. 83

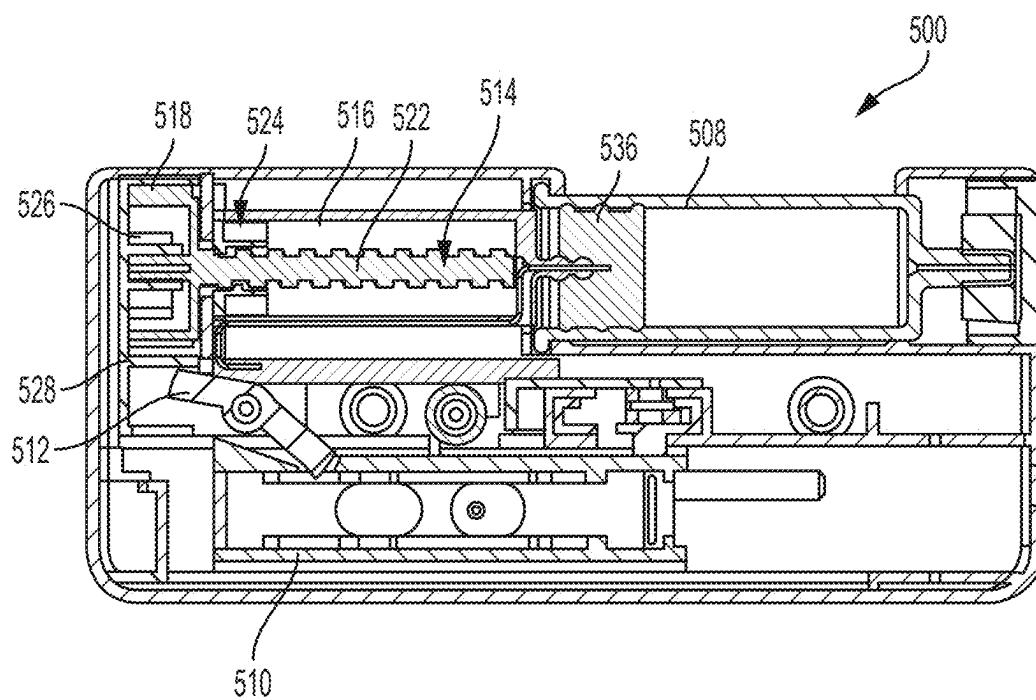


FIG. 84

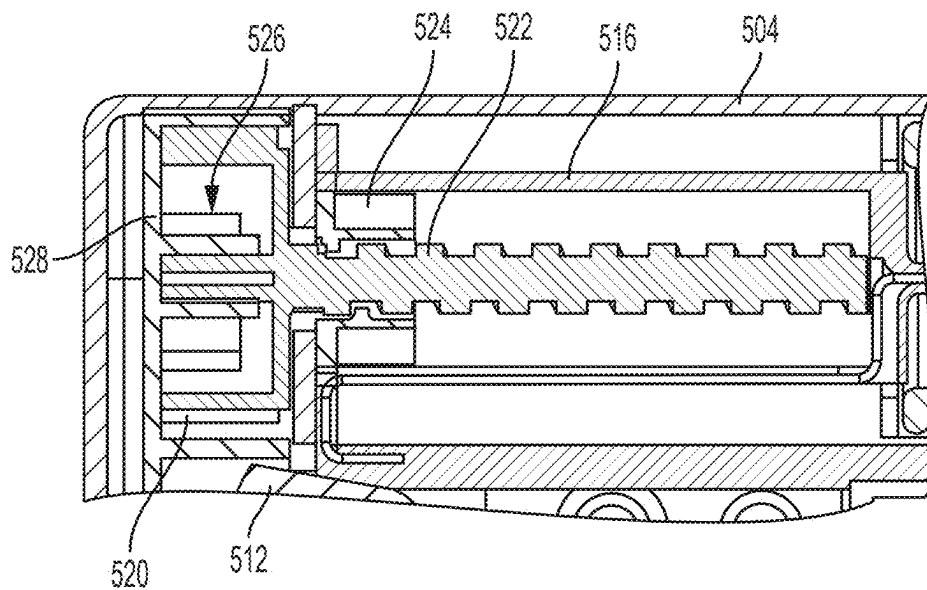


FIG. 85

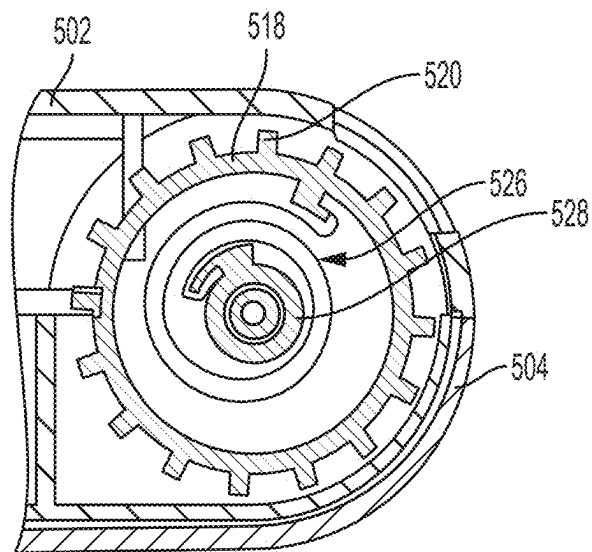


FIG. 86

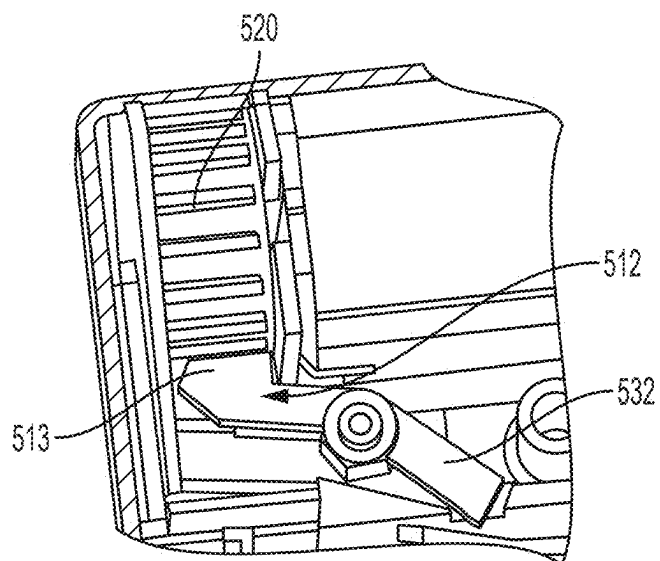


FIG. 87

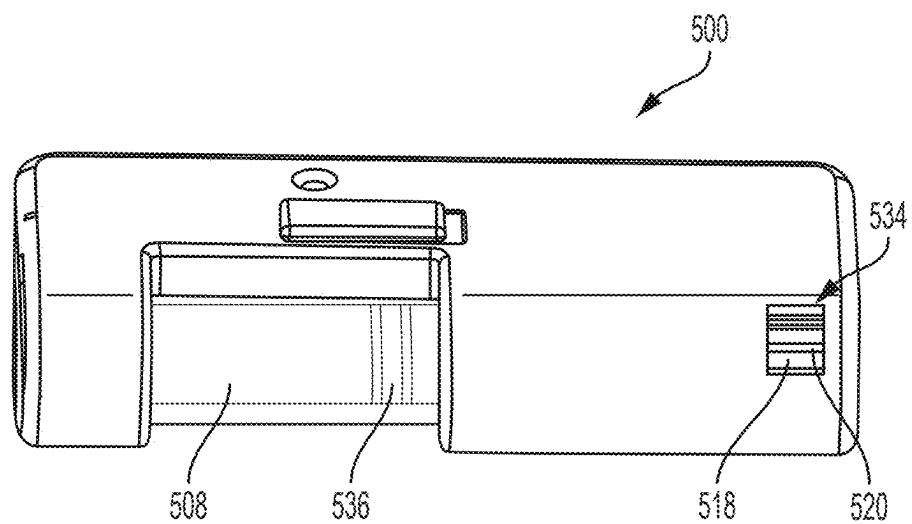


FIG. 88

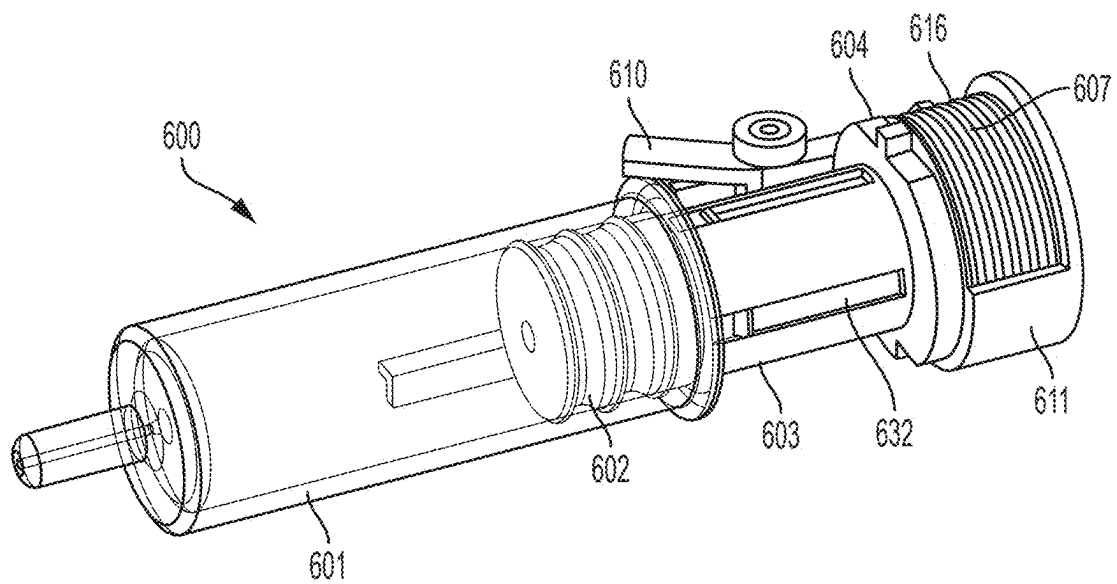


FIG. 89

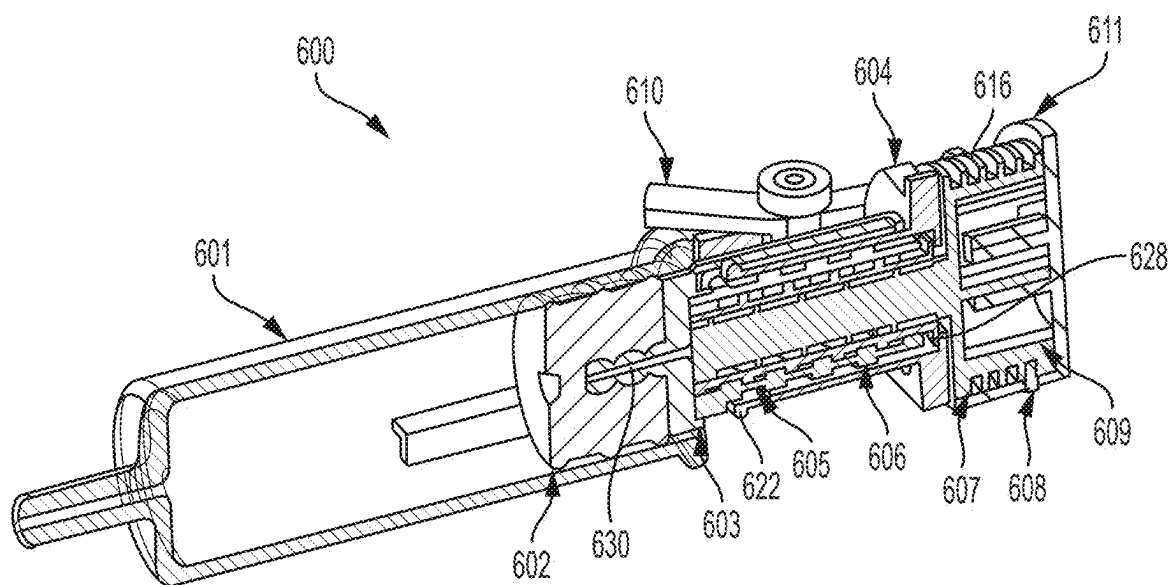


FIG. 90

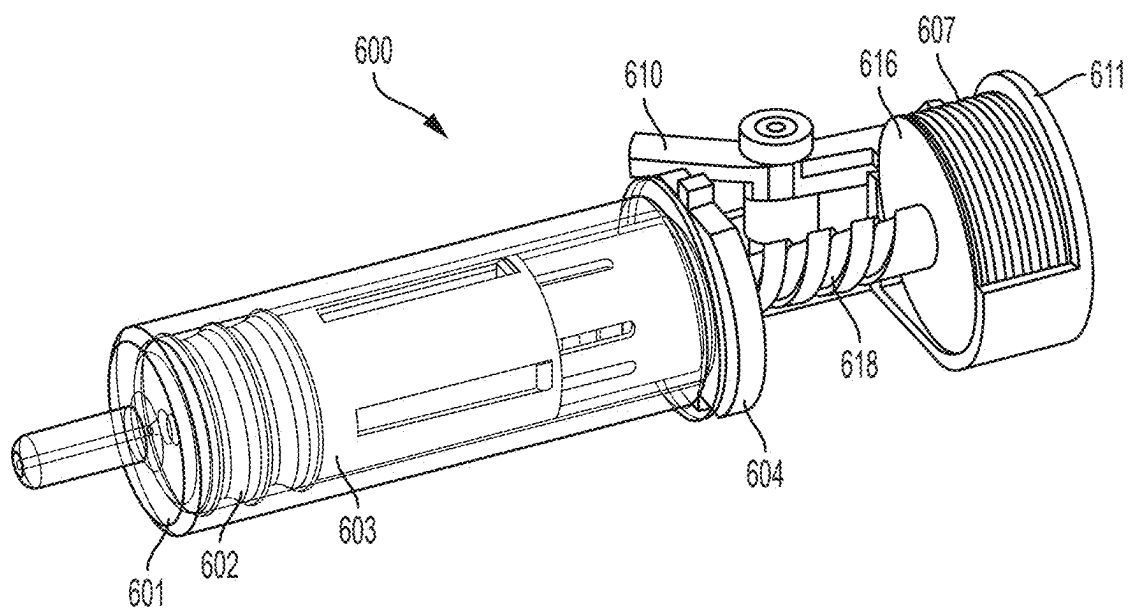


FIG. 91

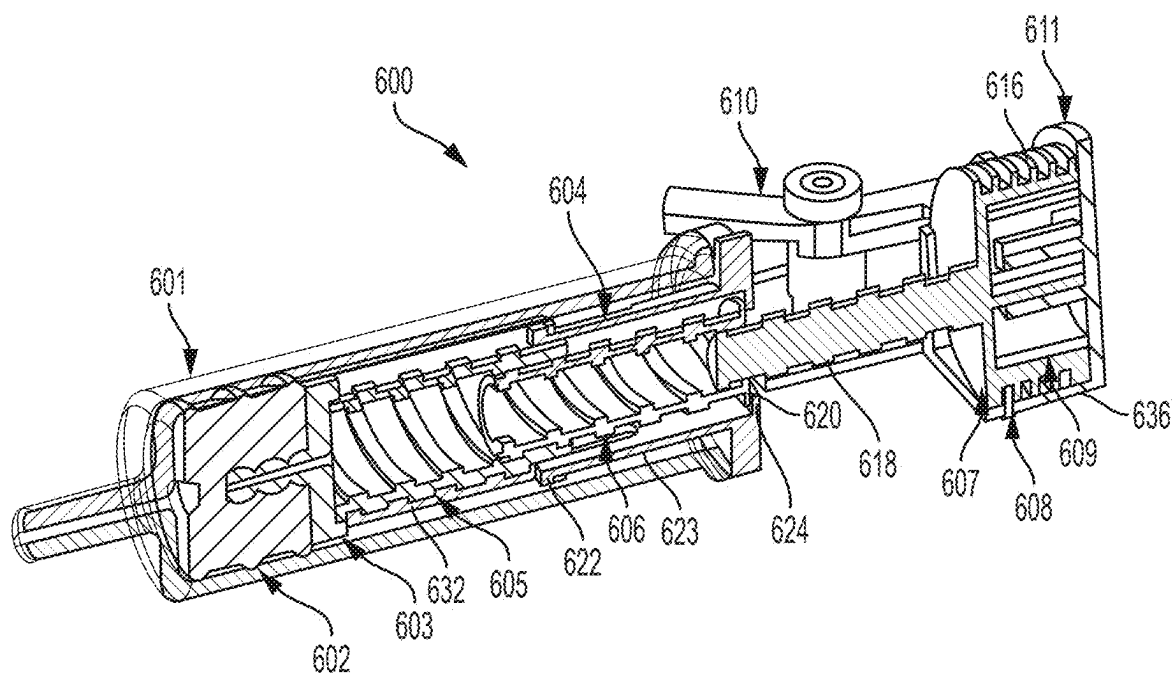


FIG. 92

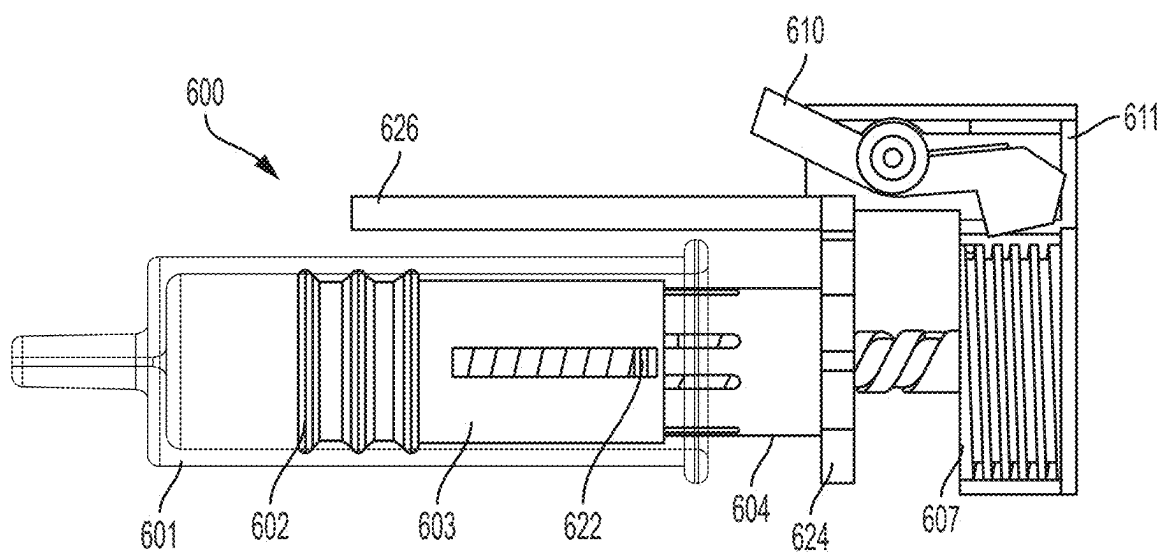


FIG. 93

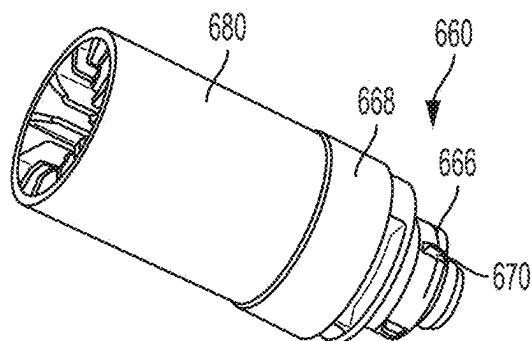


FIG. 94

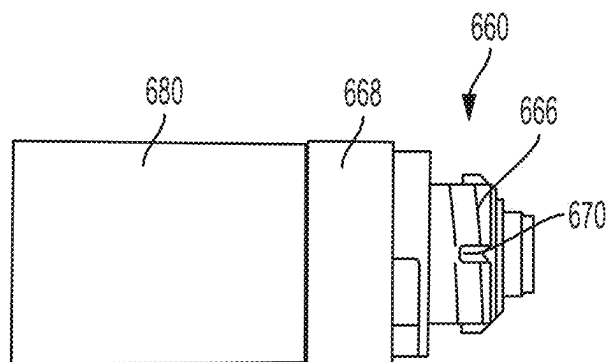


FIG. 95

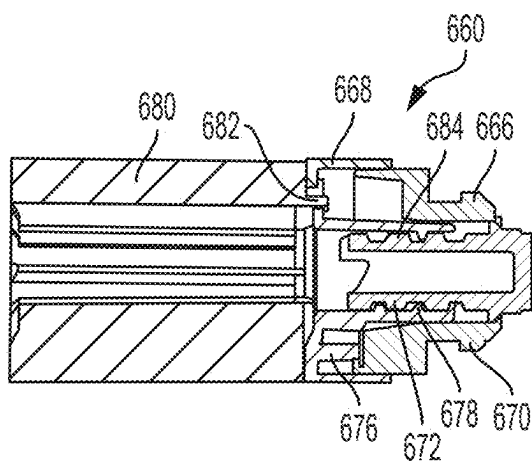


FIG. 96

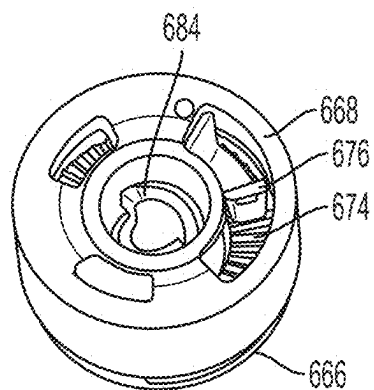


FIG. 97

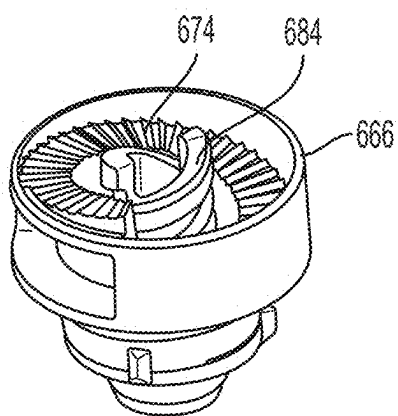


FIG. 98

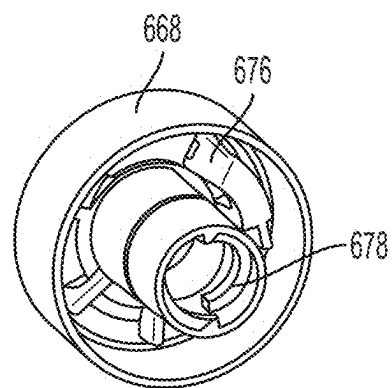


FIG. 99

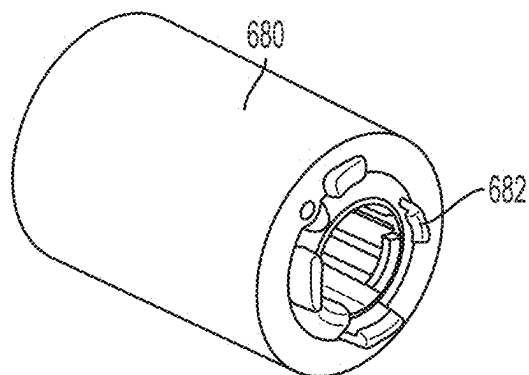


FIG. 100

NEEDLE ACTUATOR ASSEMBLY FOR DRUG DELIVERY SYSTEM

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application is a continuation of U.S. patent application Ser. No. 18/134,841, Apr. 14, 2023, which is a continuation of U.S. patent application Ser. No. 16/805,996, filed Mar. 2, 2020 (now U.S. Pat. No. 11,654,248), which is a continuation of U.S. patent application Ser. No. 15/616,212, filed Jun. 7, 2017 (now U.S. Pat. No. 10,603,445), which claims priority to U.S. Provisional Application Ser. No. 62/347,921, filed Jun. 9, 2016, the entire disclosures of which are hereby incorporated by reference in their entirety.

BACKGROUND OF THE INVENTION

Field of the Disclosure

[0002] The present disclosure relates generally to an injector device and method for delivering a fluid into the body of a patient by injection.

Description of the Related Art

[0003] Various types of automatic injection devices have been developed to allow drug solutions and other liquid therapeutic preparations to be administered by untrained personnel or to be self-injected. Generally, these devices include a reservoir that is pre-filled with the liquid therapeutic preparation, and some type of automatic needle-injection mechanism that can be triggered by the user. When the volume of fluid or drug to be administered is generally below a certain volume, such as 1 mL, an auto-injector is typically used, which typically has an injection time of about 10 to 15 seconds. When the volume of fluid or drug to be administered is above 1 mL, the injection time generally becomes longer resulting in difficulties for the patient to maintain contact between the device and the target area of the patient's skin. Further, as the volume of drug to be administered becomes larger, increasing the time period for injection becomes desirable. The traditional method for a drug to be injected slowly into a patient is to initiate an IV and inject the drug into the patient's body slowly. Such a procedure is typically performed in a hospital or outpatient setting.

[0004] Certain devices allow for self-injection in a home setting and are capable of gradually injecting a liquid therapeutic preparation into the skin of a patient. In some cases, these devices are small enough (both in height and in overall size) to allow them to be "worn" by a patient while the liquid therapeutic preparation is being infused into the patient. These devices typically include a pump or other type of discharge mechanism to force the liquid therapeutic preparation to flow out of a reservoir and into the injection needle. Such devices also typically include a valve or flow control mechanism to cause the liquid therapeutic preparation to begin to flow at the proper time and a triggering mechanism to initiate the injection.

SUMMARY OF THE INVENTION

[0005] In one aspect, a needle actuator assembly for a drug delivery system includes a needle actuator body having a guide surface, a needle shuttle having a cam surface, with the needle shuttle moveable along a vertical axis between a

first position and a second position. The needle shuttle is configured to move between the first and second position through engagement between the guide surface of the needle actuator body and the cam surface of the needle shuttle. The assembly further including a needle received by the needle shuttle.

[0006] The needle shuttle may include a biasing member configured to move the needle shuttle from the first position to the second position with the guide surface of the needle actuator body disengaged from the cam surface of the needle shuttle. The assembly may include a guide post, where the needle shuttle moves along the guide post. The guide post may be linear. The cam surface of the needle shuttle may include a first cam member and a second cam member spaced from the first cam member, where the guide surface is non-linear and includes a first side and a second side positioned opposite from the first side. The second cam member of the needle shuttle may be configured to engage the second side of the guide surface to move the needle shuttle from the first position to the second position, where the first cam member of the needle shuttle is configured to engage the first side of the guide surface to move the needle shuttle from the second position to the first position.

[0007] In a further aspect, a drug delivery system for injecting a medicament includes a housing, a needle actuator assembly received within the housing, where the needle actuator assembly includes a needle actuator body having a guide surface, with the needle actuator body moveable between a first position and a second position, and a needle shuttle having a cam surface. The needle shuttle is moveable along a vertical axis between a first position and a second position, where the needle shuttle is configured to move between the first and second position through engagement between the guide surface of the needle actuator body and the cam surface of the needle shuttle. The needle actuator assembly also includes a needle received by the needle shuttle. The system further including a button actuator at least partially received by the housing, where movement of the button actuator is configured to move the needle actuator body from the first position to the second position, and where movement of the needle actuator body from the first position to the second position is configured to cause movement of the needle shuttle from the first position to the second position.

[0008] The needle shuttle may include a biasing member configured to move the needle shuttle from the first position to the second position with the guide surface of the needle actuator body disengaged from the cam surface of the needle shuttle. The system may include a guide post extending from the housing, where the needle shuttle moves along the guide post. The guide post may extend about perpendicular from the housing.

[0009] The system may include a pad configured to engage the needle when the needle actuator is in the second position. The pad may be received by a pad arm having a cam surface configured to engage a corresponding cam track on the housing to move the pad beneath the needle.

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] The above-mentioned and other features and advantages of this disclosure, and the manner of attaining them, will become more apparent and the disclosure itself will be better understood by reference to the following

descriptions of embodiments of the disclosure taken in conjunction with the accompanying drawings, wherein:

[0011] FIG. 1 is a perspective view of a drug delivery system according to one aspect of the present invention.

[0012] FIG. 2 is a perspective, cross-sectional view of the drug delivery system of FIG. 1 according to one aspect of the present invention.

[0013] FIG. 3 is a front, cross-sectional view of the drug delivery system of FIG. 1 according to one aspect of the present invention.

[0014] FIG. 4 is a top view of the drug delivery system of FIG. 1 according to one aspect of the present invention, showing a top portion of the housing removed and the drug delivery system in a pre-use position.

[0015] FIG. 5 is a top, cross-sectional view of the drug delivery system of FIG. 1 according to one aspect of the present invention, showing the drug delivery system in a pre-use position.

[0016] FIG. 6 is a front, cross-sectional view of the drug delivery system of FIG. 1 according to one aspect of the present invention, showing the drug delivery system in a pre-use position.

[0017] FIG. 7 is a top view of the drug delivery system of FIG. 1 according to one aspect of the present invention, showing a top portion of the housing removed and the drug delivery system in an initial actuation position.

[0018] FIG. 8 is a top, cross-sectional view of the drug delivery system of FIG. 1 according to one aspect of the present invention, showing the drug delivery system in an initial actuation position.

[0019] FIG. 9 is a front, cross-sectional view of the drug delivery system of FIG. 1 according to one aspect of the present invention, showing the drug delivery system in an initial actuation position.

[0020] FIG. 10 is a top view of the drug delivery system of FIG. 1 according to one aspect of the present invention, showing a top portion of the housing removed and the drug delivery system in a use position.

[0021] FIG. 11 is a top, cross-sectional view of the drug delivery system of FIG. 1 according to one aspect of the present invention, showing the drug delivery system in a use position.

[0022] FIG. 12 is a front, cross-sectional view of the drug delivery system of FIG. 1 according to one aspect of the present invention, showing the drug delivery system in a use position.

[0023] FIG. 13 is a top view of the drug delivery system of FIG. 1 according to one aspect of the present invention, showing a top portion of the housing removed and the drug delivery system in a post-use position.

[0024] FIG. 14 is a top, cross-sectional view of the drug delivery system of FIG. 1 according to one aspect of the present invention, showing the drug delivery system in a post-use position.

[0025] FIG. 15 is a front, cross-sectional view of the drug delivery system of FIG. 1 according to one aspect of the present invention, showing the drug delivery system in a post-use position.

[0026] FIG. 15A is a front, cross-sectional view of the drug delivery system of FIG. 1 according to one aspect of the present invention, showing a pad with the drug delivery system in a pre-use position.

[0027] FIG. 15B is a perspective, cross-sectional view of the drug delivery system of FIG. 1 according to one aspect

of the present invention, showing a pad with the drug delivery system in a pre-use position.

[0028] FIG. 15C is a perspective, cross-sectional view of the drug delivery system of FIG. 1 according to one aspect of the present invention, showing a pad with the drug delivery system in a pre-use position.

[0029] FIG. 16 is a partial cross-sectional view of the drug delivery system of FIG. 1 according to one aspect of the present invention, showing a valve assembly.

[0030] FIG. 17 is a perspective view of a drive assembly for a drug delivery system according to one aspect of the present invention.

[0031] FIG. 18 is a cross-sectional view of the drive assembly of FIG. 17 according to one aspect of the present invention, showing a pre-use position of the drive assembly.

[0032] FIG. 19 is a cross-sectional view of the drive assembly of FIG. 17 according to one aspect of the present invention, showing a use position of the drive assembly.

[0033] FIG. 20 is a cross-sectional view of the drive assembly of FIG. 17 according to one aspect of the present invention, showing a post-use position of the drive assembly.

[0034] FIG. 21 is a perspective view of a plunger actuation member of the drive assembly of FIG. 17 according to one aspect of the present invention.

[0035] FIG. 22 is a perspective view of a first plunger member of the drive assembly of FIG. 17 according to one aspect of the present invention.

[0036] FIG. 23 is a perspective view of a plunger actuation member and first plunger member of the drive assembly of FIG. 17 according to one aspect of the present invention, showing the plunger actuation member engaged with the first plunger member.

[0037] FIG. 24 is a perspective view of a plunger actuation member and first plunger member of the drive assembly of FIG. 17 according to one aspect of the present invention, showing the plunger actuation member disengaged from the first plunger member.

[0038] FIG. 25 is a perspective view of a plunger actuation member and first plunger member of the drive assembly of FIG. 17 according to one aspect of the present invention, showing the plunger actuation member disengaged from and axially displaced relative to the first plunger member.

[0039] FIG. 26 is a front view of a first plunger member and a second plunger member of the drive assembly of FIG. 17 according to one aspect of the present invention.

[0040] FIG. 27 is a top view of a drive assembly for a drug delivery system according to a further aspect of the present invention.

[0041] FIG. 28 is a perspective view of the drive assembly of FIG. 27 according to one aspect of the present invention.

[0042] FIG. 29 is a cross-sectional view of the drive assembly of FIG. 27 according to one aspect of the present invention, showing a pre-use position of the drive assembly.

[0043] FIG. 30 is a perspective view of the drive assembly of FIG. 27 according to one aspect of the present invention, showing the drive assembly received by a bottom portion of a housing.

[0044] FIG. 31 is a perspective view of the housing of FIG. 30 according to one aspect of the present invention.

[0045] FIG. 32 is a top view of the drive assembly of FIG. 27 according to one aspect of the present invention, showing engagement of the drive assembly with a portion of a needle actuator in an initial actuation position of the drive assembly.

[0046] FIG. 33 is an enlarged perspective view of the drive assembly of FIG. 27 according to one aspect of the present invention, showing engagement of the drive assembly with a portion of a needle actuator in an initial actuation position of the drive assembly.

[0047] FIG. 34 is a front view of a needle actuator assembly according to one aspect of the present invention.

[0048] FIG. 35 is a left side perspective view of a needle shuttle of the needle actuator assembly of FIG. 34 according to one aspect of the present invention.

[0049] FIG. 36 is a right side perspective view of a needle shuttle of the needle actuator assembly of FIG. 34 according to one aspect of the present invention.

[0050] FIG. 37A is a front view of the needle actuator assembly of FIG. 34 according to one aspect of the present invention, showing the needle actuator assembly in a pre-use position.

[0051] FIG. 37B is a front view of the needle actuator assembly of FIG. 34 according to one aspect of the present invention, showing the needle actuator assembly in a use position.

[0052] FIG. 37C is a front view of the needle actuator assembly of FIG. 34 according to one aspect of the present invention, showing the needle actuator assembly in an initial post-use position.

[0053] FIG. 37D is a front view of the needle actuator assembly of FIG. 34 according to one aspect of the present invention, showing the needle actuator assembly in a post-use position.

[0054] FIG. 38A is a perspective view of the needle actuator assembly of FIG. 34 according to one aspect of the present invention, showing the needle actuator assembly in a use position.

[0055] FIG. 38B is a perspective view of the needle actuator assembly of FIG. 34 according to one aspect of the present invention, showing the needle actuator assembly in an initial post-use position.

[0056] FIG. 39 is a perspective view of an actuator button and the needle actuator assembly of FIG. 34 according to one aspect of the present invention, showing the needle actuator assembly in an initial post-use position.

[0057] FIG. 40A is a cross-sectional view of an actuator button and the needle actuator assembly of FIG. 34 according to one aspect of the present invention, showing the needle actuator assembly in an initial post-use position.

[0058] FIG. 40B is a perspective view of an actuator button and the needle actuator assembly of FIG. 34 according to one aspect of the present invention, showing the needle actuator assembly in a post-use position.

[0059] FIG. 41 is a perspective view of a drive assembly for a drug delivery system according to a further aspect of the present invention.

[0060] FIG. 42 is a perspective view of the drive assembly of FIG. 41 according to one aspect of the present invention, showing a top portion of a housing removed.

[0061] FIG. 43 is a cross-sectional view of the drive assembly of FIG. 41 according to one aspect of the present invention.

[0062] FIG. 44 is a perspective view of the drive assembly of FIG. 41 according to one aspect of the present invention.

[0063] FIG. 45 is a cross-sectional view of the drive assembly of FIG. 41 according to one aspect of the present invention, showing the drive assembly in a pre-use position.

[0064] FIG. 46 is a cross-sectional view of the drive assembly of FIG. 41 according to one aspect of the present invention, showing the drive assembly in a pre-use position.

[0065] FIG. 47 is a top view of the drive assembly of FIG. 41 according to one aspect of the present invention, showing the drive assembly in a pre-use position.

[0066] FIG. 48 is a top view of the drive assembly of FIG. 41 according to one aspect of the present invention, showing the drive assembly in an initial actuation position.

[0067] FIG. 49 is a top view of the drive assembly of FIG. 41 according to one aspect of the present invention, showing the drive assembly in an initial actuation position.

[0068] FIG. 50 is a top view of the drive assembly of FIG. 41 according to one aspect of the present invention, showing the drive assembly in an initial actuation position.

[0069] FIG. 51 is a top view of the drive assembly of FIG. 41 according to one aspect of the present invention, showing the drive assembly in a use position.

[0070] FIG. 52 is a top view of the drive assembly of FIG. 41 according to one aspect of the present invention, showing the drive assembly in a use position.

[0071] FIG. 53 is a cross-sectional view of the drive assembly of FIG. 41 according to one aspect of the present invention, showing the drive assembly in a use position.

[0072] FIG. 54 is a top view of the drive assembly of FIG. 41 according to one aspect of the present invention, showing the drive assembly in a use position.

[0073] FIG. 55 is a cross-sectional view of the drive assembly of FIG. 41 according to one aspect of the present invention, showing the drive assembly in a use position.

[0074] FIG. 56 is a cross-sectional view of the drive assembly of FIG. 41 according to one aspect of the present invention, showing the drive assembly in a use position.

[0075] FIG. 57 is a top view of the drive assembly of FIG. 41 according to one aspect of the present invention, showing the drive assembly in a use position.

[0076] FIG. 58 is a top view of the drive assembly of FIG. 41 according to one aspect of the present invention, showing the drive assembly in an initial post-use position.

[0077] FIG. 59 is a perspective view of the drive assembly of FIG. 41 according to one aspect of the present invention, showing the drive assembly in an initial post-use position.

[0078] FIG. 60 is a top view of the drive assembly of FIG. 41 according to one aspect of the present invention, showing the drive assembly in a post-use position.

[0079] FIG. 61 top view of the drive assembly of FIG. 41 according to one aspect of the present invention, showing the drive assembly in a post-use position.

[0080] FIG. 62 is a cross-sectional view of the drive assembly of FIG. 41 according to one aspect of the present invention, showing the drive assembly in a pre-use position.

[0081] FIG. 63 is a cross-sectional view of the drive assembly of FIG. 41 according to one aspect of the present invention, showing the drive assembly in a use position.

[0082] FIG. 64 is a perspective view of a drive assembly according to a further aspect of the present invention.

[0083] FIG. 65A is a front view of a needle actuator assembly according to one aspect of the present invention, showing the needle actuator assembly in a use position.

[0084] FIG. 65B is a front view of the needle actuator assembly of FIG. 65A according to one aspect of the present invention, showing the needle actuator assembly in a use position.

[0085] FIG. 65C is a front view of the needle actuator assembly of FIG. 65A according to one aspect of the present invention, showing the needle actuator assembly in an initial post-use position.

[0086] FIG. 65D is a front view of the needle actuator assembly of FIG. 65A according to one aspect of the present invention, showing the needle actuator assembly in a post-use position.

[0087] FIG. 65E is a front view of the needle actuator assembly of FIG. 65A according to one aspect of the present invention, showing the needle actuator assembly in a pre-use position.

[0088] FIG. 65F is a cross-sectional view of the needle actuator assembly of FIG. 65A according to one aspect of the present invention, showing the needle actuator assembly in a pre-use position.

[0089] FIG. 65G is a front view of the needle actuator assembly of FIG. 65A according to one aspect of the present invention, showing the needle actuator assembly in a pre-use position with a button actuator axially displaced.

[0090] FIG. 65H is a cross-sectional view of the needle actuator assembly of FIG. 65A according to one aspect of the present invention, showing the needle actuator assembly in a pre-use position with a button actuator axially displaced.

[0091] FIG. 66 is a perspective view of a button spring of the needle actuator assembly of FIG. 65A according to one aspect of the present invention.

[0092] FIG. 67 is a perspective view of an actuator button of the needle actuator assembly of FIG. 65A according to one aspect of the present invention.

[0093] FIG. 68 is a cross-sectional view of a button spring and actuator button of the needle actuator assembly of FIGS. 65A according to one aspect of the present invention.

[0094] FIG. 68A is a perspective view of an actuator button of the needle actuator assembly of FIG. 65A according to a further aspect of the present invention.

[0095] FIG. 68B is a bottom view of an actuator button of the needle actuator assembly of FIG. 65A according to a further aspect of the present invention.

[0096] FIG. 68C is a front view of an actuator button of the needle actuator assembly of FIG. 65A according to a further aspect of the present invention.

[0097] FIG. 68D is a top view of an actuator button of the needle actuator assembly of FIG. 65A according to a further aspect of the present invention, showing the actuator button in a pre-use position.

[0098] FIG. 68E is a front view of an actuator button of the needle actuator assembly of FIG. 65A according to a further aspect of the present invention, showing the actuator button in a pre-use position.

[0099] FIG. 68F is a top view of an actuator button of the needle actuator assembly of FIG. 65A according to a further aspect of the present invention, showing the actuator button in a use position.

[0100] FIG. 68G is a front view of an actuator button of the needle actuator assembly of FIG. 65A according to a further aspect of the present invention, showing the actuator button in a use position.

[0101] FIG. 69 is a top view of an actuator button of the needle actuator assembly of FIG. 65A according to one aspect of the present invention.

[0102] FIG. 70A is a schematic view of a drive assembly according to one aspect of the present invention, showing the drive assembly in a pre-use position.

[0103] FIG. 70B is a schematic view of the drive assembly of FIG. 70A according to one aspect of the present invention, showing the drive assembly in a use position.

[0104] FIG. 70C is a schematic view of the drive assembly of FIG. 70A according to one aspect of the present invention, showing the drive assembly in a use position.

[0105] FIG. 70D is a schematic view of the drive assembly of FIG. 70A according to one aspect of the present invention, showing the drive assembly in a use position.

[0106] FIG. 70E is a schematic view of the drive assembly of FIG. 70A according to one aspect of the present invention, showing the drive assembly in a use position.

[0107] FIG. 70F is a schematic view of the drive assembly of FIG. 70A according to one aspect of the present invention, showing the drive assembly in a post-use position.

[0108] FIG. 70G is a schematic view of the drive assembly of FIG. 70A according to one aspect of the present invention, showing the drive assembly in a post-use position.

[0109] FIG. 71 is a perspective view of a spacer assembly for a drug delivery system according to one aspect of the present invention, showing an assembled, pre-use position of the spacer assembly.

[0110] FIG. 72 is a perspective view of the spacer assembly of FIG. 71 according to one aspect of the present invention, showing a use position of the spacer assembly.

[0111] FIG. 73 is a perspective view of the spacer assembly of FIG. 71 according to one aspect of the present invention, showing an initial post-use position of the spacer assembly.

[0112] FIG. 74 is a perspective view of a restriction member according to one aspect of the present invention.

[0113] FIG. 75 is a front view of a spacer assembly for a drug delivery system according to a further aspect of the present invention.

[0114] FIG. 76 is a top view of a spacer assembly for a drug delivery system according to one aspect of the present invention.

[0115] FIG. 77 is a perspective view of the spacer assembly of FIG. 76 according to one aspect of the present invention.

[0116] FIG. 78 is a cross-sectional view of the spacer assembly of FIG. 76 according to one aspect of the present invention.

[0117] FIG. 79 is a perspective view of a spacer assembly for a drug delivery system according to a further aspect of the present invention.

[0118] FIG. 80 is a perspective view of a spacer assembly for a drug delivery system according to another aspect of the present invention.

[0119] FIG. 81A is a cross-sectional view of the spacer assembly of FIG. 80 according to one aspect of the present invention, showing a pre-assembly position of the spacer assembly.

[0120] FIG. 81B is a cross-sectional view of the spacer assembly of FIG. 80 according to one aspect of the present invention, showing an assembled position of the spacer assembly.

[0121] FIG. 82 is a perspective view of a drive assembly for a drug delivery system according to one aspect of the present invention.

[0122] FIG. 83 is a perspective view of the drive assembly of FIG. 82 according to one aspect of the present invention, showing a top portion of a housing removed.

[0123] FIG. 84 is a cross-sectional view of the drive assembly of FIG. 82 according to one aspect of the present invention, showing a pre-use position of the drive assembly.

[0124] FIG. 85 is an enlarged cross-sectional view of the drive assembly of FIG. 82 according to one aspect of the present invention, showing a pre-use position of the drive assembly.

[0125] FIG. 86 is a top view of a biasing member of the drive assembly of FIG. 82 according to one aspect of the present invention.

[0126] FIG. 87 is a perspective view of the drive assembly of FIG. 82 according to one aspect of the present invention, showing a restriction member engaged with the drive assembly.

[0127] FIG. 88 is a perspective view of a drive assembly for a drug delivery system according to one aspect of the present invention.

[0128] FIG. 89 is a perspective view of the drive assembly of FIG. 88 according to one aspect of the present invention, showing a pre-use position of the drive assembly.

[0129] FIG. 90 is a cross-sectional view of the drive assembly of FIG. 88 according to one aspect of the present invention.

[0130] FIG. 91 is a perspective view of the drive assembly of FIG. 88 according to one aspect of the present invention, showing a post-use position of the drive assembly.

[0131] FIG. 92 is a cross-sectional view of the drive assembly of FIG. 88 according to one aspect of the present invention, showing a pre-use position of the drive assembly.

[0132] FIG. 93 is a front view of the drive assembly of FIG. 88 according to one aspect of the present invention, showing a use position of the drive assembly.

[0133] FIG. 94 is a perspective view of a spacer assembly for a drug delivery system according to one aspect of the present invention.

[0134] FIG. 95 is a front view of the spacer assembly of FIG. 94 according to one aspect of the present invention.

[0135] FIG. 96 is a cross-sectional view of the spacer assembly of FIG. 94 according to one aspect of the present invention.

[0136] FIG. 97 is a perspective view of the spacer assembly of FIG. 94 according to one aspect of the present invention, showing a shim removed.

[0137] FIG. 98 is a perspective view of a fixed spacer of the spacer assembly of FIG. 94 according to one aspect of the present invention.

[0138] FIG. 99 is a perspective view of an adjustable spacer of the spacer assembly of FIG. 94 according to one aspect of the present invention.

[0139] FIG. 100 is a perspective view of a shim of the spacer assembly of FIG. 94 according to one aspect of the present invention.

[0140] Corresponding reference characters indicate corresponding parts throughout the several views. The exemplifications set out herein illustrate exemplary aspects of the disclosure, and such exemplifications are not to be construed as limiting the scope of the disclosure in any manner.

DETAILED DESCRIPTION

[0141] The following description is provided to enable those skilled in the art to make and use the described embodiments contemplated for carrying out the invention. Various modifications, equivalents, variations, and alternatives, however, will remain readily apparent to those skilled

in the art. Any and all such modifications, variations, equivalents, and alternatives are intended to fall within the spirit and scope of the present invention.

[0142] For purposes of the description hereinafter, the terms “upper”, “lower”, “right”, “left”, “vertical”, “horizontal”, “top”, “bottom”, “lateral”, “longitudinal”, and derivatives thereof shall relate to the invention as it is oriented in the drawing figures. However, it is to be understood that the invention may assume various alternative variations, except where expressly specified to the contrary. It is also to be understood that the specific devices illustrated in the attached drawings, and described in the following specification, are simply exemplary embodiments of the invention. Hence, specific dimensions and other physical characteristics related to the embodiments disclosed herein are not to be considered as limiting.

[0143] Referring to FIGS. 1-16, a drug delivery system 10 according to one aspect of the present invention includes a drive assembly 12, a container 14, a valve assembly 16, and a needle actuator assembly 18. The drive assembly 12, the container 14, the valve assembly 16, and the needle actuator assembly 18 are at least partially positioned within a housing 20. The housing 20 includes a top portion 22 and a bottom portion 24, although other suitable arrangements for the housing 20 may be utilized. In one aspect, the drug delivery system 10 is an injector device configured to be worn or secured to a user and to deliver a predetermined dose of a medicament provided within the container 14 via injection into the user. The system 10 may be utilized to deliver a “bolus injection” where a medicament is delivered within a set time period. The medicament may be delivered over a time period of up to 45 minutes, although other suitable injection amounts and durations may be utilized. A bolus administration or delivery can be carried out with rate controlling or have no specific rate controlling. The system 10 may deliver the medicament at a fixed pressure to the user with the rate being variable. The general operation of the system 10 is described below in reference to FIGS. 1-16 with the specifics of the drive assembly 12, needle actuator assembly 18, and other features of the system 10, discussed below in connection with FIGS. 17-93.

[0144] Referring again to FIGS. 1-16, the system 10 is configured to operate through the engagement of an actuation button 26 by a user, which results in a needle 28 of the needle assembly 18 piercing the skin of a user, the actuation of the drive assembly 12 to place the needle 28 in fluid communication with the container 14 and to expel fluid or medicament from the container 14, and the withdrawal of the needle 28 after injection of the medicament is complete. The general operation of a drug delivery system is shown and described in International Publication Nos. 2013/155153 and 2014/179774, which are hereby incorporated by reference in their entirety. The housing 20 of the system 10 includes an indicator window 30 for viewing an indicator arrangement 32 configured to provide an indication to a user on the status of the system 10 and a container window 31 for viewing the container 14. The indicator window 30 may be a magnifying lens for providing a clear view of the indicator arrangement 32. The indicator arrangement 32 moves along with the needle actuator assembly 18 during use of the system 10 to indicate a pre-use status, use status, and post-use status of the system 10. The indicator arrangement 32 provides visual indicia regarding the status, although

other suitable indicia, such an auditory or tactile, may be provided as an alternative or additional indicia.

[0145] Referring to FIGS. 4-6, during a pre-use position of the system 10, the container 14 is spaced from the drive assembly 12 and the valve assembly 16 and the needle 28 is in a retracted position. During the initial actuation of the system 10, as shown in FIGS. 7-9, the drive assembly 12 engages the container 14 to move the container 14 toward the valve assembly 16, which is configured to pierce a closure 36 of the container 14 and place the medicament within the container 14 in fluid communication with the needle 28 via a tube (not shown) or other suitable arrangement. The drive assembly 12 is configured to engage a stopper 34 of the container 14, which will initially move the entire container 14 into engagement with the valve assembly 16 due to the incompressibility of the fluid or medicament within the container 14. The initial actuation of the system 10 is caused by engagement of the actuation button 26 by a user, which releases the needle actuator assembly 18 and the drive assembly 12 as discussed below in more detail. During the initial actuation, the needle 28 is still in the retracted position and about to move to the extended position to inject the user of the system 10.

[0146] During the use position of the system 10, as shown in FIGS. 10-12, the needle 28 is in the extended position at least partially outside of the housing 20 with the drive assembly 12 moving the stopper 34 within the container 14 to deliver the medicament from the container 14, through the needle 28, and to the user. In the use position, the valve assembly 16 has already pierced a closure 36 of the container 14 to place the container 14 in fluid communication with the needle 28, which also allows the drive assembly 12 to move the stopper 34 relative to the container 14 since fluid is able to be dispensed from the container 14. At the post-use position of the system 10, shown in FIGS. 13-15, the needle 28 is in the retracted position and engaged with a pad 38 to seal the needle 28 and prevent any residual flow of fluid or medicament from the container 14. The container 14 and valve assembly 16 may be the container 14 and valve assembly 16 shown and described in International Publication No. WO 2015/081337, which is hereby incorporated by reference in its entirety.

[0147] Referring to FIGS. 15A-15C, the pad 38 is biased into the pad as the needle actuator body 96 moves from the use position to the post-use position. In particular, the pad 38 is received by a pad arm 122 having a cam surface 124 that cooperates with a cam track 126 on the bottom portion 24 of the housing 20. The pad arm 122 is connected to the needle actuator body 96 via a torsion bar 128. The cam surface 124 is configured to engage the cam track 126 to deflect the pad arm 122 downwards thereby allowing the pad 38 to pass beneath the needle 28 before being biased upwards into the needle 28. The torsion bar 128 allows the pad arm 122 to twist about a pivot of the needle actuator body 96. The pad 38 may be press-fit into an opening of the pad arm 122, although other suitable arrangements for securing the pad 38 may be utilized.

[0148] Referring to FIGS. 1-33, the drive assembly 12 according to one aspect of the present invention is shown. As discussed above, the drive assembly 12 is configured to move the container 14 to pierce the closure 36 of the container 14 and also to move the stopper 34 within the container 14 to dispense fluid or medicament from the container 14. The drive assembly 12 shown in FIGS. 17-33

is configured to engage and cooperate with a spacer assembly 40 received by the stopper 34 of the container 14. The spacer assembly 40 includes a spacer 42 and a spacer holder 44. The spacer holder 44 is received by the stopper 34 and the spacer 42 is received by the spacer holder 44. The spacer holder 44 includes a first threaded portion 46 that engages a corresponding threaded portion of the stopper 34, although other suitable arrangements may be utilized. The spacer 42 also includes a threaded portion 48 that engages a corresponding second threaded portion 50 of the spacer holder 44 for securing the spacer 42 to the spacer holder 44, although other suitable arrangements may be utilized. The drive assembly 12 is configured to dispense a range of predetermined fill volumes of the container 14 while maintaining the functional features of the system 10 described above, including, but not limited to, retraction of the needle 28 after the end of the dose and providing an indication of the status of the system 10 while also minimizing abrupt engagement of the stopper 34 by the drive assembly 12. The drive assembly 12 is configured to dispense a plurality of discrete fill volume ranges by utilizing a plurality of sizes of the spacers 42. In one aspect, twelve fill volume ranges and twelve spacer 42 sizes are provided. In one aspect, the length of the spacer 42 is changed to accommodate different fill volumes in the container 14. Alternatively, a single size spacer 42 may be utilized with a plurality of fill volumes in the container 14 accommodated by utilizing a plurality of shims that are received by the spacer 42.

[0149] Referring to FIGS. 17-26, the drive assembly 12 includes a first plunger member 52, a second plunger member 54 received by the first plunger member 52, a first biasing member 56, a second biasing member 58, a plunger actuation member 60, and an index member 62. The first plunger member 52 is moveable from a pre-use position (shown in FIG. 18), to a use position (shown in FIG. 19), to a post-use position (shown in FIG. 20) with the first plunger member 52 configured to engage the spacer assembly 40 and move the stopper 34 within the container 14 to dispense medicament from the container 14. The first plunger member 52 is configured to move axially. The second plunger member 54 and the first plunger member 52 form a telescoping arrangement with the second plunger 54 configured to move axially after the first plunger member 52 moves a predetermined axial distance. The movement of the first and second plunger members 52, 54 is provided by the first and second biasing members 56, 58, which are compression springs, although other suitable arrangements for the biasing members 56, 58 may be utilized.

[0150] The first biasing member 56 is received by the second plunger member 54 and is constrained between the plunger actuation member 60 (and index member 62) and a first spring seat 64 of the second plunger member 54. The second biasing member 58 is positioned radially inward from the first biasing member 56 and received by the second plunger member 54. The second biasing member 58 is constrained between a second spring seat 66 of the second plunger member 54 and the first plunger member 52. The second biasing member 58 is configured to bias the first plunger 52 member towards the container 14 from the pre-use position, to the use position, and to the post-use position. The first biasing member 56 is configured to bias the second plunger member 54 towards the container 14, which, in turn, biases the first plunger member 52 towards the container 14 from the pre-use position, to the use

position, and to the post-use position. More specifically, the second biasing member 58 is configured to drive the first plunger member 52 against the spacer assembly 40 or stopper 34 to move the container 14 into engagement the valve assembly 16 thereby piercing the closure 36 of the container 14 and placing the container 14 in fluid communication with the needle 28. The first biasing member 56 is configured to move the stopper 34 within the container 14 to dispense the medicament within the container 14. The second biasing member 58 has a different spring constant than the first biasing member 56. In particular, the second biasing member 58 is stiffer than the first biasing member 56 to provide a high force for piercing the closure 36 of the container 14 while the first biasing member 56 provides a lower force for dispensing as appropriate for the viscosity of the fluid or medicament within the container 14.

[0151] Referring again to FIGS. 17-26, the plunger actuation member 60 has an annular portion 68 and a spindle portion 70. The plunger actuation member 60 is rotationally moveable relative to the first plunger member 52 between a first rotational position and a second rotational position spaced from the first rotational position. The first rotational position may be 15 degrees from the second rotational position, although other suitable positions may be utilized. The annular portion 68 includes a drive surface 72 including a plurality of gears 74, although other suitable arrangements may be utilized for the drive surface 72. The spindle portion 70 includes an actuator locking surface 76 configured for engagement and release from a plunger locking surface 78 of the first plunger member 52. The plunger locking surface 78 includes a plurality of projections 80 configured to be received by a plurality of slots or cutouts 81 defined by the actuator locking surface 76.

[0152] As shown in FIGS. 18 and 23, in the first rotational position of the plunger actuation member 60, the plurality of projections 80 and the plurality of slots or cutouts 81 are out of alignment such that the plunger actuation member 80 is engaged with the first plunger member 52 to prevent movement of the first and second plunger members 52, 54 with the first and second biasing members 56, 58 biasing the first and second plunger members 52, 54 away from the plunger actuation member 60. As shown in FIGS. 19 and 24, in the second rotational position of the plunger actuation member 60, the plurality of projections 80 and the plurality of slots or cutouts 81 are aligned with each other such that the plunger actuation member 60 is disengaged with the first plunger member 52 to allow movement of the first and second plunger members 52, 54 thereby starting the dispensing process from the container 14.

[0153] Referring to FIGS. 7 and 33, the drive surface 72 of the plunger actuation member 60 is configured to be engaged by a portion of the needle actuator assembly 18. After engagement of the actuator button 26 and release of the needle actuator assembly 18, which is discussed in more detail below, the needle actuator assembly 18 moves within the housing 20 from the pre-use position, to the use position, and to the post-use position. During the initial movement of the needle actuator assembly 18, a portion of the needle actuator assembly 18 engages the drive surface 72 of the plunger actuation member 60 to move the plunger actuation member 60 from the first rotational position to the second rotational position. As shown in FIG. 33, an angled blade portion 82 of the needle actuator assembly 18 engages the

drive surface 72 of the plunger actuation member 60 to cause rotation of the plunger actuation member 60.

[0154] Referring to FIGS. 11, 13, and 26, the second plunger member 52 includes a plurality of coded projections 84 with a preselected one of the plurality of coded projections 84 configured to engage a restriction member 86 of the system 10. As discussed in more detail below, the restriction member 86 cooperates with the needle actuator assembly 18 and restricts movement of the needle actuator assembly 18 from the use position to the post-use position until a predetermined end-of-dose position of the stopper 34 is reached. In one aspect, the restriction member 86 is configured to restrict axial movement of the needle actuator assembly 18 from the use position through engagement between the restriction member 86 and a portion of the needle actuator assembly 18. Such engagement between the restriction member 86 and the needle actuator assembly 18 is released by rotation of the restriction member 86 when the stopper 34 reaches the end-of-dose position. During the use position of the needle actuator assembly 18, the restriction member 86 is biased in a rotational direction with the rotation of the restriction member 86 being prevented through engagement between the restriction member 86 and one of the plurality of coded projections 84 of the second plunger member 54. The plurality of coded projections 84 may be axial ribs of varying length, although other suitable arrangements may be utilized. Each coded projection 84 defines a point at which the restriction member 86 is able to rotate thereby releasing the needle actuator assembly 18. The smooth portion of the second plunger member 52 may also provide a further “code” for determining when the system 10 transitions to the end-of-dose position.

[0155] As discussed above, the indicator arrangement 32 moves with different portions of the indicator arrangement 32 visible through the indicator window 30 as the system 10 moves from the pre-use, use, and post-use or end-of-dose positions. More specifically, the indicator arrangement 32 engages a portion of the restriction member 86 and moves along with the restriction member 86 through the various stages of the system 10 to provide an indication to the user regarding the state of the system 10.

[0156] During assembly of the system 10, the dosage of the container 14 is matched with a specific spacer 42 having a set length and a corresponding one of the plurality of coded projections 84 is aligned with the restriction member 86. Accordingly, as discussed above, the container 14 may be provided with a plurality of dosage volumes with each volume corresponding to a specific spacer 42 and coded projection 84. Thus, even for different dosage volumes, the system 10 is configured to inject the needle 28 into the user to deliver a dose of medicament from the container 14, retract the needle 28 after the end of the dose, and provide an indication of the status of the system 10 while minimizing abrupt engagement of the stopper 34 by the drive assembly 12. In particular, the size of the stopper 34 may be selected to minimize the distance between the first plunger member 52 and the spacer assembly 40 and does not require the use of damping.

[0157] Referring to FIGS. 27-33, a drive assembly 12A according to a further aspect of the present invention is shown. The drive assembly 12A shown in FIGS. 27-33 is similar to and operates in the same manner as the drive assembly 12 shown in FIGS. 17-26 and described above. In the drive assembly of FIGS. 27-33, however, the first

plunger member 52 is received by the second plunger member 54 and extends from the second plunger member 54 during axial movement from the pre-use position to the use position. Further, the first plunger member 52 includes an extension portion 88 configured to engage the second plunger member 54 after the first plunger member 52 moves predetermined axial distance such that the first and second plunger members 52, 54 move together. The first and second biasing members 56, 58 engage and act on the first and second plunger members 52, 54 in the same manner as the drive assembly 12 of FIGS. 17-26.

[0158] Referring to FIGS. 27-32, the index member 62 is positioned about the first and second plunger members 52, 54 and includes a plurality of ratchet teeth 90 configured to engage a flexible tab 92 positioned on the bottom portion 24 of the housing 20. When the drive assembly 12, 12A is installed into the bottom portion 24 of the housing 20, the engagement of the ratchet teeth 90 of the index member 62 with the flexible tab 92 of the housing 20 provide a one-way rotation of the index member 62. The index member 62 is configured to rotate to align one of the coded projections 84 of the second plunger member 52 with the restriction member 86 based on the dosage volume and spacer 42 size as discussed above. The index member 62 may provide the drive assembly 12, 12A with 24 rotational positions of which 12 may have unique dose values associated with them.

[0159] Referring to FIGS. 1-16 and 34-40B, the needle actuator assembly 18 according to one aspect of the present invention is shown. The needle actuator assembly 18 includes a needle actuator body 96 having guide surfaces 98, a needle shuttle 102 having cam surfaces 104, and the needle 28 received by the needle shuttle 102 and configured to be in fluid communication with the container 14 as discussed above. The needle actuator body 96 is generally rectangular with the guide surfaces 98 protruding radially inward. The needle shuttle 102 is received within the needle actuator body 96. As described above, the needle actuator body 96 is moveable within the housing 20 from a pre-use position (shown in FIGS. 4-6), an initial actuation position (FIGS. 7-9), a use position (FIGS. 10-12), and a post-use position (FIGS. 13-15). The needle actuator body 96 is biased from the pre-use position to the post-use position via an extension spring 106, although other suitable biasing arrangements may be utilized. The needle actuator body 96 is released and free to move from the pre-use position to the use position upon engagement of the actuator button 26, which is discussed in more detail below. The needle actuator body 96 moves from the use position to the post-use position after rotation of the restriction member 86 as discussed above in connection with FIGS. 17-33.

[0160] Referring to FIGS. 34-40B, the needle shuttle 102 is moveable along a vertical axis between a retracted position where the needle 28 is positioned within the housing 20 and an extended position where at least a portion of the needle 28 extends out of the housing 20. The needle shuttle 102 is configured to move between the retracted position and the extended position through engagement between the guide surfaces 98 of the needle actuator 96 and the cam surfaces 104 of the needle shuttle 102. The cam surfaces 104 are provided by first and second cam members 108, 110, with the first cam member 108 spaced from the second cam member 110. The housing 20 includes a guide post 112 having recess configured to receive a T-shaped projection 114 on the needle shuttle 102, although other shapes and

configurations may be utilized for the guide post 112 and T-shaped projection 114. The needle shuttle 102 moves along the guide post 112 between the retracted and extended positions. The guide post 112 is linear and extends about perpendicular from the housing 20, although other suitable arrangements may be utilized. The guide surfaces 98 of the needle actuator body 86 are non-linear and each include a first side 116 and a second side 118 positioned opposite from the first side 116.

[0161] As discussed below, the guide surfaces 98 of the needle actuator body 96 cooperate with the cam members 108, 110 of the needle shuttle 102 to move the needle shuttle 102 vertically between the retracted and extended positions as the needle actuator body 96 moves axially from the pre-use position to the post-use position. The needle shuttle 102 also includes a shuttle biasing member 120 configured to engage the housing 20 or the actuator button 26. In particular, the shuttle biasing member 120 engages the housing 20 or actuator button 26 and provides a biasing force when the needle actuator body 96 is transitioning from the use position to the post-use position. When the needle actuator body 96 is fully transitioned to the post-use position, the cam members 108, 110 of the needle shuttle 102 are disengaged from the guide surfaces 98 of the needle actuator body 96 and the shuttle biasing member 120 biases the needle shuttle 102 downward such that the needle 28 engages the pad 38, as discussed above. As discussed above in connection with FIGS. 1-16, however, the pad 38 may also be biased into the needle 28 rather than biasing the needle shuttle 102 downwards via the shuttle biasing member 120. The needle actuator body 96 may interact with the actuator button 26 to prevent the actuator button 26 from popping back up until the post-use position is reached, which is discussed below in more detail.

[0162] Referring to FIGS. 37A-40B, in a pre-use position (FIG. 37A), the needle shuttle 102 is in the retracted position with the cam members 108, 110 spaced from the guide surface 98 of the needle actuator body 96. As the needle actuator body 96 moves to the use position (FIGS. 37B and 38A), the second cam member 110 of the needle shuttle 102 engages the second side 118 of the guide surfaces 98 to move the needle shuttle 102 from the retracted position to the extended position. During the transition from the use position to the post-use position of the needle actuator body 96 (FIG. 37C), the first cam member 108 of the needle shuttle 102 is engaged with the first side 116 of the guide surfaces 98 to move the needle shuttle 102 from the second position to the first position. After the needle actuator body 96 is fully transitioned to the post-use position (FIGS. 37D and 38B), the shuttle biasing member 120 biases the needle shuttle 102 downward as the cam members 108, 110 disengage from the guide surfaces 98 of the needle actuator body 96 with the needle 28 engaging the pad 38. The transition of the needle actuator body 96 and the corresponding position of the needle shuttle 102 is also shown in FIGS. 39-40B. The interaction between the actuator button 26 and the needle actuator body 96 is discussed in detail in connection with FIGS. 65A-67. Referring to FIGS. 41-64, a drug delivery system 200 according to a further embodiment is shown. The system 200 includes a housing 202 having an upper housing 204 and a lower housing 206. The housing has a proximal end 205 and a distal end 207. The upper housing 204 has a status view port 208 so that a user can view the operating status of the system 200. The system 200 also

includes a valve assembly 212, a tube 214 fluidly connecting the valve assembly 214 with a patient needle 215 that is disposed in a proximal end of a needle arm 216. A spring 218 biases a needle actuator 220 distally.

[0163] As shown in FIGS. 42-46, the system 200 additionally includes a container or medicament container 222 with a stopper 224 movably disposed therein, although the stopper 224 is omitted from various figures to aid clarity. Preferably, the distal end of the medicament container 222 has a septum assembly 228 that is spaced apart from the valve assembly 212 prior to actuation of the device 222, as best shown in FIG. 47.

[0164] For manufacturing purposes, using one size for a medicament container is often desirable, even if multiple fill volumes or dosages are contemplated for use with the container. In such cases, when medicament containers are filled, the differing fill volumes result in different positions of the stopper. To accommodate such different stopper positions, as well as accommodate manufacturing differences of the stoppers, aspects of the present invention include a bespoke or custom spacer 226 disposed in a proximal end of the container 222, proximal to the stopper 224. In other words, the bespoke spacer 226 provides an option that allows dispensing of a range of manufacturer-set pre-defined fill volumes by selection of different spacers 226, and reduces or eliminates the need for assembly configuration operations. The size of the spacer 226 can be employed to account for under-filled volumes of the container 222, and provide a consistent bearing surface at the proximal end of the container.

[0165] The spacer 226 is selected from a plurality of different size spacers 226 to occupy space from a proximal end of the stopper 224 to a proximal end of the container 222. According to one embodiment, as shown in FIGS. 45-47, the spacer 226 is selected to be substantially flush with the proximal end of the container 222. Additionally, according to one embodiment, the spacer 226 has a “top hat” shape, which includes a central column 230 and a distal flange 232, as best shown in FIG. 45.

[0166] Returning to FIGS. 44-47, the system 200 also includes a drive assembly 234 for displacing the container 222 distally to establish the fluid connection between the container 222 and the patient needle 215, as well as dispensing the medicament from the container 222. In more detail, the drive assembly 234 includes an inner spring 236 disposed within a central plunger 238, an outer plunger 240, an outer spring 242 disposed between the central plunger 238 and the outer plunger 240, a telescoping member 244, and a release gate 246.

[0167] Preferably, the inner spring 236 has a greater spring constant than the outer spring 242, and is therefore, stronger or stiffer than the outer spring 242. The inner spring 236 is disposed inside the central plunger 238, and pushes between a spring flange 248 in the lower housing (best shown in FIG. 46) and the central plunger 238, which bears directly on the proximal end of the spacer 226 subsequent to device activation. The outer spring 242 is disposed inside outer plunger 240, and pushes between a proximal external flange 250 of the central plunger 238 and a distal internal flange 252 of the outer plunger 240. Thus, the inner and outer springs 236 and 242 are nested, and can provide a more compact drive assembly (and thus, a more compact system 200) than employing a single spring.

[0168] According to one aspect, the inner spring 236 acts only to displace the container 222 to establish the fluid connection with the patient needle 215, and the outer spring 242 acts only to subsequently dispense the medicament from the container 222. According to another aspect, the inner spring 236 acts to displace the container 222 to establish the fluid connection with the patient needle 215, and also acts to begin dispensing the medicament from the container 222, and the outer spring 242 acts to complete dispensing the medicament. In a further aspect, the inner spring 236 causes the initial piercing of the container 222 with the outer spring 242 completing the piercing and dispensing of the medicament from the container 222.

[0169] As shown in FIGS. 44-47, and as subsequently described in greater detail, the outer plunger 240 includes a pair of proximal flanges or feet 254 that each have a slanted surface that interacts with a corresponding slanted surface (or surfaces) on the release gate to retain and subsequently release the power module subsequent to actuation of the device 200.

[0170] As best shown in FIGS. 46 and 47, as initially assembled, the container 222 is disposed in clearance from the drive assembly 234 and the valve assembly 212. A lateral flange 256 on the needle actuator 220 axially retains the medicament container 222, and the needle actuator 220 prevents the release gate 246 from displacing laterally. According to one embodiment, a spring (not shown) biases the needle actuator 220 distally, but the actuation button 210 (and/or its associated assembly) prevents distal displacement of the needle actuator 220 prior to actuation of the device 200. A status bar 258 is disposed on the needle actuator 220, and has a top surface that is visible through the status view port 208. According to one embodiment, the top surface of the status bar has a plurality of colors or patterns, and when the device is in a pre-actuated state, a first color or pattern, such as yellow, is visible through the status view port 208.

[0171] FIGS. 48-52 are top views of the system 200 illustrating the operation of events at and subsequent to actuation of the system 200. In FIG. 47, a user slides the actuation button 210 proximally and then displaces the button 210 vertically into the housing 202, thereby freeing the needle actuator 220 to displace distally under the influence of the spring (omitted for clarity). As shown in FIG. 49, as the needle actuator displaces distally, tracks 260 on the needle actuator 220 interact with lateral bosses 262 on the needle arm 216 to insert the patient needle 215. Preferably at this stage, the proximal end of the needle actuator 220 has not yet cleared the release gate 246, and thus, the drive assembly 234 has not yet been released. But the lateral flange 256 has displaced distally and therefore, the container 222 is unrestrained.

[0172] Subsequently, as shown in FIGS. 50 and 51, with continued distal displacement, the proximal end of the needle actuator 220 clears the release gate 246 (thereby releasing the drive assembly 234). The needle actuator 220 comes to temporarily rest against a feature on a rotatable release flipper 264, driving the release flipper 264 against an outrigger 266 (best shown in FIGS. 44 and 59) of the telescoping member 244. The needle actuator 220 remains in this position until the medicament has been dispensed. In this position, preferably, a second color or pattern of the status bar 258, such as green, is visible through the status view port 208.

[0173] At this stage, the force of the springs 236 and 242 and the interaction of the angled surfaces of the proximal flanges or feet 254 with the corresponding angled surface (or surfaces) on the release gate 246 causes the release gate 246 to displace laterally, thereby freeing the outer plunger 240 from restraining interaction with the release gate 246. Up to this point, the outer plunger 240 has been restraining the central plunger 238.

[0174] Referring to FIGS. 52 and 53 (the inner spring 236 is omitted from FIG. 52 for clarity), the stiff inner spring 236 distally drives central plunger 238 to contact the spacer 226. Because the medicament container 222 is filled with a substantially incompressible fluid, the continued distal displacement of the central plunger 238 distally displaces the spacer 226, the stopper 224, and the container 222 relative to the housing 202. This distal displacement causes the septum assembly 228 to be pierced by the valve assembly 212, establishing fluid communication between the container 222 and the patient needle 215. The central plunger 238 travels distally until its proximal external flange 250 (best shown in FIG. 59) contacts a flange on the lower housing 206, thereby limiting the “piercing travel.” Preferably, another flange on the lower housing 206 and/or the lateral flange 256 of the needle actuator 220 limits distal travel of the container 222.

[0175] Subsequently, because the inner spring 236 can no longer distally displace the central plunger 238, the lighter outer spring 242 distally displaces the outer plunger 240 relative to the central plunger 238 to contact the distal flange 232 of the spacer 226, as shown in FIGS. 54 and 55. As subsequently described in greater detail, preferably, the contact between the outer plunger 240 and the spacer 226 is damped to minimize the impact force. Further expansion of the outer spring 242 distally displaces the outer plunger 240 to dispense the medicament.

[0176] As shown in FIGS. 56 and 57, as the outer spring 242 continues to expand and distally displace the outer plunger 240, upon a predetermined distal displacement of the outer plunger 240 relative to the telescoping member 244, an external feature or flange 268 of the outer plunger 240 interacts with an internal distal feature or flange 270 of the telescoping member 244 to “pick up” the telescoping member 244. This ensures that further distal displacement of the outer plunger 240 causes corresponding distal displacement of the telescoping member 244. This paired distal displacement continues until the end of the medicament dispensing.

[0177] As previously noted, the outrigger 266 is disposed on the telescoping member 244. The axial length of the outrigger and the distal travel of the telescoping member 144 controls the timing of the disengagement of the outrigger 266 with the release flipper 264. As shown in FIGS. 58 and 59, at the end of medicament dispensing, the proximal end of the outrigger 266 bypasses the release flipper 264. This allows the release flipper 264 to rotate out of engagement with the needle actuator 220 (FIG. 60), and allows the needle actuator 220 to continue its distal displacement and withdraw the patient needle 215 (FIG. 61). At this stage, another color or pattern of the status bar 258, such as red, is visible through the status view port 208, signifying that the device 200 has completed operation.

[0178] As previously noted, the contact between the outer plunger 240 and the spacer 226, as illustrated in FIGS. 62 and 63, is preferably damped to minimize the impact force.

The highest level of energy dissipation is desirable for under-filled syringes containing viscous fluid, as the outer spring 242 will be stiffer to provide desired dispense rates. The lowest level of energy dissipation is desirable for maximum-filled syringes containing low-viscosity fluid, as the outer spring can be less stiff to provide desired dispense rates. Various methods can be employed to adjust damping levels, such as air damping, or closed-cell foam damping.

[0179] As another method of damping the impact force, FIG. 64 illustrates an embodiment of a spacer 226 in which one or more axial interface ribs 272 are circumferentially arrayed about the central column 230 of the spacer 226. In this embodiment, the outer plunger 240 must drive past the interference ribs 272, which provide frictional resistance to the distal displacement of the outer plunger 240 relative to the spacer 226. The frictional force created by the interference between interference ribs 272 and the outer plunger 240 is independent of plunger speed. Preferably, the frictional force does not exceed the minimum dispense spring load, to avoid stalling weaker springs. The interference can be tuned to give the desired level of frictional resistance. For different fluid viscosities, there can be different sizing (axial and/or radial) of the interference ribs 272. This could mean a bespoke or custom spacer for each viscosity and fill-level combination, or, depending on the number of springs required for a viscosity range, there can be a number of tuned positions, whereby the spacer can be set to a particular position for a particular modular spring (the position have had the interference/damping tuned for that particular spring load/viscosity scenario).

[0180] Referring to FIGS. 65A-69, an actuator button arrangement 280 for actuating the system 10 according to one aspect of the present invention is shown. The actuator button arrangement 280 includes the actuator button 26, a button spring 284, and a needle actuator body 286. The needle actuator body 286 may be similar to the needle actuator bodies 96, 220 discussed above and configured to move within the housing 20 to transition the needle shuttle 102 or needle 28 between retracted and extended positions. As shown in FIG. 69, the actuator button 26 includes a user interface portion 288 for interacting with a user. Preferably, the user interface portion 288 is about 22 mm long and about 10 mm wide, although other suitable dimensions may be utilized. The actuator button 26 includes two pairs of lockout arms 290, 292 that interact with button contacting surfaces 294, 296 on the needle actuator body 286 prior to device actuation to prevent the needle actuator body 286 from rocking upward. As shown in FIG. 65H, an overlap between the needle actuator body 286 and the housing 20 prevents premature actuation. Referring to FIG. 66, the button spring 284 includes a first bearing surface 298 and a second bearing surface 300 spaced from the first bearing surface 298, and a cantilevered central spring arm 302 surrounded by a pair of outer arms 304 that are joined by the first bearing surface 298.

[0181] The actuation button arrangement 280 is configured to provide one or more of the following features, which are discussed in more detail below: one-way axial displacement or sliding of the actuator button 26; transverse movement (raised and depressed positions) of the actuator button 26 where the actuator button 26 remains depressed during the use position of the needle actuator body 286; and lockout of the actuator button 26 in the post-use position of the

needle actuator body 286 such that the button 26 is in the raised position and cannot be depressed by a user.

[0182] To actuate the system 10 using the actuator button 26, the user first slides the user interface portion 288 in a first axial direction, shown as being to the right in FIGS. 65G and 65H. The user may be required to slide the user interface portion 288 about 10 mm or about 8 mm, although other suitable distances may be utilized. Moving the actuator button 26 axially moves the lockout arms 290, 292 to clear the button contact surfaces 294, 296 on the needle actuator body 286 to allow movement of the actuator button 26 from the raised position to the depressed position.

[0183] As the user distally slides the user interface portion 288, the central spring arm 302 of the button spring 284 rides over a spring arm 306 bearing surface on the housing 20 while the first and second bearing surfaces 298, 300 engage first and second bearing ramps 308, 310 on the housing 20. The forces on the button spring 284 are balanced through the engagement with the spring arm bearing surface 306 and the first and second bearing ramps 308, 310 to provide a smooth axial displacement or sliding of the actuator button 26.

[0184] As the actuator button 26 and the button spring 284 reach the end of their axial sliding travel, the central spring arm 302 and the first bearing surface 298 pass the end of a respective stops 312, 314 to prevent the actuator button 26 from sliding backward to its original position, as shown in FIG. 65H. Further, when the actuator button 26 and the button spring 284 reach the end of their axial sliding travel, the user engages the user interface portion 288 to move the actuator button 26 downward to its depressed position. The actuator button 26 may be depressed about 2 mm and the minimum force required to depress the actuator button 26 is about 3 N, and most preferably, about 2.8 N, although other suitable distances and minimum forces may be utilized.

[0185] As the user depresses the user interface portion 288, shown in FIGS. 65A and 65B, the actuator button 26 rotates the needle actuator body 286 to release the needle actuator body 286 thereby allowing the needle actuator body 286 to move from the pre-use position to the use position. As shown in FIG. 65B, as the needle actuator body 286 travels to the use position, the lockout arms 290, 292 run along the underside of the button contact surfaces 294, 296 to prevent the actuator button 26 springing upward. After the medicament has been delivered and as the needle actuator body 286 is transitioning from the use position to the post-use position, shown in FIG. 65C, the lockout arms 290, 292 are disengaged from the button contact surfaces 294, 296 allowing the actuator button 26 to spring back up under the influence of the button spring 284. Once the needle actuator body 286 fully transitions to the post-use position, shown in FIG. 65D, the actuator button 26 has finished moving from the depressed position to the raised position due to the biasing force of the button spring 284. When the needle actuator body 286 is in the post-use position, a spring arm 316 on the needle actuator body 286 engages the actuator button 26 to prevent the actuator button 26 from moving to the depressed position while axial movement is still restricted by the engagement of the spring arm 302 with the stops 312, 314. Thus, the actuator button 26 is locked after delivery of the medicament is complete to provide a clear indication between a used system and an unused system.

[0186] Furthermore, if the user holds down the actuator button 26 during dispensing of the medicament, proper dosing and needle retraction will still complete, but the

actuator button 26 will not spring back up to the raised position until the button 26 is released.

[0187] In one aspect, the button spring 284 is made of plastic. The button spring 284 may also be a pressed metal spring could be used instead, although any other suitable material may be utilized.

[0188] Referring to FIGS. 68A-68G, rather than providing a separate actuator button 26 and button spring 284, the spring may be provided integrally with the button 26. More specifically, an actuator button 320 according to a further aspect of the present invention includes an integral spring arm 322. The actuator button 320 also includes lockout arms 324, retention arms 326, and a rear pivot 328. As shown in FIGS. 68D and 68E, the spring arm 322 engages prongs 330 in the top portion 22 of the housing 20. During transition of the system 10 from the pre-use position to the use position, the spring arm 322 slides past a detent of the prongs 330 providing an axial spring force. The end of the spring arm 322 engages a portion of the top portion 22 of the housing 20 to provide the vertical spring force as the spring arm 322 deflects. The actuator button 320 is configured to a fluid motion between the sliding and depression movements of the button 320 even though two separate motions are occurring, which is similar to the operation of the button 26 discussed above. During transition between the pre-use position and the use position, the button 320 pivots about the rear pivot 328 with the retention arm 326 engaging a portion of the needle actuator body 286 thereby maintaining a depressed position of the button 320 until the end-of-dose position is reached in a similar manner as actuator button 26. The lockout arms 324 deflect inwards and engages a portion of the needle actuator body 286 as the needle actuator body 286 moves to the end-of-dose position thereby preventing further movement of the actuator button 320 in a similar manner as the actuator button 26 discussed above.

[0189] Aspects of the present invention provide improvements over previous button designs. For example, the actuation button arrangement 280 provides multiple surfaces to hold the needle actuator body 286 in place against a needle actuator spring 106 prior to actuation, thereby reducing the likelihood of premature actuation during a drop impact. The actuation button arrangement 280 physically prevents the needle actuator body 286 from moving prior to actuation by holding it in a tilted (locked) state in such a way that the surfaces have no room to separate and pre-activate.

[0190] In addition, button slide forces of the actuation button arrangement 280 are controlled more precisely by utilizing a flexing arm rather than using a simple bump detent. This permits longer sliding strokes of the button 26 with better force control, resulting in a more ergonomically effective design. Further, the actuation button arrangement 280 causes the button 26 to pop back out at the end of injection, giving the user an additional visual, audible, and tactile indication that the medicament delivery is completed.

[0191] According to one aspect, the fluid delivery volume of the system 10 is determined by the end position of a plunger relative to a point inside the housing regardless of actual fill volume, container inner diameter, and stopper starting position and length. The dosing accuracy variability can be significant because the tolerances of the factors above can be quite large. Aspects of the present invention allow for the elimination of some or all of these tolerances from the dosing equation, resulting in a more precise and less variable injection volume of medicament.

[0192] Referring to FIGS. 70A-70G, a spacer assembly 400 for use in connection with a drive assembly according to one aspect of the present invention is shown.

[0193] Elements in a chain of tolerances in the stopper spacer assembly 400 include a thickness (A) of a flange 402 of an inner plunger 404, an internal length (B) of an outer plunger 406 between an internal proximal end 408 and an internal shoulder 410, and an initial offset distance (C_1) between the inner plunger flange 402 and the internal proximal end 408 of the outer plunger. This initial offset distance (C_1) is preferably greater than a gap distance (C_2) between outer plunger 406 and the proximal end of the medicament barrel 412. The chain of tolerances in the stopper spacer assembly 400 also includes the internal barrel diameter (D). Once assembled, the stopper spacer 414 and the outer plunger 406 are unique for a given medicament volume.

[0194] FIGS. 70B-70G illustrate operation of the stopper spacer assembly 400. As shown in FIG. 70B, when the system is actuated, the both inner and outer plungers 404 and 406 are released. An outer spring 416 pushes the outer plunger 406 into the barrel 412, compressing damping material 418, and an inner spring 420. The stopper 422 does not yet move relative to the barrel 412 due to the fluid column of medicament.

[0195] Next, as shown in FIGS. 70C, the outer spring 416 distally displaces the outer plunger 406 and the barrel 412 to open a valve (not shown) at the distal end of the barrel 412 that establishes fluid communication with the needle (not shown). Due to the incompressibility of the liquid medicament, the stopper 422 cannot displace relative to the barrel 412 until the valve is opened and the fluid path to the patient needle is established.

[0196] Subsequently, as shown in FIGS. 70D and 70E, the inner spring 420 displaces the inner plunger 404, the stopper spacer 414, and the stopper 422, to dispense the fluid.

[0197] FIG. 70F illustrates the end of medicament delivery when the proximal flange 402 of the inner plunger 404 contacts the internal shoulder 410 of the outer plunger 406, thereby ceasing displacement of the inner plunger 404 (and the stopper spacer 414 and stopper 422) relative to the medicament barrel 212 and stopping the flow of medicament.

[0198] According to one aspect, as shown in FIG. 70G, the cessation of displacement of the inner plunger 404 relative to the medicament barrel 412 triggers an end-of-dose indicator for the system.

[0199] Referring to FIGS. 71 and 72, a collapsible spacer assembly 430 includes a forward spacer portion 432 secured to a stopper 434, an inner plunger 436, a rear spacer portion 438, and a rotating shuttle 440. The inner plunger 436 can translate relative to the forward spacer portion 432, but not rotate relative thereto. Similarly, the rear spacer portion 438 can also move axially relative to the forward spacer portion 432, but not rotate relative to the forward spacer portion 432. As subsequently described in greater detail, the rotating shuttle 440 first rotates, and subsequently translates.

[0200] According to one aspect, forward spacer portion 432 is fixedly secured to the stopper 434. One skilled in the art will understand that many methods can be employed to secure the forward spacer portion 432 to the stopper 434, for example, adhesive, mechanical fasteners, or any other suit-

able arrangement. Preferably, the forward spacer portion 432 includes threads that engage mating threads in the stopper 434.

[0201] When the stopper spacer assembly 430 is screwed into the stopper 434, an axial load is applied through access openings 442 in the rear spacer portion 438. This force can be used to push the stopper 434 forward, applying pressure to the fluid medicament. This pressure causes the front (distal) face of the stopper 434 to deflect and press proximally, pushing back on the rear spacer portion 438 and rotating the rotating shuttle into its "as assembled" condition. In other words, when a medicament barrel is filled with medicament and the system's plunger is applying axial force to the medicament via the spacer assembly 430, the distal face of the stopper 434 is deformed by the pressure of the medicament. During medicament delivery, pressure is applied by a drive assembly (via the plunger) to the rear spacer portion 438, which in turn applies a rotational torque to the rotating shuttle 440 via helical faces 444 of the rear spacer portion 438. But the stopper deformation from the medicament provides a rearward or proximal force on the inner plunger 436, which prevents rotation of the rotating shuttle 440.

[0202] According to one aspect, an axial reaction load on the inner plunger 436 can be increased by increasing the length of the inner plunger 436.

[0203] Once the medicament delivery is complete, as shown in FIG. 73, the pressure on the stopper 434 decreases, thereby permitting the distal end of the inner plunger 436 to displace distally. This distal displacement permits the rotating shuttle 440 to rotate. The continued axial force applied by the drive assembly rotates and distally displaces the rotating shuttle 440 due to interaction of the helical faces 444 in the rear spacer portion 438 with corresponding cam-faced arms 446 of the rotating shuttle 440. According to one aspect, this final movement of the rotating shuttle 440 causes the drive assembly to trigger needle retraction.

[0204] Referring to FIGS. 74 and 75, a restriction member 452 according to one aspect of the present invention is disposed with the drive assembly. The restriction member 452 governs the timing of the final displacement of the needle actuator bodies 96, 220 subsequent to the completion of the medicament dose. Instead of rotating about a fixed post, the restriction member 452 floats freely. Once a plunger displaces sufficiently distally for a gap to align with the restriction member 452 (as shown in FIGS. 74 and 75), the restriction member 452 displaces laterally into the gap because of the force of the spring on the needle actuator 96, 220 and the angled face 454 on the rear of the arm of the restriction member 174 that engages the needle actuator body (best shown in FIG. 75). Once the restriction member no longer retains the needle actuator body 96, 220, the needle actuator body 96, 220 is free to complete the axial movement to the post-use position. Further, as shown in FIG. 75, the restriction member 452 is biased onto the rear of the barrel portion of the container 14, which minimizes the tolerance chain of the various components and improves dose accuracy.

[0205] Referring to FIGS. 76-78, a spacer assembly 460 according to a further aspect of the present invention is shown. The spacer assembly 460 shown in FIGS. 76-78 allows for the removal of the effect of manufacturing toler-

ance build up through adjustment of the spacer assembly thereby allowing each system to inject the same amount of medicament.

[0206] As shown in FIG. 77, the spacer assembly 460 includes a stopper 462 and a stopper spacer 464. The stopper spacer 464 includes a fixed spacer piece or fixed spacer 466 that is fixedly connected with the stopper 462, and an adjustable spacer piece or adjustable spacer 468 that is rotationally displaceable in one direction relative to the fixed spacer 466.

[0207] One skilled in the art will understand that many methods can be employed to secure the fixed spacer 466 to the stopper 462, for example, adhesive, mechanical fasteners, or any other suitable arrangement. Preferably, the fixed spacer 466 includes one or more external threads that engage one or more mating threads in the stopper 462. According to one aspect, the adjustable spacer 468 has a distal stem with an external thread 470. The distal stem thread 470 engages an internal thread 472 in the fixed spacer 466 (best shown in FIG. 78) to rotationally control axial displacement of the adjustable spacer 468 relative to the fixed spacer 466.

[0208] As shown in FIGS. 76 and 77, the fixed spacer 466 includes radially spaced detents 474 and the adjustable spacer 468 includes a spring detent arm 476, the free end of which engages a selected one of the detents 474 to prevent rotation and axial displacement of the adjustable spacer 468 toward the fixed spacer 466. The free end of the spring detent arm 476 is shaped to pass over the detents 474 in one direction, thereby permitting rotation and proximal axial displacement of the adjustable spacer 468 away from the fixed spacer 466.

[0209] Despite variations in the dimensions of stoppers and containers, the adjustable spacer 468 can be adjusted relative to the fixed spacer 466 to provide a consistent axial length of the stopper assembly 460.

[0210] As shown in FIG. 78, once the container is filled, an axial load, such as a load that would be encountered when installed in the system 10, 200, can be applied to the adjustable spacer 468 (and thus, the fixed spacer 466 and the stopper 462). Once the axial load is applied, the adjustable spacer 468 can be proximally backed out to ensure a consistent gap 478 between the proximal end of a medicament barrel 480 and the proximal face of the adjustable spacer 468, thereby accounting for variations in the medicament barrel glass and the compressibility of any entrapped air. In other words, the spacer assembly 460 allows the adjustable spacer 468 to have a predetermined set position relative to the container 14 independent of the variables of the container 14 and stopper length. Accordingly, the start position of the spacer assembly 460 is a predetermined distance from the container 14 and the end position of the spacer assembly 460 is also a predetermined distance from the container 14 such that the travel of the stopper 462 is defined by the effective length of the plungers 52, 54 of the drive assembly 12.

[0211] Referring to FIGS. 79 and 80, a base column 482 and a cap 484 of an automatically adjusting spacer 486 according to one aspect of the present invention is shown. The base column 482 includes a base portion 488 and an axially extending column 490. According to one embodiment, the base column 482 includes a plurality of columnar protrusions 491 that each have a plurality of ratchet teeth 492 disposed on a proximal portion thereof. A locking barb 493 is disposed at the proximal end of each of the plurality

of ratchet teeth 492. The cap 484 is hollow, and a distal end of the cap 484 includes one or more axial springs 494. According to one aspect, the axial springs 494 are bent, cantilevered arms formed during molding of the cap 484. According to another aspect, a separate biasing member, such as a compression spring can be employed in the automatically adjusting spacer 486. When assembled with the base column 482, the springs 494 engage the base portion 488 and maintain an initial spacing between the base column 482 and the cap 484. According to one aspect, the springs 494 are omitted. The cap 484 also includes a plurality of flexible cantilevered arms or tabs 496, which each have a free proximal portion with a plurality internal of ratchet teeth 497. The proximal end of each flexible tab 496 includes a foot 498.

[0212] FIG. 81B illustrates the cap of the automatically adjusting spacer deployed within a proximal recess of a stopper 494 at a proximal portion of a medicament barrel. The base column 482 is assembled into the hollow cap 484 with the base portion 482 engaging the stopper 494 and the feet 498 disposed outside the proximal end of the barrel.

[0213] In operation, as shown in FIGS. 81A and 81B, the cap 484 displaces distally relative to the base column 482 (as well as the stopper 494 and the barrel) until the proximal end of the cap 484 is flush with the end of the medicament barrel. This action causes the feet 498 to engage the internal surface of the barrel and displace radially inward, thereby forcing the ratchet teeth 492 into locking engagement with the ratchet teeth 497. The locking barb 493, the engagement of the ratchet teeth 492 and 497, and the engagement of the feet 498 with the internal surface of the barrel prevents the displacement of the cap 484 relative to the base column 482. Thus, the automatically adjusting spacer 486 can accommodate differences in stoppers, barrel diameters, and medicament fill volumes, to automatically provide a bearing surface flush the proximal end of the medicament barrel.

[0214] One aspect of the present invention is a spacer assembly 486 that is situated against the stopper in the container within the system. The spacer design is such that its effective length can be adjusted in order to allow the dispensing of a precise quantity of medicament. The length adjustment is intended to compensate for manufacturing tolerances within the container, the fill volume, and especially the stopper length, which can add up to 1/3 of the variability in a delivered dose using a non-adjustable spacer. The spacer length can be adjusted through several techniques, depending on the specific aspect. The spacer length can be self adjusting based on its location to the back of the container, it can be adjusted by assembly equipment at the time of final assembly of the primary container into the subassembly, and it can be made an integral part of the stopper and adjusted as a subassembly prior to filling. The adjustable spacer 486 allows a more precise volume of fluid to be injected compared to a non-adjustable stopper.

[0215] Referring to FIGS. 82-87, a drive assembly 500 for a drug delivery system according to one aspect of the present invention is shown. The drive assembly 500 includes an actuation button 506, a container 508, a needle actuator assembly 510, an actuation release or flipper 512, a lead screw 514, and a plunger 516. The lead screw includes a drum portion 518 with external radially-protruding vanes 520, and, as best shown in FIGS. 84 and 85 and subsequently described in greater detail, a screw thread portion 522. Prior to activation, as best shown in FIGS. 83 and 86 one end 513

of the actuation release **512** engages one of the vanes **520** to prevent rotation of the lead screw **514**.

[0216] According to one aspect, as shown in FIGS. **84-86**, the screw thread portion **522** of the lead screw **514** engages internal threads of a nut **524** connected with the plunger **516**. According to another aspect, the nut and its internal threads are integrally formed with the plunger as a unitary structure. Additionally, a constant force spring **526** is received within the drum portion **518** and biases the lead screw **514** in a rotational direction. According to one aspect, the spring **526** is secured to the base cover **504**. According to another aspect, as shown in FIGS. **84-86**, a drive assembly housing **528** is disposed within the system and the spring **526** is secured to the power pack housing **528**.

[0217] Unlike a helical spring, such as a compression spring, which has a force profile proportional to its displacement, the constant force spring **526** and the like maintain a relatively flat or even force profile over a long working length. The even force profile advantageously provides an injection force that is proportional to the spring force. This will provide a flat or even injection force, and thus, a substantially constant injection rate for the medicament. Although the spring **526** is illustrated in FIG. **86** as having only two turns of material, one skilled in the art will appreciate that fewer or greater numbers of turns can be employed. Preferably, an assembler winds the spring **526** when the drive assembly **500** is assembled, and the spring **526** is stored in the wound position until the time of actuation.

[0218] Upon actuation of the system, the needle actuator assembly **510** is released to axially displace (to the right in FIGS. **82-85**) from the pre-use position to the post-use position under the influence of a biasing member **530** (best shown in FIG. **83**). During this displacement, the needle actuator assembly **510** bears against a second end **532** of the actuation release **512** and rotates the release **512** counter-clockwise, as shown in FIG. **87**. This counter-clockwise rotation of the actuation release **512** frees the first end **513** thereof from engagement with the vane **520**. Subsequent to the disengagement of the first end **513** from the vane **520**, the spring **526** unwinds and drives rotation of the lead screw **514**, which, in combination with the nut **524**, advances the plunger **514** to dispense the medicament.

[0219] As the lead screw **514** is rotating, the rotation of the drum portion **518** and the vanes **520** is visible through a window **534** in the housing. This window **534** indicates progress of the screw in a way that is much more apparent than viewing the linear movement of the stopper **536** in the container **508**. In fact, this rotational movement is many times more sensitive than the linear movement. One skilled in the art will appreciate that the exact amount of advantage or increase depends on the pitch of screw thread portion **522** of the lead screw **514**, the diameter of the drum portion **518**, and number of vanes **520** on the drum portion **518**.

[0220] Referring to FIGS. **88-93**, a drive assembly **600** for a drug delivery system according to a further aspect of the present invention is shown. The drive assembly **600** acts to store a spring's mechanical energy and to activate it when triggered. The drive assembly **600** includes a medicament barrel **601**, a stopper **602** slidably disposed in the barrel **601**, a first valve plunger **603**, a second valve plunger **604**, a first revolve nut **605**, and a second revolve nut **606**. The drive assembly **600** also includes a rotary indicator **607**, a locking element **608**, a constant force spring **609** disposed within the

rotary indicator **607**, and an actuation release or flipper **610**. The drive assembly **600** is at least partially disposed within a housing **611** that can be assembled into a drug delivery system.

[0221] The constant force spring **609** is contained between the housing **611** and the rotary indicator **607** within a drum portion **616** of the rotary indicator **607**. The drive assembly's inactive state is such that energy is applied by uncoiling the spring **609** and harnessing this energy geometrically with the housing **611**, rotary indicator **607**, and actuation release **610**. When the drive assembly **600** is deactivated, the spring recoils and translates the mechanical energy into rotational motion of the rotary indicator.

[0222] The telescoping multi-part plunger is oriented along a force axis between the medicament barrel **601** and the rotary indicator **607**. The rotary indicator **607** features a threaded shaft **618**. According to one aspect, the threads are dual lead, and are either square or rectangular in nature. The multi-part telescoping plunger includes a two-part threaded nut (first revolve nut **605** and second revolve nut **606**) and a two-part plunger (first valve plunger **603** and second valve plunger **604**). The second revolve nut **606** is a threaded shaft that mates with the rotary indicator **607** and first revolve nut **605** and features matching threads on its inner and outer surfaces (internal and external threads, respectively) to mate with them. The second revolve nut **606** also has a circular collar **620** (best shown in FIG. **92**) on its proximal end that bottoms down on the second valve plunger **604**. The second revolve nut **606** is free to spin along the force axis. The first revolve nut **605** is also a threaded shaft that features threads on its inner diameter corresponding to the external threads of the second revolve nut **606** to mate with the second revolve nut **606**.

[0223] According to one aspect, on one end, the first revolve nut **605** has a hexagonal collar that press fits on the first valve plunger **603** to fixedly connect the first valve plunger **603** with the first revolve nut **605**. In the drive assembly **600**, the first revolve nut is not free to rotate and will only translate when the power module subassembly is actuated.

[0224] The second valve plunger **604** is a hollow cylindrical component with a small collar **622** on its distal end, a large collar **624** on its proximal end, and an extended L-shaped arm **626** (best shown in FIG. **93**) protruding from the large proximal collar **624**. According to one embodiment, the small collar **622** is discontinuous and features four leaf cantilevered arms or leaf springs **623** that allow the collar to bend and mate with the first valve plunger **603**. The inner surface of the second valve plunger **604** has an undercut through its length terminating at its proximal end a radially inward protruding shelf **628** of the large collar **624**. The shelf **628** engages the second revolve nut **606** within the telescoping assembly.

[0225] The first valve plunger **603** attaches to the stopper **602** and is also a hollow cylindrical component that mates with the second valve plunger **604**. More specifically, the first valve plunger **603** features a cylindrical protrusion **630** on its distal end to mate with the stopper **602**. According to one aspect, as best shown in FIG. **89**, four thru slots **632** are disposed on the proximal quadrants of the first valve plunger **603** to mate with the leaf springs or arms **623** and small collar portion **622** of the second valve plunger **604**. Both the first and second valve plungers **603** and **604** are free to slide.

[0226] Telescoping is achieved when the constant force spring 609 recoils and the rotary indicator 607 starts spinning. The threaded attachment between the rotary indicator 607 and the second revolve nut 606 causes second revolve nut 606 to rotate. But because the second revolve nut 606 is threaded to the first revolve nut 605, which cannot rotate and experiences resistance to distal translation due to the pressure caused by medicament in the barrel 601, the second revolve nut 606 will displace proximally and bottom out on the second valve plunger's radially inward protruding shelf 628. The second valve plunger 604 is prevented from displacing proximally by the housing 611. Subsequently, and with continued rotation of the rotary indicator 607, because the second revolve nut 606 is threaded with the first revolve nut 605 (which cannot rotate) the first revolve nut 605 translates distally to push the first valve plunger 603 (and the stopper 602) to dispense medicament from the barrel 601.

[0227] The first valve plunger 603 displaces distally relative to the second valve plunger 604 until the small collar sections 622 (respectively disposed on the distal ends of the leaf springs or arms 623 of the second valve plunger 604) engage the corresponding proximal ends of the slots 632 of the first valve plunger 603. This locks the relative position of the first and second valve plungers 603 and 604, with continued rotation of the rotary indicator 607, both valve plungers translate distally while also pushing the second revolve nut along (because of its proximal engagement with the shelf 624).

[0228] The initial and final positions of the telescoping plunger, and thus the medicament dose, are controlled by the rectangular thread form of the threaded shaft 618 of the rotary indicator 607, a threaded shaft on the drum portion 616 of the rotary indicator 607, and a stepped pin that acts as the locking element 608. According to one aspect, threaded shaft on the drum portion 616 of the rotary indicator 607 is single lead, and because the rest of the components in the telescoping chain have dual lead threads, the axial travel of the other threaded components is twice the axial travel of the lock 608 relative to the rotary indicator.

[0229] According to one embodiment, the lock 608 is cylindrical and features a domed tip on one end and a cylindrical collar on the other. The threads on the exterior of the rotary indicator's drum portion 616 along with a slot and undercut 636 at the bottom of the housing 611 captures the lock 608 in place, allowing it to slide parallel to the force axis. Thus, as the spring 609 is released and the rotary indicator 607 turns, the lock 608 translates as well and creates a positive stop when the distal end of the thread on the exterior of the rotary indicator's drum portion 616 is reached.

[0230] One benefit of aspects of the drive assembly 600 include the use of a constant force spring 609, the mechanical energy of which is converted into substantially constant linear force to the medicament in the barrel 601. In turn, this creates a uniform medicament delivery rate. Another benefit is that employing the telescoping plunger driven by a thread form, the drive assembly can create in-line space savings of up to 0.75 inches compared to other plunger designs. Additionally, the drive assembly provides a controlled medicament dose through an initial and final mechanical constraint within the same component.

[0231] As previously noted, other drug delivery systems utilize a compressed coil spring, which exerts a maximum force at actuation that eventually decreases as the spring

expands. A decreasing force at the plunger translates into variable medicament delivery time and medicament exit pressure. By using a constant force spring, the force exerted on the plunger is constant from the beginning to the end of the dosage. In addition, the distance a coil spring has to travel in addition to the length of a static plunger that needs translate inside the drug container can create a long assembly. In contrast, in embodiments of the present invention, the constant force spring is contained radially and does not require any additional space before or after activation. Furthermore, the aspects of the telescoping plunger allow that the plunger length of the can be significantly reduced in comparison to the length of a static plunger.

[0232] Previous drug delivery systems have variable dose accuracy performance because the mechanical components enabling the drug delivery create a geometric dependence by bottoming down on the container, which cannot be fabricated with tight tolerances. Some embodiments of the present invention create a control to the start and end times of the translating plunger via a thread form in the rotary indicator and the use of the constant force spring.

[0233] The drive assembly creates a space saving geometry in addition to well-controlled time, volume and pressure for the drug delivery device, which translates to a more attractively compact and precise drug delivery device.

[0234] Some aspects of the drive assembly implement three rotating threaded shafts to create a linear space savings of about 0.75 inch. In other aspects, the same concept can be employed using two rotating threaded shafts and result in a space savings of about 0.5 inch. Some aspects of the present invention convert the rotational energy of a constant force spring to a translational force motion of a plunger.

[0235] Referring to FIGS. 94-100, a spacer assembly 660 according to a further aspect of the present invention is shown. The spacer assembly 660 is similar to the spacer assembly 460 discussed above and shown in FIGS. 76-78 and operates in a similar manner to achieve similar advantages. The spacer assembly 660 includes a fixed spacer 666 and an adjustable spacer 668. The fixed spacer 666 is configured to be received by the stopper 462 with lugs 670 engaging the stopper 462 to secure the fixed spacer 666 within the stopper 462, although other suitable securing arrangements, such as threads, may be utilized. The fixed spacer 666 includes interior threads 672 that receive exterior threads 678 of the adjustable spacer 668. The fixed spacer 666 includes a plurality of detents 674 positioned on a helical portion of the fixed spacer 666. The adjustable spacer 668 includes a spring detent arm 676 that engages one of the detents 674 to prevent rotation and axial displacement of the adjustable spacer 668 relative toward the fixed spacer 666. The spring detent arm 676 is shaped and configured to pass over the detents 674 in one direction to allow rotation and axial displacement of the adjustable spacer 668 away from the fixed spacer 666. The adjustable spacer 668 may be initially secured to the fixed spacer 666 via the threads 672, 678 by applying a force to the top of the spring detent arm 676, which biases the spring detent arm 676 away from the detents 674 to allow the spacers 666, 668 to be secured to each other. Accordingly, in the same manner as discussed above in connection with spacer assembly 460, the adjustable spacer is free to rotate in one axial direction to adjust the length of the spacer assembly 660.

[0236] Referring again to FIGS. 94-100, the spacer assembly 660 further includes a shim 680 configured to be

received and secured to the adjustable spacer **668**. Rather than providing a plurality of sizes of adjustable spacers **468**, **668**, a plurality of shim **680** sizes can be provided to accommodate a plurality of different fill volumes within the container **14**. The shim **680** may be secured to the adjustable spacer **668** via a connector **682** extending from the shim **680** that is received by the adjustable spacer **668** using a snap-fit, although other suitable securing arrangements may be utilized. A center portion **684** of the fixed spacer **666** is configured to be engaged while the adjustable spacer **668** is rotated relative to the fixed spacer **666** to prevent rotation of the fixed spacer **666** along with the adjustable spacer **268**. The center portion **684** of the fixed spacer **666** is accessible through an opening in the shim **680**.

[0237] Elements of one disclosed aspect can be combined with elements of one or more other disclosed aspects to form different combinations, all of which are considered to be within the scope of the present invention.

[0238] While this disclosure has been described as having exemplary designs, the present disclosure can be further modified within the spirit and scope of this disclosure. This application is therefore intended to cover any variations, uses, or adaptations of the disclosure using its general principles. Further, this application is intended to cover such departures from the present disclosure as come within known or customary practice in the art to which this disclosure pertains and which fall within the limits of the appended claims.

What is claimed is:

1. A drug delivery system comprising:
 - a container configured to hold a medicament;
 - a valve assembly;
 - a stopper configured to move within the container to dispense the medicament from the container; and
 - a drive assembly configured to move the stopper within the container, the drive assembly comprising:
 - a first plunger member configured to move axially between a pre-use position, a use position, and a post-use position;
 - a second plunger member received by the first plunger member;
 - a first biasing member received by the second plunger member and configured to move the stopper within the container; and
 - a second biasing member received by the second plunger member and configured to drive the first plunger member against the stopper to move the container into engagement with the valve assembly, wherein the second plunger member is configured to move axially after the first plunger member has moved a predetermined axial distance.
2. The drug delivery system according to claim 1, wherein the drive assembly further comprises a plunger actuation member configured for engagement and release from the first plunger member.
3. The drug delivery system according to claim 2, wherein the plunger actuation member is rotationally moveable relative to the first plunger member between a first rotational position and a second rotational position.
4. The drug delivery system according to claim 3, wherein the first rotational position is 15 degrees from the second rotational position.
5. The drug delivery system according to claim 2, wherein the plunger actuation member comprises a plurality of

projections configured to be received by a plurality of slots or cutouts defined by the first plunger member.

6. The drug delivery system according to claim 5, wherein, in a first rotational position of the plunger actuation member, the plurality of projections and the plurality of slots or cutouts are out of alignment such that the plunger actuation member is engaged with the first plunger member to prevent movement of the first plunger member and the second plunger member, with the first biasing member and the second biasing members biasing the first plunger member and the second plunger member away from the plunger actuation member.

7. The drug delivery system according to claim 6, wherein, in a second rotational position of the plunger actuation member, the plurality of projections and the plurality of slots or cutouts are aligned with each other such that the plunger actuation member is disengaged with the first plunger member to allow movement of the first plunger member and the second plunger member.

8. The drug delivery system according to claim 2, further comprising a needle actuator assembly configured to engage a drive surface of the plunger actuation member to move the plunger actuation member from the first rotational position to the second rotational position.

9. The drug delivery system according to claim 8, wherein the drive surface comprises a plurality of gears.

10. The drug delivery system according to claim 1, wherein the first plunger member forms a telescoping arrangement with the second plunger member.

11. The drug delivery system according to claim 1, wherein each of the first biasing member and the second biasing member comprises a compression spring.

12. The drug delivery system according to claim 1, wherein the first biasing member is received by the second plunger member and is constrained between the plunger actuation member and a first spring seat of the second plunger member.

13. The drug delivery system according to claim 1, wherein the second biasing member is constrained between a second spring seat of the second plunger member and the first plunger member.

14. The drug delivery system according to claim 1, wherein the second biasing member is stiffer than the first biasing member.

15. A drug delivery system comprising:

- a container configured to hold a medicament;
- a needle;
- a valve assembly configured to pierce a closure of the container to place the container in fluid communication with the needle;
- a stopper configured to move within the container to dispense the medicament from the container to the needle; and
- a drive assembly configured to move the stopper within the container, the drive assembly comprising:
 - a first plunger member configured to move axially between a pre-use position, a use position, and a post-use position;
 - a second plunger member received by the first plunger member;
 - a plunger actuation member configured for engagement and release from the first plunger member;
 - a first biasing member positioned between the second plunger member and the plunger actuation member,

the first biasing member configured to move the stopper within the container; and
a second biasing member positioned between the second plunger member and the first plunger member, the second biasing member configured to drive the first plunger member against the stopper to move the container into engagement with the valve assembly, wherein the second plunger member is configured to move axially after the first plunger member has moved a predetermined axial distance.

16. The drug delivery system according to claim **15**, wherein the plunger actuation member is rotationally moveable relative to the first plunger member between a first rotational position and a second rotational position.

17. The drug delivery system according to claim **16**, wherein the plunger actuation member comprises a plurality of projections configured to be received by a plurality of slots or cutouts defined by the first plunger member.

18. The drug delivery system according to claim **17**, wherein, in the first rotational position of the plunger actuation member, the plurality of projections and the plurality of slots or cutouts are out of alignment such that the plunger actuation member is engaged with the first plunger member to prevent movement of the first plunger member and the second plunger member, with the first biasing member and the second biasing members biasing the first plunger member and the second plunger member away from the plunger actuation member.

19. The drug delivery system according to claim **18**, wherein, in the second rotational position of the plunger actuation member, the plurality of projections and the plurality of slots or cutouts are aligned with each other such that the plunger actuation member is disengaged with the first plunger member to allow movement of the first plunger member and the second plunger member.

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